Proper Base Change
Detailed Proof Extraction, Dependency Ledger, and Exact Gaps

Extracted from the current Weil-conjectures proof blueprint

11 August 2026

Status contract. This document extracts the complete printed S0–S10 proof process from the frozen manuscript. Every original `\uses` declaration is preserved and visible. The extracted prose is not treated as an unconditional closure certificate: the corrected-v4 interfaces listed in the gap ledger remain `OPEN`. A statement called a proposition in the extracted text is therefore read as a conditional reassembly whenever one of its exact inputs is still open.

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Chapter 1

Target, scope, and proof spine

Fix a complete discrete valuation ring Λ with maximal ideal $\mathfrak{m}$, put $\Lambda_n = \Lambda/\mathfrak{m}^n$ for $n \geq 1$, and assume that ℓ is invertible on every scheme under consideration. All schemes in the ordinary finite-level statement are noetherian and finite-dimensional. The functors are first formed in the ambient derived categories and are restricted to constructible objects only through the separately named constructibility inputs.

Open interface 1.0.1 (Proper base change target)**.** For a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array} \qquad (f \text{ proper and finitely presented}),$$

and for $K \in D_c^+(X, \Lambda_n)$, construct the Beck–Chevalley mate

$$\mathrm{Ex}_{\square,K} : Lg^* Rf_* K \longrightarrow Rf'_* Lg'^* K,$$

prove that it is an isomorphism, and prove naturality, identity-square, horizontal-pasting, vertical-pasting, coefficient, localization, projection formula, and Frobenius compatibilities in the coefficient layer where each expression is typed.

Uses: `OBL-BA31-S0-GLOBAL-MATE`, `OBL-BA31-S1-FINITE`, `OBL-BA31-S2-STABILITY`, `OBL-BA31-S3-PROJECTIVE-REDUCTION`, `OBL-BA31-S4-HENSELIAN-P1`, `OBL-BA31-S5-STALK-COMPARISON`, `OBL-BA31-S6-SHEAF-ASSEMBLY`, `OBL-BA31-S7-BOUNDED-BELOW`

Open obligation: `OBL-PROPER-BASE-CHANGE`

Conditional theorem 1.0.2 (Reassembly represented by S0–S10)**.** Assume every exact terminal input in Appendix A, together with every imported interface whose manifest status is not closed. Then the ordered S0–S7 argument proves finite-torsion proper base change in $D_c^+(X_{\mathrm{et}})$; S8 transports the comparison to finite-level pro-etale objects; and S9–S10 prove the completed comparison under the displayed perfect-quotient, projection-formula, reconstruction, and $R\varprojlim$ hypotheses.

Uses: `OBL-BA31-FINITE-PBC`, `OBL-SIX-FUNCTOR-ADIC`

Proof. The detailed construction and every intermediate proof are reproduced in Chapters 3 and 5. Appendix A records the exact unproved hypotheses, so this theorem does not promote the historical owner. □

1.1 Ordered dependency spine

S1. Construct the single global unit–Cartesian-identification–counit mate.

S2. Prove the finite-morphism fibre formula and exactness.

S3. Prove composition stability and proper-surjective cancellation.

S4. Use Chow reduction and the finite symmetric-product map to reduce to $\mathbf{P}^1$.

S5. Prove the strict-henselian $\mathbf{P}^1$ comparison from degree zero, affine Gabber comparison, and finite-etale torsor lifting.

S6. Identify the global stalk map using strict henselization and proper-curve invariance under separably closed extension.

S7. Assemble prime and arbitrary torsion coefficients, then pass to D^+ by a bounded-below hypercohomology spectral sequence.

S8. Transport finite levels to the pro-etale site.

S9. Reconstruct completed coefficients by perfect reduction and $R\varprojlim$.

S10. Verify naturality, identity, pasting, and typed Frobenius coherence.

Status. The proof order above is a reduction order. Formal status may be promoted only in the reverse dependency direction: exact leaves first, then S0–S7, then finite PBC, and only later the completed and compact-support exports.

Chapter 2

Corrected typed finite-level interface

Status. This chapter is the authoritative typing boundary for the ordinary finite-level proof. It replaces the ill-typed historical sectionwise S0 formula. Every named S0–S7 middle owner remains OPEN/INCOMPLETE, and every atomic input is listed in Appendix A.

For a Cartesian square of noetherian finite-dimensional schemes

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y, \end{array}$$

and $K \in D_c^+(X, \Lambda_n)$, first construct in the ambient category the single derived morphism

$$\begin{aligned} \mathrm{BC}_{f,g,K} : Lg^* Rf_* K &\xrightarrow{\eta_{f'}} Rf'_* Lf'^* Lg^* Rf_* K \\ &\xrightarrow{\text{cart}} Rf'_* Lg'^* Lf^* Rf_* K \xrightarrow{Rf'_*(Lg'^* \varepsilon_f)} Rf'_* Lg'^* K. \end{aligned} \tag{BA31-MATE}$$

Replacement independence and naturality concern this fixed map. Its identity, horizontal-pasting, and vertical-pasting equations must be proved by adjunction and mate calculus.

Two stalk facts have disjoint types and disjoint uses:

1. `CAT-BA31-SHEAF-STALK-FAITHFULNESS`: if $u, v : \mathcal{F} \to \mathcal{G}$ are maps in the abelian category of Λ_n-sheaves and $u_{\bar{y}} = v_{\bar{y}}$ at every geometric point, then $u = v$. It is consumed only in S5, after the maps of cohomology sheaves have already been constructed.
2. `CAT-BA31-DERIVED-STALK-ISO-CONSERVATIVITY`: if $u : A \to B$ is an already constructed morphism in $D^+(Y'_{\acute{e}t}, \Lambda_n)$ and every $H^i(u)_{\bar{y}}$ is invertible, then u is invertible. It is consumed only in S6.

No stalk principle in this fragment detects equality of arbitrary derived morphisms or constructs a derived inverse from pointwise inverses.

S0–S7 middle interfaces. The following are literal middle owners and all remain `OPEN/INCOMPLETE`.

- `OBL-BA31-S0-GLOBAL-MATE` constructs (BA31-MATE) and proves replacement independence, naturality, identity, and both pasting laws from the three named categorical atoms.
- `OBL-BA31-S1-FINITE` proves exact finite pushforward and the finite geometric-fibre product formula, then identifies the latter with the stalk of the already constructed mate.
- `OBL-BA31-S2-STABILITY` proves typed composition and proper-surjective cancellation diagrams for the actual Rf_* pseudofunctor; no inverse is supplied as input data.
- `OBL-BA31-S3-PROJECTIVE-REDUCTION` states a proper finitely presented Chow datum and a finite-surjective retract/cancellation theorem reducing projective space to products of projective lines.
- `OBL-BA31-S4-HENSELIAN-P1` assembles independently owned degree-zero, affine, and torsor/degree-one theorems into the strict-henselian projective-line comparison.

- `OBL-BA31-S5-STALK-COMPARISON` combines the strict-henselization stalk formula and proper-curve field invariance and identifies their comparison with $H^i(\mathrm{BC}_{f,g,K})$. Sheaf-stalk faithfulness is used only for this equality of maps of H^i-sheaves.
- `OBL-BA31-S6-SHEAF-ASSEMBLY` applies only derived-stalk isomorphism conservativity to the fixed mate after S1–S5 have proved all its cohomology-stalk maps invertible.
- `OBL-BA31-S7-BOUNDED-BELOW` uses the selected bounded-below Postnikov/Cartan–Eilenberg leaf, including convergence and identification of the filtered comparison with (BA31-MATE).

The root `OBL-BA31-FINITE-PBC` remains `OPEN/INCOMPLETE` until these middle nodes and every listed atom are actually proved. The historical finite stalk-descent certificate is retired and has no edge into this graph.

Chapter 3

Detailed S0–S10 proof text

Status. The following block preserves the mathematical S0–S10 body and every original \usesdeclaration. The sole source-level correction is the ill-typed S0 sectionwise formula, replaced in place by the categorical mate and an explicit erratum. The remaining propositions are conditional reassembly steps whenever their gap-ledger inputs are OPEN.

Chapter 4

Stacks route: comparison through the strictly henselian projective line

Status. This file is the S0–S4 core of the A3 DAG `research/a3_stacks_proper_base_change_dag.md`. The coefficient sheaf is a torsion sheaf on the small etale site. A Stacks tag identifies the source paragraph to be audited; it is never used as an admitted theorem.

4.1 S0. Comparison data and injective acyclicity

Definition 4.1.1 (Cartesian square and chosen pullback identification)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-cartesian-square`

For schemes and morphisms

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y, \end{array}$$

we write $\square = (f, f', g, g', \alpha)$ when $\alpha : X' \xrightarrow{\sim} X \times_Y Y'$ is a specified isomorphism and $f \circ g' = g \circ f'$ as maps to Y. The notation $g'^{-1}F$ always means inverse image using this specified identification.

Definition 4.1.2 (Underived exchange map: corrected categorical construction)**.** Lean: `conditional-interface:def:weil-pbc-underived-exchange`

Uses: `def:weil-pbc-cartesian-square`

Let F be a sheaf of abelian groups on X_{et}. The chosen Cartesian identification gives a canonical isomorphism of inverse-image functors

$$f'^{-1}g^{-1} \xrightarrow{\sim} g'^{-1}f^{-1}.$$

Define $\mathrm{Ex}_{\square,F}$ to be the adjoint of the composite

$$f'^{-1}g^{-1}f_*F \xrightarrow{\sim} g'^{-1}f^{-1}f_*F \xrightarrow{g'^{-1}\varepsilon_f} g'^{-1}F.$$

Thus

$$\mathrm{Ex}_{\square,F} : g^{-1}f_*F \longrightarrow f'_*g'^{-1}F$$

is a globally constructed morphism of sheaves. Naturality, identity, and pasting are properties of this fixed mate; no family of stalk maps is used to construct it.

Warning 4.1.3 (Erratum to the historical sectionwise presentation). The frozen draft represented a local section by a section on $V_i \times_{Y'} Y$. The square only supplies $g : Y' \to Y$, not a map $Y \to Y'$, so that fibre product is not defined. The categorical unit–Cartesian-identification–counit construction above is the replacement used throughout this extraction. The removed historical block has SHA-256 `E982FFC21CEAF8ABCD945A58DDFADBA4EB898F882D56A00A70BAC9F6365679A0` and is not a proof input.

Lemma 4.1.4 (Sectionwise exchange is natural)**.** *Lean:* `conditional-interface:lemma:weil-pbc-underived-exchange-naturality`

Uses: `def:weil-pbc-underived-exchange`

For every morphism $u : F \to G$ the square

$$\begin{array}{ccc} g^{-1}f_*F & \xrightarrow{\mathrm{Ex}_{\square,F}} & f'_*g'^{-1}F \\ {\scriptstyle g^{-1}f_*u}\downarrow & & \downarrow{\scriptstyle f'_*g'^{-1}u} \\ g^{-1}f_*G & \xrightarrow{\mathrm{Ex}_{\square,G}} & f'_*g'^{-1}G \end{array}$$

commutes on every etale object of Y'.

Proof. Uses: `def:weil-pbc-underived-exchange`

On a representative section s_i, both paths first apply u and then pull back along the same two maps. Since inverse image is a functor, the two resulting sections of $g'^{-1}G$ are equal on each V_i and therefore after descent on V. Equality for all V is equality of sheaf morphisms. □

Lemma 4.1.5 (Finite refinement of a surjective etale morphism)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-finite-refinement-etale-cover`

Let $U \to T$ be surjective and etale, with T quasi-compact and quasi-separated. There is a finite, finitely presented, surjective morphism $r : T' \to T$ and a finite open covering $T' = \bigcup_a W_a$ such that every $W_a \to T$ factors through $U \to T$.

Proof. Uses:

Choose finitely many affine opens $U_a \subset U$ whose images cover T and replace U by their disjoint union. The resulting map is quasi-compact, separated, etale, and has a finite upper bound d on the cardinalities of its geometric fibres. We induct on d. If $d = 1$, the map is radicial and etale, hence an open immersion; surjectivity makes it an isomorphism.

For $d > 1$, apply the affine form of Zariski's Main Theorem to the separated quasi-finite map $U \to T$: adjoining to $\mathcal{O}_T$ the integral elements of the quasi-coherent algebra $u_*\mathcal{O}_U$ gives an integral $\pi : \widetilde{T} \to T$ and a quasi-compact open immersion $j : U \hookrightarrow \widetilde{T}$. Replace $\widetilde{T}$ by the scheme-theoretic closure of U, so U is schematically dense. The diagonal section gives an open-and-closed decomposition

$$U \times_T \widetilde{T} = U \amalg W.$$

The image V of $W \to \widetilde{T}$ is open and contains $\widetilde{T} \setminus U$. Over a point of the dense open U, the fibre of $W \to V$ is the fibre of $U \to T$ with the diagonal point removed, so it has fewer than d points. The number of points in a separated etale fibre cannot increase under specialization: after writing the map locally as a finite disjoint union of Jacobian etale algebra presentations, distinct idempotents at a specialization lift to distinct idempotents at a generization. Thus every geometric fibre of $W \to V$ has cardinality $< d$.

By induction there is an integral surjection $Z \to V$, locally factoring through W. Apply the same integral-closure construction to $Z \to \widetilde{T}$ and take the closure Z' of Z; then $Z' \times_{\widetilde{T}} V = Z$. If C is the reduced closed complement $\widetilde{T} \setminus V$, then $C \subset U$, and $Z' \amalg C \to T$ is integral and surjective. Near a point of Z it factors through $W \to U$; near a point of $Z' \setminus Z$ it factors through $C \subset U$. Hence it locally factors through the original cover.

Finally write this integral morphism as the inverse limit of its finite finitely presented T-subalgebras. Choose finitely many affine opens on which the factorization through U is defined. Each open, each map to U, the equalities over T, and surjectivity descend to a common finite stage: the first three assertions are finite-presentation data, while failure of surjectivity has quasi-compact open image and would persist after pullback to the limit. That common stage is the required T'. This is the construction located at Tag `09Z0`, with the degree induction and final finite stage made explicit. □

Lemma 4.1.6 (Degree-zero comparison over a strictly henselian base)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-proper-strict-henselian-h0`

Let $(A, \mathfrak{m})$ be strictly henselian local, let $h : T \to \operatorname{Spec} A$ be proper, and let T_0 be the closed geometric fibre. For every etale sheaf of sets E on T, restriction is a bijection

$$\Gamma(T, E) \xrightarrow{\sim} \Gamma(T_0, E|_{T_0}).$$

The bijection is functorial in E and in finite morphisms over T.

Proof. Uses: `lemma:weil-pbc-finite-refinement-etale-cover`

We first prove the corresponding assertion for Zariski sheaves, uniformly after every finite $T' \to T$. The space T' is spectral and $T' \to \operatorname{Spec} A$ is proper. For an irreducible closed reduced $C \subset T'$, Stein factorization gives

$$C \longrightarrow \operatorname{Spec} B \longrightarrow \operatorname{Spec} A,$$

where $A \to B$ is integral and the first map has geometrically connected fibres. The pair $(B, \mathfrak{m}B)$ is henselian: an idempotent of a finite B-algebra modulo $\mathfrak{m}$ satisfies $e^2 - e = 0$, and the simple roots $0, 1$ lift uniquely by the henselian property of A after restriction of scalars along the integral algebra B. Consequently the closed subset $\operatorname{Spec} B \cap V(\mathfrak{m})$ is connected and nonempty. Its inverse image $C \cap T'_0$ is connected because the map is closed, surjective onto its image, and has connected fibres.

For completeness, let G be a Zariski sheaf on T'. Every point specializes to T'_0, so a global section is determined by its restriction. For existence, write G as the filtered union of subsheaves generated by finitely many sections on quasi-compact opens. Such a finitely generated subsheaf embeds into a finite product $\prod_b (i_b)_* \underline{M_b}$, where $i_b : C_b \hookrightarrow T'$ is closed and M_b is a finite set: take the finitely many values of the generators and separate their equalizer loci. Sections of $(i_b)_*\underline{M_b}$ are locally constant maps $C_b \to M_b$. Connectedness of every $C \cap T'_0$ says that the connected components of C_b and $C_b \cap T'_0$ correspond, so such maps extend uniquely. Finite products, intersections with the embedded subsheaf, and filtered colimits preserve this equality of section sets. Thus $\Gamma(T', G) = \Gamma(T'_0, G|)$ for every Zariski sheaf and every finite $T' \to T$.

Now let E be an etale sheaf. The map from the restriction to the Zariski site followed by closed-fibre inverse image to the Zariski restriction of $E|_{T_0}$ is stalkwise injective. Indeed, an elementary etale neighbourhood $V \to T$ whose closed fibre has a local section has open image $V_0 \subset T$, and $V \to V_0$ is an etale cover; hence

$$E(V_0) \longrightarrow E(V) \rightrightarrows E(V \times_{V_0} V)$$

is an equalizer. Taking the colimit over such neighbourhoods proves the stalkwise injection. The Zariski result therefore proves injectivity of $\Gamma(T, E) \to \Gamma(T_0, E|)$, and the same argument applies after every finite $T' \to T$.

For surjectivity, take $t_0 \in \Gamma(T_0, E|)$. By the definition of inverse image and sheafification there are finitely many etale maps $V_a \to T_0$ covering T_0, etale $U_a \to T$, maps $V_a \to U_a \times_T T_0$, and sections $s_a \in E(U_a)$ representing $t_0|_{V_a}$. Shrink U_a to the open image of V_a in $U_a \times_T T_0$ and then in U_a; the sheaf equalizer shows that $s_a|_{U_a \times_T T_0} = t_0|_{U_a \times_T T_0}$. Every nonempty closed subset of T meets T_0 by the connectedness argument above, so $\coprod_a U_a \to T$ is surjective.

Choose the finite surjective refinement $r : T' \to T$ of the preceding lemma and opens $W_a \subset T'$ mapping to U_a. Pulling back s_a gives local sections on W_a. On $W_a \cap W_b$ their closed-fibre restrictions both equal $r_0^{-1}t_0$; the already proved injectivity

for the finite scheme $T' \to T$ makes them equal. They glue to $s' \in \Gamma(T', r^{-1}E)$. The two pullbacks of s' to $T' \times_T T'$ have the same closed-fibre restriction, and injectivity for the finite morphism $T' \times_T T' \to T$ makes them equal.

It remains to spell out descent along r, which need not be etale. At a geometric point $\bar{x} \to T$, choose a geometric $\bar{x}' \to T'$ above it. The germ of s' at $\bar{x}'$ is an element of the stalk $E_{\bar{x}}$. The equalizer equality on $T' \times_T T'$ makes this element independent of $\bar{x}'$. Represent it on an etale neighbourhood of $\bar{x}$ in T. After shrinking, its pullback equals s' near every point over $\bar{x}$; finiteness of r makes that inverse image quasi-compact, so one shrinking works for all of them. These local descended sections agree on overlaps because equality can be checked at geometric stalks, and hence glue to $s \in \Gamma(T, E)$. Its restriction is t_0 by injectivity on T_0. All pullbacks used are the functorial sheafification units, so the argument also proves functoriality in E and compatibility with finite pullback. The source locators are Tags `09ZG` and `09ZF`; neither is used as an unexpanded conclusion. □

Lemma 4.1.7 (Underived proper exchange is an isomorphism)**.** Lean: `conditional-interface:lemma:weil-pbc-proper-underived-exchange`

Uses: `def:weil-pbc-underived-exchange`

If f is proper, then $\mathrm{Ex}_{\square,F} : g^{-1}f_*F \to f'_*g'^{-1}F$ is an isomorphism for every sheaf F.

Proof. Uses: `lemma:weil-pbc-underived-exchange-naturality,lemma:weil-pbc-proper-strict-henselian-h0`

Fix a geometric point $\bar{y}' \to Y'$ and let $\bar{y} \to Y$ be its composite with g. Let $\mathcal{N}^{\mathrm{aff}}_{\bar{y}}$ be the category of pointed affine etale neighbourhoods $(V, \bar{v})$ of $\bar{y}$. It is cofiltered: an affine open containing the lifted point in a fibre product refines two objects, and an affine neighbourhood of the equalizer of two maps makes parallel maps equal. It is cofinal among all pointed etale neighbourhoods because every lifted point has an affine open neighbourhood. Thus

$$(f_*F)_{\bar{y}} = \varinjlim_{(V,\bar{v})} \Gamma(X \times_Y V, F|).$$

Writing $V = \operatorname{Spec} A_V$, the strict henselization is $Y_{(\bar{y})} = \operatorname{Spec}(\varinjlim A_V)$ and $X_{(\bar{y})} = X \times_Y Y_{(\bar{y})} = \varprojlim(X \times_Y V)$. The natural map from the displayed colimit to $\Gamma(X_{(\bar{y})}, F|)$ is bijective. For surjectivity, represent a section on the limit by sections on a finite cover by quasi-compact etale objects. Each object, each covering map, and each section representative descends to a stage because etale morphisms are locally of finite presentation. The finitely many equality conditions on the pairwise fibre products descend after enlarging that stage. The sheaf equalizer then glues the descended representatives. For injectivity, represent two sections at one stage; equality on the limit is witnessed on a finite etale cover, and the cover together with its finitely many equality witnesses descends to a later stage. Hence the two representatives agree in the colimit.

The base $Y_{(\bar{y})}$ is strictly henselian local and $X_{(\bar{y})}$ is proper over it. The preceding lemma identifies its sections with those on the closed geometric fibre:

$$(f_*F)_{\bar{y}} \xrightarrow{\sim} \Gamma(X_{(\bar{y})}, F|) \xrightarrow{\sim} \Gamma(X_{\bar{y}}, F|_{X_{\bar{y}}}).$$

The identical construction for f' gives the target stalk. Since $\bar{y}$ is the composite of the same $\operatorname{Spec}\Omega \to Y'$ with g, the Cartesian isomorphism identifies $X_{\bar{y}}$ and $X'_{\bar{y}'}$ with $X \times_Y \operatorname{Spec}\Omega$.

Finally, a germ represented by $s \in \Gamma(X \times_Y V, F|)$ travels around the comparison by pulling s first to $X' \times_{Y'} (V \times_Y Y')$ and then to the geometric fibre. The two stalk identifications above pull s directly to that same fibre. Hence the fibre bijection is the stalk of $\mathrm{Ex}_{\square,F}$, not an unnamed isomorphism. Geometric stalks are conservative, so the exchange map is an isomorphism. This closes the compatibility omission at Tag `0A3U`. □

Definition 4.1.8 (Derived exchange map)**.** Lean: `conditional-interface:def:weil-pbc-derived-exchange`

Uses: `def:weil-pbc-underived-exchange`

For $K \in D^+(X_{\mathrm{et}})$ choose an injective resolution $K \xrightarrow{\simeq} I^\bullet$. Apply $\mathrm{Ex}_{\square,I^r}$ in every degree and use the naturality lemma to obtain a morphism of complexes

$$g^{-1}f_*I^\bullet \longrightarrow f'_*g'^{-1}I^\bullet.$$

Choose a K-injective replacement $g'^{-1}I^\bullet \xrightarrow{\simeq} J^\bullet$ in the Grothendieck category of etale abelian sheaves on X'. Compose the displayed map with $f'_*g'^{-1}I^\bullet \to f'_*J^\bullet$ and define

$$\mathrm{Ex}_{\square,K} : g^{-1}f_*I^\bullet \longrightarrow f'_*J^\bullet$$

to be its class in $D^+(Y'_{\mathrm{et}})$. The target complex is a model for $Rf'_*g'^{-1}K$, since $J^\bullet$ is K-injective; the source is a model for $g^{-1}Rf_*K$ because g^{-1} is exact. Its class in $D^+(Y'_{\mathrm{et}})$ is

$$\mathrm{Ex}_{\square,K} : g^{-1}Rf_*K \longrightarrow Rf'_*g'^{-1}K.$$

Derived Exchange Comparison Chain

Definition 4.1.9 (Derived-exchange presentation)**.** Lean: `conditional-interface:def:weil-pbc-derived-exchange-presentation-v5`

Uses: `def:weil-pbc-derived-exchange`

For a bounded-below complex K, a presentation consists of an injective resolution $u : K \to I$, a K-injective complex J, a quasi-isomorphism $q : g'^{-1}I \to J$, and the complex map

$$E_I : g^{-1}f_*I \longrightarrow f'_*g'^{-1}I$$

obtained degreewise from the underived exchange. Write $E_{I,J} := f'_*q\,E_I$.

Definition 4.1.10 (Hom complex and K-injective test)**.** Lean: `conditional-interface:def:weil-pbc-derived-exchange-hom-test-v5`

Uses: `def:weil-pbc-derived-exchange-presentation-v5`

For complexes A, B, let $\underline{\mathrm{Hom}}^r(A,B)$ be the degree-r graded maps, with differential $d(h) = d_B h - (-1)^r h d_A$. A complex B is K-injective when this Hom complex is acyclic for every acyclic A.

Lemma 4.1.11 (The Hom differential squares to zero)**.** Lean: `blueprint-target:lemma:weil-pbc-derived-exchange-hom-differential-v5`

Uses: `def:weil-pbc-derived-exchange-hom-test-v5`

The displayed Hom differential satisfies $d^2 = 0$.

Proof. Uses: `def:weil-pbc-derived-exchange-hom-test-v5`

For homogeneous h of degree r, expanding $d(dh)$ gives $d_B^2 h - (-1)^r d_B h d_A - (-1)^{r+1} d_B h d_A - h d_A^2 = 0$. □

Lemma 4.1.12 (Bounded-below injectives are K-injective)**.** Lean: `blueprint-target:lemma:weil-pbc-derived-exchange-bounded-injective-kinjective-v5`

Uses: `def:weil-pbc-derived-exchange-hom-test-v5`

Every bounded-below complex whose terms are injective is K-injective.

Proof. Uses: `def:weil-pbc-derived-exchange-hom-test-v5,lemma:weil-pbc-derived-exchange-hom-differential-v5`

After shifting the target, it suffices to treat a degree-zero Hom cocycle $f : C \to I$. Choose m with $I^n = 0$ for $n < m$, and construct maps $s^n : C^n \to I^{n-1}$ upward from $n = m$ so that

$$f^n = ds^n + s^{n+1}d. \tag{*}$$

Assume $s^m, \dots, s^n$ are known (with $s^m = 0$ at the initial step). On $B^{n+1} = \mathrm{im}(d : C^n \to C^{n+1}) = \ker d$, define

$$t(dx) := f^n(x) - ds^n(x) \in I^n.$$

This is well-defined: if $dx = 0$, acyclicity gives $x = dy$, and the cocycle equation together with (*) in degree $n-1$ gives $f^n(x) = df^{n-1}(y) = ds^n(dy) = ds^n(x)$. Injectivity extends the map $t : B^{n+1} \to I^n$ along the monomorphism $B^{n+1} \hookrightarrow C^{n+1}$ to $s^{n+1} : C^{n+1} \to I^n$. Its defining equality is (*) in degree n. Thus $f = ds$, so every Hom cocycle is a coboundary. □

Definition 4.1.13 (Target-replacement comparison problem)**.** Lean: `conditional-interface:def:weil-pbc-derived-exchange-target-comparison-v5`

Uses: `def:weil-pbc-derived-exchange-presentation-v5`

For two target replacements $q : A \to J$ and $q' : A \to J'$ with J, J' K-injective, a comparison is a chain map $c : J \to J'$ together with a homotopy $cq \simeq q'$.

Lemma 4.1.14 (Target-replacement comparisons exist)**.** Lean: `blueprint-target:lemma:weil-pbc-derived-exchange-target-comparison-exists-v5`

Uses: `def:weil-pbc-derived-exchange-target-comparison-v5, def:weil-pbc-derived-exchange-hom-test-v5`

Every target-replacement comparison problem has a comparison.

Proof. Uses: `def:weil-pbc-derived-exchange-target-comparison-v5,def:weil-pbc-derived-exchange-hom-test-v5`

The cone $C(q)$ is acyclic because q is a quasi-isomorphism. K-injectivity of J' makes $\underline{\mathrm{Hom}}(C(q), J')$ acyclic. The cone triangle gives the exact segment

$$H^0\underline{\mathrm{Hom}}(J, J') \longrightarrow H^0\underline{\mathrm{Hom}}(A, J') \longrightarrow H^1\underline{\mathrm{Hom}}(C(q), J') = 0.$$

Thus $[q']$ lifts to $[c]$; equality $[cq] = [q']$ is exactly a homotopy $cq \simeq q'$. □

Lemma 4.1.15 (Target comparisons are unique up to homotopy)**.** Lean: `blueprint-target:lemma:weil-pbc-derived-exchange-target-comparison-unique-v5`

Uses: `def:weil-pbc-derived-exchange-target-comparison-v5, def:weil-pbc-derived-exchange-hom-test-v5`

Two comparisons $c, d : J \to J'$ are homotopic. Comparisons in both directions are inverse to one another in the homotopy category.

Proof. Uses: `def:weil-pbc-derived-exchange-target-comparison-v5,def:weil-pbc-derived-exchange-hom-test-v5`

The difference $(c - d)q$ is null-homotopic. Since $\underline{\mathrm{Hom}}(C(q), J')$ is acyclic, the cone exact segment

$$H^0\underline{\mathrm{Hom}}(C(q), J') = 0 \longrightarrow H^0\underline{\mathrm{Hom}}(J, J') \xrightarrow{q^*} H^0\underline{\mathrm{Hom}}(A, J')$$

makes q^* injective; hence $c - d$ is null-homotopic. If $c : J \to J'$ and $c' : J' \to J$ are comparisons, both $c'c$ and 1_J compare q with itself, and both cc' and $1_{J'}$ compare q' with itself. The first assertion therefore gives $c'c \simeq 1_J$ and $cc' \simeq 1_{J'}$. □

Definition 4.1.16 (Source-resolution comparison problem)**.** Lean: `conditional-interface:def:weil-pbc-derived-exchange-source-comparison-v5`

Uses: `def:weil-pbc-derived-exchange-presentation-v5`

For injective resolutions $u : K \to I$ and $u' : K \to I'$, a comparison is a chain map $a : I \to I'$ with $au \simeq u'$.

Lemma 4.1.17 (Source-resolution comparisons exist and invert). *Lean:* `blueprint-target:lemma:weil-pbc-derived-exchange-source-comparison-v5`

Uses: `def:weil-pbc-derived-exchange-source-comparison-v5`, `def:weil-pbc-derived-exchange-hom-test-v5`

There are comparisons $a : I \to I'$ and $b : I' \to I$, unique up to homotopy, with $ba \simeq 1_I$ and $ab \simeq 1_{I'}$.

Proof. Uses: `def:weil-pbc-derived-exchange-source-comparison-v5`, `lemma:weil-pbc-derived-exchange-bounded-injective-kinjective-v5`, `lemma:weil-pbc-derived-exchange-target-comparison-exists-v5`, `lemma:weil-pbc-derived-exchange-target-comparison-unique-v5`

The resolutions are bounded below, so the preceding lemma makes I' and I K-injective. Apply Lemma 4.1.14 with $A = K$, $q = u$, $q' = u'$, $J = I$, and $J' = I'$ to obtain a. Interchanging the two resolutions gives b. Now apply Lemma 4.1.15: ba and 1_I are two comparisons of u with itself, while ab and $1_{I'}$ compare u' with itself. It gives uniqueness up to homotopy and the two displayed inverse equations. □

Lemma 4.1.18 (Exchange naturality is a homotopy square). *Lean:* `blueprint-target:lemma:weil-pbc-derived-exchange-naturality-square-v5`

Uses: `def:weil-pbc-derived-exchange-presentation-v5`, `def:weil-pbc-derived-exchange-target-comparison-v5`

Let (I, J, q) and (I', J', q') be presentations and let $a : I \to I'$ be a source comparison. A target comparison $c : J \to J'$ for $q'g'^{-1}a$ and cq gives

$$f'_* c\, E_{I,J} \simeq E_{I',J'}\, g^{-1} f_* a.$$

Proof. Uses: `def:weil-pbc-derived-exchange-presentation-v5`, `lemma:weil-pbc-underived-exchange-naturality`, `lemma:weil-pbc-derived-exchange-target-comparison-exists-v5`

Underived naturality gives the strict degreewise equality $f'_* g'^{-1} a\, E_I = E_{I'} g^{-1} f_* a$. The defining homotopy $cq \simeq q'g'^{-1}a$ remains a homotopy after f'_*; compose it with E_I and substitute the strict equality. □

Definition 4.1.19 (Identity and pasted exchange presentations). Lean: `conditional-interface:def:weil-pbc-derived-exchange-pasting-presentations-v5`

Uses: `def:weil-pbc-derived-exchange-presentation-v5`

For an identity or a horizontally or vertically pasted Cartesian square, use one injective resolution and form the degreewise exchange maps by their unit–pullback-identification–counit formula before selecting targets.

Lemma 4.1.20 (Identity exchange equation). *Lean:* `blueprint-target:lemma:weil-pbc-derived-exchange-identity-v5`

Uses: `def:weil-pbc-derived-exchange-pasting-presentations-v5`

The exchange for an identity square is the identity in the derived category.

Proof. Uses: `def:weil-pbc-derived-exchange-pasting-presentations-v5`, `lemma:weil-pbc-derived-exchange-target-comparison-unique-v5`

The unit and counit for the identity pullback are identities on every sheaf term. Thus the degreewise exchange is 1; changing targets leaves its class unchanged by the target-comparison uniqueness argument above. □

Lemma 4.1.21 (Horizontal and vertical exchange pasting). *Lean:* `blueprint-target:lemma:weil-pbc-derived-exchange-pasting-v5`

Uses: `def:weil-pbc-derived-exchange-pasting-presentations-v5`, `def:weil-pbc-underived-exchange`, `def:weil-pbc-cartesian-square`

The exchange for a horizontal or vertical pasted square equals the composite of the exchanges for its two constituent squares in the derived category.

Proof. Uses: `def:weil-pbc-derived-exchange-pasting-presentations-v5`, `def:weil-pbc-underived-exchange`, `def:weil-pbc-cartesian-square`, `lemma:weil-pbc-derived-exchange-target-comparison-unique-v5`

Before target replacement, expand the three degreewise maps as unit, pullback identification, and counit. Associativity of fibre products makes the two middle identifications equal, and the adjunction triangle identities cancel the adjacent unit–counit pairs. The remaining maps are equal as complex maps. Any different target replacements represent this same class by target-comparison uniqueness. □

Lemma 4.1.22 (Well-definedness certificate for def:weil-pbc-derived-exchange). *Lean:* `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-derived-exchange`

Uses: `def:weil-pbc-derived-exchange`

The construction in Definition 4.1.8 is a typed raw datum. Its theorem-valued naturality, comparison, descent, or coherence fields are isolated in this property lemma.

Proof. Uses: `def:weil-pbc-derived-exchange-presentation-v5`, `lemma:weil-pbc-derived-exchange-target-comparison-unique-v5`, `lemma:weil-pbc-derived-exchange-source-comparison-v5`, `lemma:weil-pbc-derived-exchange-naturality-square-v5`, `lemma:weil-pbc-derived-exchange-identity-v5`, `lemma:weil-pbc-derived-exchange-pasting-v5`

The presentation definition constructs the complex map. Source comparisons, the naturality homotopy square, and target-comparison uniqueness identify all selected presentations in the derived category. The identity and the two pasting equations are supplied by the final two lemmas. Hence the derived exchange construction is well-defined with all fields stated in the lemma. □

Lemma 4.1.23 (Derived exchange replacement independence). Lean: `conditional-interface:lemma:weil-pbc-derived-exchange-replacement`

Uses: `def:weil-pbc-derived-exchange`

The K-injective replacement used in Definition 4.1.8 gives a well-defined derived exchange map, independent of all replacement choices.

Proof. Uses:

Given two K-injective replacements $J^\bullet$ and $J'^\bullet$, the identity of $g'^{-1}I^\bullet$ lifts to maps $J^\bullet \to J'^\bullet$ and $J'^\bullet \to J^\bullet$ because K-injective complexes are homotopically injective. Their composites are homotopic to the identities by lifting the homotopies of the replacement quasi-isomorphisms. Applying f'_* preserves chain homotopies. The two resulting maps from $g^{-1}f_*I^\bullet$ therefore define the same morphism in the derived category. A comparison of two injective resolutions of K is handled by the same lifting argument, and naturality of the underived exchange map commutes with the comparison maps. Thus the class is canonical. □

Lemma 4.1.24 (Resolution independence, identities, and pasting). Lean: `conditional-interface:lemma:weil-pbc-derived-exchange-coherence`

Uses: `def:weil-pbc-derived-exchange`

The class $\mathrm{Ex}_{\square,K}$ is independent of the injective resolution and is compatible with identity squares, horizontal pasting, and vertical pasting.

Proof. Uses: `def:weil-pbc-derived-exchange`

Given two injective resolutions, the comparison theorem supplies chain maps each way, unique up to chain homotopy, whose composites are homotopic to the identities. Naturality of the degreewise exchange map sends these homotopies to homotopies after both direct images; localization therefore gives the same derived morphism. For the identity square, the unit followed by the counit is the identity sectionwise, hence degreewise. For pasted squares, expand each exchange as "unit, pullback identification, counit"; the two middle identifications are equal by associativity of fibre products, and the two adjunction triangle identities cancel the adjacent unit–counit pairs. The same calculation with the order reversed proves vertical pasting. □

Definition 4.1.25 (Proper base-change property). Lean: `conditional-interface:def:weil-pbc-property`

Uses: `def:weil-pbc-derived-exchange`

For a proper morphism $f : X \to Y$, define $\mathsf{PBC}(f)$ to mean that for every morphism $g : Y' \to Y$, every associated Cartesian square $\square$, and every torsion sheaf F on X_{et}, the map $\mathrm{Ex}_{\square,F}$ is an isomorphism in the derived category. The assertion includes the specified map of Definition 4.1.8, not an unnamed isomorphism.

Definition 4.1.26 (Injective-acyclicity property). Lean: `conditional-interface:def:weil-pbc-injective-acyclicity-property`

Uses: `def:weil-pbc-cartesian-square`

For a proper $f : X \to Y$, define $\mathsf{IA}(f)$ to mean: for every prime ℓ, every $g : Y' \to Y$, every square $\square$, and every injective sheaf I of $\mathbf{F}_\ell$-modules on X_{et}, the sheaf $g'^{-1}I$ is f'_*-acyclic, i.e. $R^q f'_* g'^{-1} I = 0$ for all $q > 0$.

Lemma 4.1.27 (Filtered torsion union). Uses:

Lean: `conditional-interface:lemma:weil-pbc-torsion-filtered-union`

For every torsion sheaf F the inclusions $F[n] \hookrightarrow F[mn]$ form a filtered system and the canonical map $\varinjlim_n F[n] \to F$ is an isomorphism.

Proof. Uses:

The transition map sends a section killed by n to the same section, now regarded as killed by mn. The map to F is monic because a section that is zero in F is already zero in every $F[n]$. It is epic because every local section of F is killed by some positive integer and hence is locally in one of the $F[n]$; sheafification preserves this local epimorphism. □

Lemma 4.1.28 (Filtered colimits and higher direct image). Uses:

Lean: `conditional-interface:lemma:weil-pbc-direct-image-filtered-colimit`

For a quasi-compact and quasi-separated morphism of schemes $h : W \to V$ and a filtered system of torsion sheaves (F_i), the canonical map

$$\varinjlim_i R^q h_* F_i \longrightarrow R^q h_* (\varinjlim_i F_i)$$

is an isomorphism for every q.

Proof. Uses:

For $q = 0$, sections on a quasi-compact etale object are the equalizer of a finite covering diagram after a finite quasi-compact refinement; filtered colimits commute with this finite limit. For $q > 0$, represent a class by a degree-q cocycle on a split etale hypercover with finite $(q+1)$-skeleton. Each of its finitely many sections is represented in some F_i, and the finite list of face-map equalities holds at one later index because the system is filtered. This proves surjectivity of $\varinjlim_i H^q(W, F_i) \to H^q(W, \varinjlim_i F_i)$.

If a class from F_i dies after the colimit, choose a finite hypercover refinement and a degree-$(q-1)$ null-homotopy on the limit. Its finitely many sections, refinement maps, and equations descend to one index $j \geq i$; there the class is already a coboundary. Thus the map is injective. The same argument applies after every quasi-compact etale base change $U \to V$, so the cohomology-sheaf description of $R^q h_*$ identifies the displayed sheaves. This is the derived filtered-colimit argument, with no unbounded injective replacement required. □

Lemma 4.1.29 (Prime-factor induction)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-prime-factor-induction`

If a statement about torsion sheaves is stable under short exact sequences, then its verification for all $F[n]$ reduces by induction on the total number of prime factors of n to its verification for sheaves killed by one prime.

Proof. Uses:

Choose a prime $\ell \mid n$ and use the short exact sequence $0 \to F[\ell] \to F[n] \xrightarrow{\ell} \ell F[n] \to 0$. The kernel is killed by ℓ, while $\ell F[n]$ is killed by n/ℓ and therefore has fewer prime factors in its exponent. Let $C(F) = \mathrm{Cone}(\mathrm{Ex}_{\square,F})$ and use the cohomological cone convention $d_C(y,x) = (d_Y y + u(x), -d_X x)$. Applying the two derived functors to the short exact sequence gives a distinguished triangle of exchange maps and, by the octahedral axiom, a triangle

$$C(F[\ell]) \longrightarrow C(F[n]) \longrightarrow C(\ell F[n]) \xrightarrow{+1} C(F[\ell])[1].$$

The connecting map is the class of $(y,x) \mapsto y$ in the above cone convention; the sign in the last arrow is forced by $d_C^2 = 0$. Induct on $\nu(n) = \sum_\ell v_\ell(n)$. If the two end cones are acyclic, the long exact cohomology sequence of this triangle makes the middle cone acyclic. The base $\nu(n) = 1$ is the prime-torsion input. This displays the invariant and all connecting morphisms. □

Lemma 4.1.30 (Five-lemma transfer for exchange)**.** *Lean:* `conditional-interface:lemma:weil-pbc-five-lemma-transfer`

Uses: `def:weil-pbc-derived-exchange`

In a short exact sequence of torsion sheaves, if two of the three derived exchange maps are isomorphisms, then the third is an isomorphism.

Proof. Uses: `lemma:weil-pbc-prime-factor-induction`

Apply $g^{-1}Rf_*$ and $Rf'_*g'^{-1}$ to the short exact sequence and write the natural long exact cohomology sequences. Naturality of the exchange map gives a morphism between these sequences. In each degree the five-lemma applies to the five adjacent terms, with the boundary maps identified by the connecting-homomorphism construction from an injective resolution. Thus the middle exchange map is an isomorphism in every degree. □

Lemma 4.1.31 (Leray acyclicity criterion)**.** *Lean:* `conditional-interface:lemma:weil-pbc-leray-acyclicity`

Uses: `def:weil-pbc-injective-acyclicity-property`

If a complex $I^\bullet$ consists of f'_*-acyclic sheaves, then the natural map $\mathcal{H}^q(f'_*I^\bullet) \to R^q f'_* K$ is an isomorphism whenever $K \to I^\bullet$ is a quasi-isomorphism.

Proof. Uses:

Resolve K by a bounded-below Cartan–Eilenberg double complex $J^{r,s}$ with vertical differential $d_v : J^{r,s} \to J^{r+1,s}$ resolving I^s and horizontal differential $d_h : J^{r,s} \to J^{r,s+1}$ lifting the differential of $I^\bullet$. Set $d_{\mathrm{tot}} = d_v + (-1)^r d_h$. Applying f'_* and filtering the total complex by r gives

$$E_1^{r,s} = R^r f'_* I^s, \qquad E_2^{r,s} = R^r f'_* \mathcal{H}^s(K) \Longrightarrow \mathcal{H}^{r+s}(Rf'_* K).$$

The assumed acyclicity makes $E_1^{r,s} = 0$ for $r > 0$, so the edge map from the $r = 0$ column is the cohomology map of the ordinary complex $f'_* I^\bullet$. The sign $(-1)^r$ makes the square with the horizontal differential commute, and the edge morphism is therefore exactly the natural map $\mathcal{H}^q(f'_*I^\bullet) \to R^q f'_* K$. Collapse of the first column proves it is an isomorphism. □

Lemma 4.1.32 (Prime-torsion devissage)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-prime-torsion-devissage`

For a torsion sheaf F, write $F[n] = \ker(n : F \to F)$. Then $F = \varinjlim_n F[n]$; for $m \mid n$ the inclusion $F[m] \subseteq F[n]$ is monic, and if $n = \prod_i \ell_i^{e_i}$, repeated use of

$$0 \longrightarrow F[\ell_i] \longrightarrow F[n] \xrightarrow{\ell_i} \ell_i F[n] \longrightarrow 0$$

reduces assertions about $F[n]$ to assertions about ℓ_i-torsion.

Proof. Uses: `lemma:weil-pbc-torsion-filtered-union,lemma:weil-pbc-direct-image-filtered-colimit,lemma:weil-pbc-prime-factor-induction,lemma:weil-pbc-five-lemma-transfer`

The filtered union is Lemma 4.1.27. The exact sequence is the kernel–image sequence for multiplication by ℓ_i, since the kernel in $F[n]$ is $F[\ell_i]$. Lemma 4.1.29 and the five-lemma transfer then prove the assertion for every $F[n]$, and Lemma 4.1.28 passes to F. □

Lemma 4.1.33 (Equivalence of PBC and IA)**.** *Lean:* `conditional-interface:lemma:weil-pbc-property-iff-acyclicity`

Uses: `def:weil-pbc-property,def:weil-pbc-injective-acyclicity-property`

For proper f, $\mathsf{PBC}(f)$ holds if and only if $\mathsf{IA}(f)$ holds.

Proof. Uses: `def:weil-pbc-property,def:weil-pbc-injective-acyclicity-property,lemma:weil-pbc-prime-torsion-devissage,lemma:weil-pbc-proper-underived-exchange,lemma:weil-pbc-leray-acyclicity`

Assume $\mathsf{PBC}(f)$. For an injective $\mathbf{F}_\ell$-module sheaf I and a base change g, apply the isomorphism to the complex concentrated in degree zero. The source has no higher cohomology because I is injective, so the target has none; this is the required acyclicity. Conversely assume $\mathsf{IA}(f)$. Resolve any torsion F by injectives $I^\bullet$. The inverse image complex $g'^{-1}I^\bullet$ computes $Rf'_* g'^{-1}F$ because each term is acyclic. For an $\mathbf{F}_\ell$-module sheaf E, choose an injective resolution $E \to I^\bullet$ in the

$\mathbf{F}_\ell$-module category. IA makes every $g'^{-1}I^r$ f'_*-acyclic, so Lemma 4.1.31 identifies the target of the derived exchange map with the cohomology of $f'_* g'^{-1} I^\bullet$. In each degree, the underived proper exchange lemma gives an isomorphism

$$g^{-1} f_* I^r \xrightarrow{\sim} f'_* g'^{-1} I^r,$$

and the replacement-independence lemma makes these maps a morphism of the chosen resolutions. Thus $\mathrm{Ex}_{\square,E}$ is an isomorphism. For $F[n]$, the prime-factor induction lemma and the five-lemma transfer propagate this isomorphism through the exact sequences $0 \to F[\ell] \to F[n] \xrightarrow{\ell} \ell F[n] \to 0$; the induction parameter is $\nu(n) = \sum_\ell v_\ell(n)$, with the prime-torsion case just proved. Finally $F = \varinjlim_n F[n]$ by the filtered torsion-union lemma. The filtered-colimit direct-image lemma applies after every quasi-compact etale base change, so it identifies both cohomology sheaves and carries the isomorphisms for the $F[n]$ to the specified map $\mathrm{Ex}_{\square,F}$. This proves PBC for every torsion sheaf and closes the converse implication. □

4.2 S1. Finite morphisms

Lemma 4.2.1 (Geometric stalk formula for finite direct image). *Lean:* `conditional-interface:lemma:weil-pbc-finite-stalk-formula`

Uses: `def:weil-pbc-cartesian-square`

Let $h : W \to V$ be finite and $\bar v \to V$ geometric. For every sheaf F on W_{et} there is a canonical bijection

$$(h_* F)_{\bar v} \;\cong\; \prod_{\bar w \mid \bar v} F_{\bar w},$$

where the product is finite and the index runs over geometric points of W above $\bar v$ up to the natural equivalence. For an abelian sheaf the same object is canonically the finite direct sum.

Proof. Uses:
The strict henselian local scheme $V_{(\bar v)}$ has the property that a finite V-scheme becomes a finite disjoint union of finite local schemes after base change to $V_{(\bar v)}$. Sections of F over that disjoint union are products of the sections over the components. Passing through the cofiltered system of etale neighbourhoods of $\bar v$ identifies the stalk with the displayed product. Finiteness of the fibre makes product and sum equal in the abelian category. This is the stalk calculation located at Stacks Proposition 59.55.2, Tag `03QP`; no exactness is inferred until the next lemma. □

Lemma 4.2.2 (Exactness of finite direct image). *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-finite-direct-image-exact`

If h is finite, then h_* on etale abelian sheaves is exact.

Proof. Uses: `lemma:weil-pbc-finite-stalk-formula`
For a short exact sequence $0 \to F_1 \to F_2 \to F_3 \to 0$, apply the stalk formula at every geometric $\bar v$. A finite product of short exact sequences of abelian groups is exact, so the sequence of stalks of the direct images is exact. Geometric stalks are conservative for etale sheaves: injectivity and surjectivity can be checked after an etale neighbourhood of each geometric point and then sheafified. Hence the direct-image sequence is exact. Consequently $R^q h_* = 0$ for $q > 0$. □

Lemma 4.2.3 (Finite exchange isomorphism). *Lean:* `conditional-interface:lemma:weil-pbc-finite-exchange`

Uses: `def:weil-pbc-cartesian-square`

For a finite Cartesian square, the underived exchange map is an isomorphism; therefore all its derived exchange maps are isomorphisms.

Proof. Uses: `lemma:weil-pbc-finite-stalk-formula,lemma:weil-pbc-finite-direct-image-exact`
At a geometric point $\bar y' \to Y'$, the fibre of f' is the base change of the finite fibre of f along $\kappa(\bar y) \to \kappa(\bar y')$. The stalk formula identifies both sides with the same finite product indexed by these points, and the exchange map is the identity on each factor. Thus it is a stalkwise bijection and hence an isomorphism. Exactness from the preceding lemma means injective resolutions remain direct-image acyclic, so the same degreewise argument computes the derived map. □

Proposition 4.2.4 (Finite morphisms satisfy proper base change). *Uses:* `def:weil-pbc-property`

Lean: `conditional-interface:prop:weil-pbc-finite-morphism`

Every finite morphism satisfies PBC.

Proof. Uses: `def:weil-pbc-property,lemma:weil-pbc-finite-exchange,lemma:weil-pbc-property-iff-acyclicity`
The finite exchange lemma supplies the required isomorphism for every base change and every torsion sheaf, which is exactly Definition 4.1.25. Equivalently, the vanishing in the exactness lemma gives IA and Lemma 4.1.33 gives PBC. □

4.3 S2. Composition and proper-surjective cancellation

Lemma 4.3.1 (Composition of proper maps)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-composition`

Suppose $X \xrightarrow{f} Y \xrightarrow{h} Z$ are proper and both satisfy IA*. Then hf satisfies* IA *and hence* PBC*.*

Proof. Uses: `lemma:weil-pbc-property-iff-acyclicity,lemma:weil-pbc-proper-underived-exchange`

Let $a : Z' \to Z$ be a base change, $Y' = Y \times_Z Z'$, $X' = X \times_Y Y'$, and I an injective $\mathbf{F}_\ell$-module sheaf on X. Since inverse image is exact, f_* (a right adjoint) sends injectives to injectives: for an epimorphism $A \twoheadrightarrow B$, the map $\mathrm{Hom}(B, f_*I) \to \mathrm{Hom}(A, f_*I)$ identifies with $\mathrm{Hom}(f^{-1}B, I) \to \mathrm{Hom}(f^{-1}A, I)$ and is surjective. The proper underived exchange lemma identifies $f'_* a_X^{-1} I \cong b^{-1} f_* I$, where $b : Y' \to Y$ is the base change. The Grothendieck spectral sequence has terms

$$E_2^{p,q} = R^p h'_* \, R^q f'_* \, a_X^{-1} I \Longrightarrow R^{p+q}(h'f')_* a_X^{-1} I,$$

where $a_X : X' \to X$ and $h' : Y' \to Z'$ are the base changes. The first-quadrant hypotheses are: this sheaf is h'_*-acyclic by the assumed $\mathsf{IA}(h)$, and $a_X^{-1} I$ is f'_*-acyclic by $\mathsf{IA}(f)$. Thus the $q > 0$ rows vanish, then the $p > 0$ column of the surviving row vanishes, so all terms with $p + q > 0$ vanish. The spectral sequence converges because each total degree has only finitely many pairs (p, q). This is the calculation located at Tag `0A4D`. □

Lemma 4.3.2 (Cancellation along a proper surjection)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-cancellation`

Let $f : X \to Y$ be proper and surjective and $h : Y \to Z$ proper. If f and hf satisfy IA*, then h satisfies* IA*.*

Proof. Uses: `lemma:weil-pbc-property-iff-acyclicity,lemma:weil-pbc-proper-underived-exchange`

Fix a base change $a : Z' \to Z$ and an injective $\mathbf{F}_\ell$-module sheaf I on Y. Choose a monomorphism $f^{-1}I \hookrightarrow J$ with J injective on X. Applying f_* gives $\iota : I \to f_*J$. If a germ $a \in I_{\bar{y}}$ maps to zero, choose a geometric $\bar{x} \to X$ above $\bar{y}$ (surjectivity). The canonical stalk identification $(f^{-1}I)_{\bar{x}} \cong I_{\bar{y}}$ sends a to the germ of a at $\bar{x}$; the monomorphism into J makes that germ zero. The definition of the direct-image stalk then says that a vanishes on an etale neighbourhood of $\bar{y}$, so $a = 0$. Thus ι is monic. Injectivity of I supplies a retraction $r : f_*J \to I$ with $r \circ \iota = \mathrm{id}_I$, making I a direct summand. After base change, proper underived exchange gives $b^{-1} f_* J \cong f'_* a_X^{-1} J$. The spectral sequence

$$R^p h'_* R^q f'_* a_Y^{-1} I \Longrightarrow R^{p+q}(h'f')_* a_X^{-1} I$$

is computed using the injective J and the assumed acyclicity of hf. The $q > 0$ terms vanish for J by $\mathsf{IA}(f)$, and total-degree vanishing for hf forces $R^p h'_* a_Y^{-1}(f_*J) = 0$ for $p > 0$. Derived direct image preserves finite direct sums and retracts, so the same vanishing holds for the direct summand $a_Y^{-1} I$. This is Tag `0A4C`; the stalk injectivity argument is written here because surjectivity is essential. □

4.4 S3. Reduction to the projective line

Lemma 4.4.1 (Target locality)**.** *Lean:* `conditional-interface:lemma:weil-pbc-target-locality`

Uses: `def:weil-pbc-property`

For proper $f : X \to Y$, the property PBC(f) *is local for the etale topology on Y.*

Proof. Uses:

For an etale $V \to Y$ and a torsion sheaf F, sections of $R^q f_* F$ on V are the sheafification of $U \mapsto H^q(X \times_Y U, F|)$ over etale $U \to V$. This description is functorial in V and commutes with further etale localization. Therefore an exchange map is an isomorphism if and only if it is so after an etale cover of Y (and its pullback cover on Y'), because geometric stalks form a conservative family. The section description is the content located at Tag `03Q8`. □

Lemma 4.4.2 (Chow and projective factorization)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-chow-factorization`

After replacing an affine target by an etale affine cover, every proper $f : X \to Y$ admits a proper surjection $\pi : X' \to X$ and an integer n with $X' \hookrightarrow \mathbf{P}^n_Y$ a closed immersion. Moreover $X' \to Y$ factors as that closed immersion followed by $\mathbf{P}^n_Y \to Y$.

Proof. Uses: `lemma:weil-pbc-target-locality,lemma:weil-pbc-cancellation`

Chow's lemma in the form located at Tag `0202` supplies a proper surjective π for which $X' \to Y$ is projective. To make the source hypotheses explicit, first work over one affine member $Y = \mathrm{Spec}\, A$ of the selected etale cover. Write $A = \varinjlim_{\lambda \in \Lambda} A_\lambda$, where every A_λ is a finitely generated $\mathbf{Z}$-algebra and the transition maps are directed. Since a proper morphism is of finite presentation, a finite affine cover $X = \bigcup_{a=1}^{r} U_a$, the finitely presented algebras defining the U_a, the pairwise overlap maps, and their cocycle equations descend to one index λ_0. After increasing once to λ_1, the descended morphism $f_{\lambda_1} : X_{\lambda_1} \to \mathrm{Spec}\, A_{\lambda_1}$ is proper: finite presentation and separatedness descend from the finite affine data, and the valuative criterion for properness is stable under this inverse-limit presentation.

The ring A_{λ_1} is noetherian. Noetherian Chow's lemma produces a proper surjection $\pi_{\lambda_1} : X'_{\lambda_1} \to X_{\lambda_1}$ and an $f_{\lambda_1}\pi_{\lambda_1}$-ample invertible sheaf L_{λ_1}. Choose a finite affine cover $X'_{\lambda_1} = \bigcup_{b=1}^{s} W_b$. On each W_b, ampleness supplies an exponent $m_b > 0$ and finitely many global sections whose nonvanishing opens cover W_b. With $m = \prod_b m_b$, take the corresponding powers of these sections. There are then finitely many sections $s_0, \ldots, s_n$ of $L_{\lambda_1}^{\otimes m}$ whose nonvanishing opens cover X'_{λ_1}.

Increasing m by one further common multiple if necessary, the same finite affine-cover test for very ampleness separates the finitely many overlap generators; hence the map

$$[s_0 : \cdots : s_n] : X'_{\lambda_1} \longrightarrow \mathbf{P}^n_{A_{\lambda_1}}$$

is an immersion. It is proper, because X'_{λ_1} is proper over A_{λ_1} and projective space is separated over that base. A proper immersion is closed.

Base change along $A_{\lambda_1} \to A$ defines $X' = X'_{\lambda_1} \times_{A_{\lambda_1}} A$, π, and the closed immersion $X' \hookrightarrow \mathbf{P}^n_Y$. Properness and closed immersions are preserved by base change. Surjectivity is preserved because every geometric point of X lifts after base change from the surjective geometric fibre of π_{λ_1}. Thus the displayed index λ_1, exponent m, and integer n give the asserted single factorization. □

Lemma 4.4.3 (Projective embedding over X)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-chow-embedding-over-x`

Let $\iota_Y : X' \hookrightarrow \mathbf{P}^n_Y$ be the closed immersion and $\pi : X' \to X$ the Chow modification. Then

$$\iota_X = (\pi, \iota_Y) : X' \longrightarrow X \times_Y \mathbf{P}^n_Y = \mathbf{P}^n_X$$

is a closed immersion over X.

Proof. Uses: `lemma:weil-pbc-chow-factorization`

The product map is an immersion because ι_Y is an immersion and immersions are preserved by base change and composition with a graph. The map ι_X is proper over X: its source X' is proper over X, while $\mathbf{P}^n_X$ is separated over X, so the graph criterion makes every X-morphism from X' proper. A proper immersion is a closed immersion: properness makes its image closed after every base change, and an immersion identifies the source with a locally closed subscheme of that image; universal closedness forces the locally closed subset to be closed. The equality with $\mathbf{P}^n_X$ is the defining fibre-product identity. □

Lemma 4.4.4 (Symmetric product map)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-symmetric-product-p1`

For $n \geq 1$ the formula

$$((x_1 : y_1), \ldots, (x_n : y_n)) \longmapsto [F_0 : \cdots : F_n], \qquad \prod_{i=1}^{n} (x_i t + y_i) = \sum_{j=0}^{n} F_j t^j,$$

defines a finite surjective S_n-invariant morphism $\sigma_n : (\mathbf{P}^1_S)^n \to \mathbf{P}^n_S$.

Proof. Uses: `prop:weil-pbc-finite-morphism,lemma:weil-pbc-composition,lemma:weil-pbc-cancellation`

Each F_j is the sum of monomials choosing j of the x_i and the remaining y_i; hence it is homogeneous of degree one in every pair (x_i, y_i). On the chart $y_1 \cdots y_n \neq 0$, the ratios $z_i = x_i/y_i$ give elementary symmetric polynomials, so the map is induced by the universal polynomial $\prod(T - z_i)$. On a general chart, the same calculation uses the coefficient ring localized at one nonzero F_j; the coefficients generate the unit ideal because if all F_j vanished then the homogeneous polynomial $\prod_i (x_i t + y_i)$ would be zero, forcing one factor to be $(0, 0)$, impossible in projective coordinates. The coordinate ring is integral over the symmetric subring: each z_i satisfies the monic equation $T^n - e_1 T^{n-1} + \cdots + (-1)^n e_n$, and the analogous statement holds on all charts after clearing the chosen y_i or x_i. Thus σ_n is finite. Permutation of the factors leaves every F_j unchanged, proving S_n-invariance. To make finiteness independent of the abbreviated chart calculation, let $[a_0 : \cdots : a_n]$ be a geometric point of $\mathbf{P}^n_S$. Its fibre is the set of ordered n-tuples of points of $\mathbf{P}^1$ whose linear forms multiply to $\sum a_j t^j$. Over an algebraically closed field, the factorization theorem for one-variable polynomials, applied after choosing the first nonzero coefficient to decide whether $t = \infty$ is a root, gives only finitely many such tuples (at most $n!$ after the multiplicities are fixed). Hence σ_n is quasi-finite.

The source $(\mathbf{P}^1_S)^n$ is projective over S, so σ_n is proper. On an affine target chart, the quasi-finite proper map is covered by finitely many affine opens on which the coordinate rings are integral over the target ring: on the chart where the last coefficient is invertible use the monic equations for the z_i, and on the chart where the first coefficient is invertible use the reversed monic equations for $w_i = y_i/x_i$; on an intermediate chart, intersect these two covers with the finitely many opens where a selected coefficient of the binary form is invertible. The same monic equations after multiplying by that selected coefficient show that each coordinate is integral. Integral algebras glue on overlaps because the coordinate identifications are induced by the common coefficients F_j. Thus the proper quasi-finite morphism is finite. Finally a geometric binary form of degree n factors into n linear forms, including the possible factor x for $t = \infty$, so every geometric fibre is nonempty; geometric surjectivity and properness give surjectivity. This closes the chart and gluing obligation. □

Proposition 4.4.5 (Reduction of proper base change to $\mathbf{P}^1$)**.** *Uses:* `def:weil-pbc-derived-exchange`

Lean: `conditional-interface:prop:weil-pbc-reduction-to-projective-line`

If every projection $\mathbf{P}^1_S \to S$ satisfies PBC *for every base scheme S, then every proper morphism satisfies* PBC.

Proof. Uses: `def:weil-pbc-derived-exchange,lemma:weil-pbc-target-locality,lemma:weil-pbc-chow-factorization,lemma:weil-pbc-chow-embedding-over-x,lemma:weil-pbc-symmetric-product-p1,prop:weil-pbc-finite-morphism,lemma:weil-pbc-composition,lemma:weil-pbc-cancellation`

Use target locality and the Chow factorization. The closed immersion $X' \hookrightarrow \mathbf{P}^n_Y$ is proper and finite onto its image, so finite base change plus composition reduces $X' \to Y$ to $\mathbf{P}^n_Y \to Y$. The finite surjective map σ_n and cancellation reduce this projection to $(\mathbf{P}^1_Y)^n \to Y$. Repeated base change and composition reduce the product projection to the one-dimensional projection. Finally cancellation along π transfers the property from X' and $X' \to Y$ to $X \to Y$. Every comparison used is the named exchange map of Definition 4.1.8. □

4.5 S4. The projective line over a strictly henselian local base

Definition 4.5.1 (Strictly henselian setup)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-strict-henselian-setup`

Let $(A, \mathfrak{m})$ be strictly henselian local, $S = \operatorname{Spec} A$, $s = \operatorname{Spec}(A/\mathfrak{m})$, $X = \mathbf{P}^1_S$, and $X_0 = \mathbf{P}^1_s$. For a sheaf F on X_{et} write F_0 for its inverse image on $(X_0)_{\mathrm{et}}$.

Lemma 4.5.2 (Zariski restriction on a spectral space)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-spectral-restriction`

If T is spectral, $T_0 \subseteq T$ is closed, and every irreducible closed subset $C \subseteq T$ has $C \cap T_0$ connected and nonempty, then for every abelian sheaf F on T the restriction $\Gamma(T, F) \to \Gamma(T_0, F|)$ is a bijective map.

Proof. Uses:

Injectivity is pointwise. If $s \in \Gamma(T, F)$ restricts to zero, its germ vanishes at every point of T_0. The zero-germ locus is open and stable under generization. If its complement contained x, then the irreducible closed $C = \overline{\{x\}}$ would meet T_0; the specialization map from the germ at the generic point of C to the germ at x would force the latter to be zero, a contradiction.

For surjectivity, write F as the filtered colimit of subsheaves generated by finitely many sections on quasi-compact opens. Restriction to T_0 and both section functors commute with this filtered colimit, so it is enough to handle one finitely generated subsheaf. Such a subsheaf embeds into a finite product

$$F \hookrightarrow \prod_{b=1}^{r} (i_b)_* \underline{M_b},$$

with $i_b : C_b \hookrightarrow T$ closed and M_b finite: the finite germs of the chosen generators determine finitely many closed equalizer loci, and the corresponding finite value sets give the displayed embedding. If $F \subseteq G$, the injectivity already proved gives

$$\Gamma(T, F) = \Gamma(T_0, F|_{T_0}) \cap \Gamma(T, G) \subseteq \Gamma(T_0, G|_{T_0}).$$

Thus it suffices to extend sections of one finite constant pushforward.

For $i : C \hookrightarrow T$ closed, the condition on irreducible closures is inherited by C. A section of $\underline{M}$ on $C \cap T_0$ is constant on each connected component. Every component of C meets its closed subset, and that intersection is connected by hypothesis, so the same constant gives a unique section on the component. These componentwise sections agree on quasi-compact overlaps and glue by the sheaf equalizer; pushing forward gives the extension on T. Finite products, subobjects, and the filtered colimit then give an extension for F. The stalkwise injectivity proves uniqueness, so restriction is bijective. This is the expanded spectral argument at Tag `09ZG`; no etale lifting is used. □

Lemma 4.5.3 (Stein connectedness for $\mathbf{P}^1_A$)**.** *Lean: `conditional-interface:lemma:weil-pbc-stein-connectedness`*

Uses: `def:weil-pbc-strict-henselian-setup`

Every irreducible closed subset of X meets X_0 in a connected nonempty set. Consequently $\Gamma(X, F) \xrightarrow{\sim} \Gamma(X_0, F_0)$ for every Zariski sheaf F.

Proof. Uses: `lemma:weil-pbc-spectral-restriction`

Fix an irreducible closed subset and give it the reduced closed-subscheme structure $C \subseteq X$. The map $C \to S$ is proper. Its Stein factorization gives $C \xrightarrow{u} \operatorname{Spec} B \xrightarrow{v} S$, with $A \to B$ integral and u having geometrically connected fibres. Since $(A, \mathfrak{m})$ is henselian, $(B, \mathfrak{m}B)$ is a henselian pair. The image $D = v(\operatorname{Spec} B)$ is irreducible and closed because $C \to S$ is proper. Henselian connectedness makes $D \cap V(\mathfrak{m})$ connected and nonempty. The map $C \cap X_0 \to D \cap V(\mathfrak{m})$ is closed and surjective with connected fibres, so the closed-surjection connected-fibre criterion makes $C \cap X_0$ connected. This expands Tag `0A0B`; the final equality of sections is Lemma 4.5.2. □

Definition 4.5.4 (Pullback of a section along a finite morphism)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-finite-section-pullback`

Let $u : T' \to T$ be finite and F an etale sheaf on T. For an etale $V \to T$ and $s \in F(V)$, define

$$u^\sharp_V(s) \in (u^{-1}F)(V \times_T T')$$

to be the image of s under the unit from the presheaf inverse image to its sheafification.

Lemma 4.5.5 (Finite pullback identity)**.** *Lean: `conditional-interface:lemma:weil-pbc-finite-pullback-identity`*

Uses: `def:weil-pbc-finite-section-pullback`

For $u = \mathrm{id}_T$, $u^\sharp_V(s) = s$ under the canonical identity $V \times_T T = V$.

Proof. Uses:

The presheaf inverse image along the identity is the same presheaf, and its sheafification unit is the identity on an already sheafified section. □

Lemma 4.5.6 (Finite pullback composition)**.** *Lean: `conditional-interface:lemma:weil-pbc-finite-pullback-composition`*

Uses: `def:weil-pbc-finite-section-pullback`

For finite $T'' \xrightarrow{v} T' \xrightarrow{u} T$, $(uv)^\sharp_V(s) = v^\sharp_{V \times_T T'}(u^\sharp_V(s))$ under the associativity isomorphism of fibre products.

Proof. Uses: `lemma:weil-pbc-finite-pullback-identity`
Both sections are the image of the same presheaf section s under the composite inverse-image functor. Naturality of sheafification and the associativity isomorphism make the two unit maps equal. □

Lemma 4.5.7 (Finite pullback and closed-fibre restriction). *Lean:* `conditional-interface:lemma:weil-pbc-finite-pullback-closed-fibre`

Uses: `def:weil-pbc-finite-section-pullback`
For a closed immersion $i : Z \hookrightarrow T$ and $Z' = Z \times_T T'$, restriction of $u_V^\sharp(s)$ to Z' equals the finite pullback of the restriction of s to Z.

Proof. Uses: `lemma:weil-pbc-finite-pullback-composition`
Apply the composition lemma to the two equal composites $Z' \to T' \to T$ and $Z' \to Z \to T$. The Cartesian identification makes their inverse-image sheafification units equal. □

Lemma 4.5.8 (Finite descent compatibility). *Lean:* `conditional-interface:lemma:weil-pbc-finite-pullback-descent`

Uses: `def:weil-pbc-finite-section-pullback`
Let $u : T' \to T$ be finite surjective. Suppose $s' \in (u^{-1}F)(T')$ is equipped with a descended section $s \in F(T)$, meaning $u_T^\sharp(s) = s'$. Then the same equality holds after restriction to a closed fibre.

Proof. Uses: `lemma:weil-pbc-finite-pullback-composition,lemma:weil-pbc-finite-pullback-closed-fibre`
The first equality is the descent witness. Its closed-fibre restriction is the equality supplied by Lemma 4.5.7. □

Lemma 4.5.9 (Etale henselian restriction in degree zero). *Lean:* `conditional-interface:lemma:weil-pbc-etale-degree-zero`

Uses: `def:weil-pbc-strict-henselian-setup`
For every etale abelian sheaf F on X, restriction induces $\Gamma(X, F) \xrightarrow{\sim} \Gamma(X_0, F_0)$.

Proof. Uses: `lemma:weil-pbc-stein-connectedness,lemma:weil-pbc-finite-pullback-identity,lemma:weil-pbc-finite-pullback-composition,lemma:weil-pbc-finite-pullback-closed-fibre,lemma:weil-pbc-finite-pullback-descent,lemma:weil-pbc-finite-refinement-etale-cover`
The canonical map from the etale to Zariski restriction is injective on stalks because a geometric point has a strictly local etale neighbourhood. For surjectivity, lift a fibre section etale-locally; quasi-compactness of X_0 gives finitely many local lifts, and Lemma 4.1.5 gives a finite surjective refinement on which all lifts exist. The two lifts agree on the finite overlap after restriction to X_0 by the Zariski result, and the equalizer sheaf descends the lift to X. Uniqueness is stalkwise injectivity. The finite-pullback compatibility required in Stacks Tag `09ZF` is recorded as four explicit equalities: identity pullback preserves the lift, composition of finite pullbacks preserves the lift, restriction to X_0 recovers the given section, and the descended section pulls back to each local lift; these are the four preceding lemmas. The finite refinement lemma proves existence and finite-presentation stage, and its stalkwise equalizer argument proves equality of the local lifts on the self-intersection. Thus the four finite-pullback compatibilities complete the descent. □

Lemma 4.5.10 (Finite-stage killing of a class). *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-finite-stage-killing`
In an affine henselian pair, if a positive-degree class c dies on an integral strictly-local cover $T_\infty \to T_0$ represented by a filtered system $\{T_\lambda\}$ of finite finitely presented covers, then there exists λ_0 such that the pullback of c to $H^q(T_{\lambda_0}, F|)$ is zero.

Proof. Uses:
Represent the chosen cochain and its null-homotopy on the limit by finitely many sections. Each section and each equality occurs at some index because the system is filtered and the etale site is quasi-compact and quasi-separated. Choose one index dominating those finitely many indices. The descended cochain and the descended null-homotopy satisfy the coboundary equation at a further finite index; replacing by that index gives λ_0. This exposes the indexing and vanishing witness requested by Tag `09ZD`. □

Lemma 4.5.11 (Constructible approximation for effacement). *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-constructible-approximation`
For a torsion sheaf F on an affine henselian pair and a fixed class $c \in H^q(T_0, F_0)$, there is a quasi-compact etale neighbourhood $U \to T$ and a constructible sheaf C on U with a monomorphism $C \hookrightarrow F|_U$ through which a cocycle representing c factors.

Proof. Uses:
Choose a split etale hypercover $U_\bullet \to T_0$ whose $(q+1)$-skeleton is finite and quasi-compact and a cocycle $c \in \Gamma(U_q, F_0)$. Write its finite components as $U_{r,a}$ for $0 \le r \le q+1$ and denote all face and degeneracy maps by $\partial^i_{r,a}$ and $s^i_{r,a}$. Choose quasi-compact etale lifts $\widetilde{U}_{r,a} \to U \to T$ and sections of F representing the finitely many sections in c and in the equations $dc = 0$.

Enlarge the finite list by applying every $\partial^i_{r,a}$ and $s^i_{r,a}$ to every chosen section. The simplicial identities are now a finite list of equalities on finite fibre products. Shrink the lifts once so that all equalities hold on the closed fibre and all maps factor through one quasi-compact etale neighbourhood $U \to T$. On each lifted component let $C_{r,a}$ be the subsheaf of $F|_U$ generated by this finite list. The equalizer loci of pairs of generators give a finite stratification on which $C_{r,a}$ is locally constant with finite fibres, hence constructible.

Closure under all displayed face and degeneracy maps gives a simplicial constructible subsheaf $C_\bullet \subseteq F|_{U_\bullet}$. The cocycle lies in $\Gamma(U_q, C_q)$ and the equation $dc = 0$ remains true in $C_\bullet$. Take the image of the augmentation $C_0 \to F|_U$ and its finite

iterated extensions through degree $q+1$, then sheafify; this gives a constructible subsheaf $C \hookrightarrow F|_U$ through which c factors. Monicity is checked at a geometric stalk: a finite combination of the generators maps to zero in F precisely when it is already zero in the generated subsheaf. Thus every face map, cocycle, and equality used by the selected class is contained in C. □

Lemma 4.5.12 (Pushout preserves monomorphisms)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-pushout-monicity`

Given monomorphisms $A \hookrightarrow B$ and $A \hookrightarrow C$ of sheaves, the canonical map $C \to B \amalg_A C$ is monic.

Proof. Uses:

Let $P = B \amalg_A C$ be the pushout in sheaves. It is the cokernel sheaf of $A \to B \oplus C$, $a \mapsto (i_B a, -i_C a)$. If $c \in C(V)$ maps to zero in $P(V)$, then by the definition of a sheaf cokernel there is an etale cover $\{V_\alpha \to V\}$ and $a_\alpha \in A(V_\alpha)$ with $i_C(a_\alpha) = c|_{V_\alpha}$ and $i_B(a_\alpha) = 0$. Since i_B is monic, $a_\alpha = 0$, hence $c|_{V_\alpha} = 0$. The sheaf axiom gives $c = 0$. This proves monicity on arbitrary etale objects, and the same local equalizer calculation proves naturality of the pushout map. □

Lemma 4.5.13 (Constructible sheaves reduce to finite pushforwards)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-constructible-finite-pushforward`

If U is quasi-compact and quasi-separated, every constructible torsion sheaf C admits a monomorphism

$$C \longrightarrow \bigoplus_{i=1}^{r} (h_i)_* \underline{M_i},$$

where each $h_i : V_i \to U$ is finite and finitely presented and each M_i is a finite abelian group.

Proof. Uses: `lemma:weil-pbc-constructible-approximation`

Write $U = \varprojlim U_\lambda$ with affine transition maps and noetherian, quasi-compact, quasi-separated U_λ. The finite stratification of C, its finite stalk groups, all restriction maps, and their equalizer relations descend to one stage U_λ. We prove the assertion there by noetherian induction on a closed complement of a dense stratum. Choose a dense open $j : V \hookrightarrow U_\lambda$ where C_λ is finite locally constant and a finite etale cover $e : W \to V$ trivializing it. Take the integral closure of U_λ in $e_*\mathcal{O}_W$ and discard components disjoint from V; this gives a finite map $h : W' \to U_\lambda$ extending e. The adjunction unit is a monomorphism on V,

$$C_\lambda|_V \hookrightarrow e_* \underline{M} = (h_* \underline{M})|_V,$$

for a finite group M.

Let $Z = U_\lambda \setminus V$ with its reduced closed structure. By induction, $C_\lambda|_Z$ embeds into a finite sum $\bigoplus_b (h_b)_* \underline{M_b}$ with h_b finite over Z; composing with $Z \hookrightarrow U_\lambda$ keeps the maps finite. Map C_λ to

$$h_* \underline{M} \oplus \bigoplus_b (h_b)_* \underline{M_b}$$

by the dense-open unit in the first component and restriction to Z in the remaining components. At a geometric point of V the first component is injective, and at a geometric point of Z the second block is injective, so the combined map is monic. This is the localization induction, including the boundary and overlap comparison. Pull the finite diagram and monomorphism back to U. Etale inverse image is exact, and finite direct image commutes with this pullback by the finite geometric-stalk product formula. This gives the displayed monomorphism and expands Tag `09Z7`. □

Lemma 4.5.14 (Finite-cover killing and finite pushforward monicity)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-finite-cover-killing`

For a fixed $c \in H^q(T_0, C_0)$ with $q > 0$ and constructible C, there is a finite finitely presented cover $h : V \to T$ whose closed-fibre pullback $h_0^* c$ is zero; the induced map $C \hookrightarrow h_* h^* C$ is monic.

Proof. Uses: `lemma:weil-pbc-finite-stage-killing,lemma:weil-pbc-finite-stalk-formula`

Use the integral strictly-local cover killing c and represent it by the filtered system of finite, finitely presented covers $h_\lambda : V_\lambda \to T$. Lemma 4.5.10 selects an index λ_0 at which $h_{\lambda_0,0}^* c = 0$; set $h = h_{\lambda_0}$. The map to the finite pushforward is the unit of (h^*, h_*). At a geometric point its stalk is the diagonal into the finite product of stalks from Lemma 4.2.1; the factor corresponding to the chosen point above it is the identity. Hence the unit is injective on every geometric stalk and therefore monic. □

Lemma 4.5.15 (Affine Gabber effacement)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-gabber-effacement`

For an affine henselian pair (T, T_0) and a torsion sheaf F on T_{et}, every class $c \in H^q(T_0, F_0)$ with $q > 0$ admits a monomorphism $F \hookrightarrow F'$ such that the image of c in $H^q(T_0, F'_0)$ is zero.

Proof. Uses: `lemma:weil-pbc-constructible-approximation,lemma:weil-pbc-constructible-finite-pushforward,lemma:weil-pbc-finite-cover-killing,lemma:weil-pbc-pushout-monicity`

Apply constructible approximation to obtain $C \hookrightarrow F$ through which c factors, and use finite-pushforward reduction to choose

$$C \hookrightarrow D_0 = \bigoplus_{i=1}^{r} (h_i)_* \underline{M_i}.$$

Let c_i be the image of c in $H^q(T_0, (h_i)_* \underline{M_i}|_{T_0})$. Finite direct-image exactness identifies c_i with a class on the closed fibre of h_i. Apply finite-cover killing to each c_i and take the fibre product of the resulting finite covers over T; after replacing by one finite-presentation stage, call the common finite cover $h : V \to T$. Every component of $h_0^* c$ is zero. The composite unit

$$C \hookrightarrow D_0 \longrightarrow h_* h^* D_0 =: D$$

is monic by the finite stalk product formula. Form the pushout square

$$\begin{array}{ccc} C & \longrightarrow & D \\ \downarrow & & \downarrow \\ F & \longrightarrow & F' := F \amalg_C D. \end{array}$$

Pushout monicity makes $F \to F'$ injective. The image of c in $H^q(T_0, D_0)$ is carried to zero in $H^q(T_0, D)$ because its pullback to V_0 is zero and h_* is exact. The pushout square therefore carries c to zero in $H^q(T_0, F'_0)$. This displays both the common dominating cover and the null-homotopy square and closes Tag 09ZE's effacement step. □

Proposition 4.5.16 (Gabber affine comparison)**.** *Uses:*

Lean: `conditional-interface:prop:weil-pbc-gabber-affine-comparison`

For an affine henselian pair (T, T_0) and torsion F, $H^q(T, F) \xrightarrow{\sim} H^q(T_0, F_0)$ for all $q \geq 0$.

Proof. Uses:

Degree zero is the etale henselian restriction lemma applied on the affine. For $q > 0$, assume comparison in all degrees below q. To prove injectivity, take a nonzero $\xi \in H^q(T, F)$ and apply effacement with closed subscheme T itself, obtaining $F \hookrightarrow F'$ in which ξ dies. Put $Q = F'/F$. Exactness lifts ξ from $H^{q-1}(T, Q)$. The induction isomorphism in degree $q-1$ carries any such lift to T_0; if ξ restricted to zero, exactness and the commutative boundary square would force the lift to come from $H^{q-1}(T, F')$, contradicting the choice of nonzero ξ.

For surjectivity, take $\eta \in H^q(T_0, F_0)$ and use effacement with closed subscheme T_0 to choose $F \hookrightarrow F'$ killing η. Exactness lifts η from $H^{q-1}(T_0, Q_0)$. By induction this lift is the restriction of a class in $H^{q-1}(T, Q)$, whose boundary in $H^q(T, F)$ restricts to η. Thus the comparison is separately injective and surjective. This expands Tag 09ZI; no statement is imported from that tag. □

Lemma 4.5.17 (Injective restriction acyclicity bound)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-injective-restriction-acyclicity`

Let a separated scheme with affine diagonal be covered by $n+1$ affine opens. If I is injective on the scheme over a henselian pair, then the restriction to the closed fibre has vanishing cohomology in degrees $> n$.

Proof. Uses: `prop:weil-pbc-gabber-affine-comparison`

For $n = 0$ this is the affine comparison applied to the injective I. For $n > 0$, split the cover as U_0 and $V = \bigcup_{i>0} U_i$. The Mayer–Vietoris long exact sequence for $U_0 \cup V$ has terms on U_0, V, and $U_0 \cap V$; the last two have covers of size at most n and the affine comparison kills their groups above the inductive bound. The long exact sequence therefore kills the group above degree n. This is the dimension induction located at Tag 0A51. □

Lemma 4.5.18 (Finite locally constant sheaves and finite-etale schemes)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-finite-locally-constant-etale-equivalence`

For a scheme T, finite locally constant sheaves of sets on T_{et} are equivalent to finite etale T-schemes. A finite abelian group action on the sheaf corresponds to an action by finite-etale T-automorphisms.

Proof. Uses:

Given a sheaf E, choose a finite etale cover $U \to T$ trivializing it with finite fibre S. The descent isomorphism on $U \times_T U$ is a permutation of S satisfying the cocycle equation, hence descends the disjoint union $\coprod_{s \in S} U$ to a finite etale scheme P. Conversely, the sheaf of sections of a finite etale P is finite locally constant; the two constructions are inverse because both are computed after the same trivializing cover. An M-action is a compatible family of permutations, so it descends exactly when the sheaf action is compatible with the descent cocycle. □

Lemma 4.5.19 (Finite-etale lifting over a henselian pair)**.** *Lean:* `conditional-interface:lemma:weil-pbc-finite-etale-etale-henselian-lifting`

Uses: `def:weil-pbc-strict-henselian-setup`

If $(A, \mathfrak{m})$ is henselian and $Y \to \operatorname{Spec} A$ is proper, restriction from finite etale Y-schemes to finite etale Y_0-schemes is an equivalence, including unique lifting of morphisms and group actions.

Proof. Uses: `lemma:weil-pbc-finite-locally-constant-etale-equivalence`

We expand the equivalence used by Tag 0A5G; its source locator is Tag 0A48. First suppose that A is complete noetherian and put $Y_r = Y \times_A A/\mathfrak{m}^{r+1}$. Formal etaleness and nilpotent invariance of the etale site lift a finite etale $P_0 \to Y_0$ uniquely to a compatible system $P_r \to Y_r$. Finiteness can be checked modulo the nilpotent ideal: on a finite affine cover $Y_r = \bigcup_a \operatorname{Spec} R_{r,a}$, each lifted algebra is generated by lifts of a finite set of generators of the $r = 0$ algebra, and Nakayama makes the resulting algebra finite. The transition isomorphisms on the finitely many overlaps obey the cocycle equation because their restrictions do and etale morphisms are formally unramified.

Grothendieck existence applied to the compatible finite algebras $(p_r)_*\mathcal{O}_{P_r}$ algebraizes them to a coherent $\mathcal{O}_Y$-algebra, hence to a finite morphism $P \to Y$ with special fibre P_0. The etale locus of $P \to Y$ is open and contains P_0. Every point of P specializes to the closed fibre because P is proper over the local base; therefore the complementary closed non-etale locus is empty. This proves essential surjectivity in the complete noetherian case. For full faithfulness, a map $u_0 : P_0 \to Q_0$ has graph

$$\Gamma_{u_0} : P_0 \longrightarrow P_0 \times_{Y_0} Q_0,$$

which is an open-and-closed finite etale subscheme. Algebraize this graph to $W \to P \times_Y Q$. The projection $W \to P$ is finite etale and is an isomorphism on the closed fibre; its isomorphism locus is open, and the proper closed complement would meet the closed fibre if nonempty. Hence it is empty, and $W \cong P$ gives the unique lift $u : P \to Q$.

Now suppose A is a noetherian henselian local G-ring. Complete it to $\widehat{A}$. The first step gives a finite etale cover over $Y_{\widehat{A}}$. Choose a finite affine cover of Y, write the finitely many cover algebras, overlap maps, and cocycle equations with finitely many generators and relations, and descend all of them to a finite type A-algebra $A_i = A[x_1, \dots, x_N]/(f_1, \dots, f_M)$ inside $\widehat{A}$. After increasing i, the descended map is finite etale. Approximation for henselian G-rings gives a solution $(b_1, \dots, b_N) \in A^N$ of all equations with $b_j \equiv x_j \pmod{\mathfrak{m}}$. The induced retraction $A_i \to A$ base-changes the descended cover to a finite etale cover of Y with special fibre P_0. The graph argument proves full faithfulness.

For a general henselian local A, choose noetherian local G-rings A_i from finite-type $\mathbf{Z}$-models and write $A = \varinjlim_i A_i^h$. Proper finite presentation gives an index i_0 and a proper model $Y_{i_0} \to \operatorname{Spec} A_{i_0}^h$. A finite etale cover, a morphism between two such covers, and an equality of two morphisms are each finite presentation data; using a finite affine cover and the finitely many overlap equations, they descend to one common $i \geq i_0$. Thus

$$FEt_Y = \varinjlim_i FEt_{Y_i}, \qquad FEt_{Y_0} = \varinjlim_i FEt_{(Y_i)_0},$$

including the Hom sets. The equivalence at every henselian noetherian G-ring stage gives the desired equivalence after the colimit. Finally an action of a finite group consists of finitely many morphisms and finitely many multiplication, unit, and inverse equations. Full faithfulness lifts the morphisms and their equations uniquely, so group actions lift as well. □

Lemma 4.5.20 (Finite-etale torsor comparison)**.** *Lean:* `conditional-interface:lemma:weil-pbc-h1-torsor-comparison`

Uses: `def:weil-pbc-strict-henselian-setup`

For a finite abelian group M and proper $Y \to \operatorname{Spec} A$, restriction induces a bijection

$$H^1(Y, \underline{M}) \xrightarrow{\sim} H^1(Y_0, \underline{M}).$$

Proof. Uses: `lemma:weil-pbc-etale-degree-zero,lemma:weil-pbc-finite-locally-constant-etale-equivalence,lemma:weil-pbc-finite-etale-henselian-lifting`

An $\underline{M}$-torsor is a finite etale Y-scheme P with a simply transitive M-action by the finite-locally-constant equivalence. Apply the henselian lifting lemma to lift P, every automorphism $m : P \to P$, and the multiplication diagrams. Uniqueness of lifted morphisms preserves the group law and simple transitivity. Restriction and lifting are inverse on isomorphism classes, which is the displayed H^1 bijection. This is the torsor subtree of Tag `0A5G`. □

Lemma 4.5.21 (Constructible representation of an H^1 class)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-h1-constructible-representation`

For an injective sheaf I of $\mathbf{F}_\ell$-modules on $\mathbf{P}^1_A$, every class $c \in H^1(X_0, I_0)$ admits finite $\mathbf{F}_\ell$-modules M_i, finite finitely presented morphisms $h_i : V_i \to X$, and a map $\bigoplus_i (h_i)_* \underline{M_i} \to I$ such that c is the image of a tuple of classes in $\bigoplus_i H^1(X_0, (h_i)_* \underline{M_i}|_{X_0})$.

Proof. Uses: `lemma:weil-pbc-constructible-approximation,lemma:weil-pbc-constructible-finite-pushforward`

View c as the isomorphism class of an I_0-torsor. It is trivial on an etale cover; quasi-compactness of X_0 selects finitely many members $U_a \to X_0$. Choosing local origins gives a Cech cocycle $z = (z_{ab})$ with

$$z_{ab} \in I_0(U_a \times_{X_0} U_b), \qquad z_{bc} - z_{ac} + z_{ab} = 0$$

on every triple overlap. There are finitely many sections and equations. Constructible approximation therefore supplies a constructible sheaf F, a map $\alpha : F \to I$, and lifts $\widetilde{z}_{ab} \in F_0(U_{ab})$ satisfying the same equations and mapping to z_{ab}.

The finite-pushforward lemma gives a monomorphism $\iota : F \hookrightarrow F' = \bigoplus_{i=1}^r (h_i)_* \underline{M_i}$. Injectivity of I extends α across ι to a map $\psi : F' \to I$. The sheaf square and the three relevant terms of the Cech complex form the commuting three-row diagram

$$\begin{array}{ccccc}
F & \xrightarrow{\iota} & F' & \longrightarrow & \operatorname{coker}(\iota) \\
{\scriptstyle\alpha}\downarrow & & {\scriptstyle\psi}\downarrow & & \downarrow \\
I & = & I & \longrightarrow & 0 \\
\downarrow & & \downarrow & & \downarrow \\
\prod_a I_0(U_a) & \xrightarrow{d^0} & \prod_{a,b} I_0(U_{ab}) & \xrightarrow{d^1} & \prod_{a,b,c} I_0(U_{abc}).
\end{array}$$

Here the bottom vertical maps mean restriction followed by the indicated Cech cochain inclusion; commutativity in degrees $0, 1, 2$ follows from functoriality of restriction. The cocycle $\iota(\widetilde{z})$ defines $c' \in H^1(X_0, F_0')$ and $\psi_0(c') = c$.

Finite direct sums commute with H^1, while exact finite direct image and Leray give

$$H^1(X_0, F_0') = \bigoplus_{i=1}^r H^1(X_0, (h_i)_* \underline{M_i}|_{X_0}) = \bigoplus_{i=1}^r H^1((V_i)_0, \underline{M_i}).$$

Thus c' is the required finite tuple, and the displayed map $F' \to I$ sends that tuple to the original class. □

Lemma 4.5.22 (Acyclicity of injectives on $\mathbf{P}^1_s$)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-p1-injective-acyclic`

For an injective sheaf I of $\mathbf{F}_\ell$-modules on $X = \mathbf{P}^1_A$, the restriction I_0 is $\Gamma(X_0, -)$-acyclic: $H^q(X_0, I_0) = 0$ for all $q > 0$.

Proof. Uses: `lemma:weil-pbc-injective-restriction-acyclicity`, `lemma:weil-pbc-h1-torsor-comparison`, `lemma:weil-pbc-h1-constructible-representation`

The affine-cover lemma gives vanishing for $q > 1$ because $\mathbf{P}^1$ is covered by two affines. For $q = 1$, apply the representation lemma. For each finite summand, finite-direct-image exactness and Leray identify its H^1 on X_0 with $H^1((V_i)_0, \underline{M_i})$. Torsor comparison lifts every component to $H^1(V_i, \underline{M_i})$, and Leray returns a class in $H^1(X, (h_i)_* \underline{M_i})$. Their finite sum maps to zero in $H^1(X, I)$ because I is injective. Restriction of the commuting square shows that the original class is zero. The source locator is Tag `0A5H`. □

Proposition 4.5.23 (Strictly henselian $\mathbf{P}^1$ comparison)**.** Lean: `conditional-interface:prop:weil-pbc-p1-strict-henselian-comparison`

Uses: `def:weil-pbc-strict-henselian-setup`

For every torsion sheaf F on X, restriction induces isomorphisms

$$H^q(X, F) \xrightarrow{\sim} H^q(X_0, F_0) \qquad (q \geq 0).$$

Proof. Uses: `lemma:weil-pbc-etale-degree-zero`, `lemma:weil-pbc-p1-injective-acyclic`

Choose an injective resolution $F \to I^\bullet$. Exactness of inverse image along the closed immersion gives an exact complex $F_0 \to I_0^\bullet$. The preceding lemma makes each I_0^r acyclic, so Leray's acyclicity lemma identifies $R\Gamma(X_0, F_0)$ with $\Gamma(X_0, I_0^\bullet)$. Degree-zero restriction identifies this complex with $\Gamma(X, I^\bullet)$ term by term. Taking cohomology gives the displayed isomorphisms, and naturality follows from the naturality of the restriction map and of injective resolutions. The source locators are Tags `0A0C` and `0A3S`. □

Proposition 4.5.24 (Strict-henselian projective-line proper base change)**.** Uses:

Lean: `conditional-interface:prop:weil-pbc-projective-line-strict-henselian`

Let $A \to B$ be a local map of strictly henselian local rings which induces an isomorphism of their separably closed residue fields. For the Cartesian square obtained from $\mathbf{P}^1_A \to \operatorname{Spec} A$, the derived exchange map for a torsion sheaf is an isomorphism on the stalk at the closed geometric point of $\operatorname{Spec} B$.

Proof. Uses: `prop:weil-pbc-p1-strict-henselian-comparison`

Write $S = \operatorname{Spec} A$, $T = \operatorname{Spec} B$, and let $\bar{t}$ and $\bar{s}$ be their closed geometric points under the chosen residue-field identification. Choose an injective resolution $F \to I^\bullet$ on $\mathbf{P}^1_A$. A strict henselian local scheme is initial in the opposite category of pointed etale neighbourhoods of its closed geometric point: the chosen point selects a unique section of the connected component of any pointed etale neighbourhood. Consequently the pointed-neighbourhood stalk colimit has canonical restriction quasi-isomorphisms to global cohomology, and the following rectangle is defined degree by degree:

$$\begin{array}{ccc}
(Rp_*F)_{\bar{s}} & \xrightarrow{(\mathrm{Ex}_{\square,F})_{\bar{t}}} & (Rp_{T,*}F_T)_{\bar{t}} \\
{\scriptstyle \Phi_A}\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \Phi_B} \\
R\Gamma(\mathbf{P}^1_A, F) & \xrightarrow{\mathrm{res}_{A,B}} & R\Gamma(\mathbf{P}^1_B, F_T) \\
{\scriptstyle \rho_A}\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \rho_B} \qquad \sim V \\
R\Gamma(\mathbf{P}^1_{k_A}, F_{k_A}) & \xrightarrow{\mathrm{id}} & R\Gamma(\mathbf{P}^1_{k_B}, F_{k_B}).
\end{array}$$

Here Φ_A and Φ_B send a germ represented on a pointed neighbourhood to its pullback on the corresponding strict local base. To verify that they are quasi-isomorphisms without a stalk theorem as a leaf, compute a cohomology class on a finite split etale hypercover. Its $(q+1)$-skeleton, its cocycle, and any null-homotopy contain finitely many etale objects, face maps, sections, and equalities. The unique pointed sections pull this complete finite datum to the strict local base; conversely the datum descends to one pointed neighbourhood by finite presentation. Thus the stalk colimit and strict-local cohomology agree in every degree.

The upper square commutes because both routes apply to a representative $s \in I^r(\mathbf{P}^1_A \times_S V)$ by pulling it to $\mathbf{P}^1_B \times_T (V \times_S T)$ and then to the pointed limit. This is exactly the unit–counit formula defining the S0 exchange map. The lower square commutes because restriction of that same section through B and then k_B equals restriction through $k_A \cong k_B$; this also proves the compatibility with differentials of $I^\bullet$.

The maps ρ_A and ρ_B are quasi-isomorphisms by Proposition 4.5.23, and the bottom map is the identity under $k_A \cong k_B$. Hence the top exchange stalk is an isomorphism. Notice that no assertion is made here for a nontrivial extension of separably closed residue fields; that is precisely the S5.2 curve-invariance input. □

> **Status.** The unique stable finite-torsion export label is `prop:weil-proet-finite-proper-base-change`. It is deliberately not declared in this S0–S4 fragment: S5 (strict-henselization stalks and separably-closed field invariance) and S6 (global assembly and torsion devissage) must precede that declaration. The older A3 blueprint already occupies the stable label and must be replaced, not duplicated, at merge.

4.6 S5. The projective line over an arbitrary base

4.6.1 S5.1. Stalks through strict henselization

Definition 4.6.1 (Pointed etale neighbourhood category)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-pointed-etale-neighbourhoods`

Let $\bar{s} : \operatorname{Spec}\Omega \to S$ be a geometric point. Define $\mathcal{N}_{\bar{s}}$ to have objects pairs $(V, \bar{v})$ with $V \to S$ etale and $\bar{v} : \operatorname{Spec}\Omega \to V$ above $\bar{s}$. A morphism $(V, \bar{v}) \to (W, \bar{w})$ is an S-map $V \to W$ carrying $\bar{v}$ to $\bar{w}$. We index stalk colimits by $\mathcal{N}_{\bar{s}}^{\mathrm{op}}$ so that restriction maps point forward.

Lemma 4.6.2 (Affine pointed neighbourhoods are cofinal). Lean: `conditional-interface:lemma:weil-pbc-affine-neighbourhood-cofinal`

Uses: `def:weil-pbc-pointed-etale-neighbourhoods`

The full subcategory $\mathcal{N}_{\bar{s}}^{\mathrm{aff}}$ of affine pointed etale neighbourhoods is initial in $\mathcal{N}_{\bar{s}}$ and hence cofinal for the opposite-indexed stalk colimit.

Proof. Uses:

Given $(V, \bar{v})$, choose an affine open $V_0 \subseteq V$ containing the image of $\bar{v}$; then $(V_0, \bar{v}) \to (V, \bar{v})$ is a morphism. Given two affine refinements, their fibre product over V contains the lifted geometric point; an affine open around that point refines both. These two facts give nonemptiness and connectedness of every comma category required for initiality. □

Definition 4.6.3 (Strict henselization from pointed neighbourhoods). Lean: `conditional-interface:def:weil-pbc-strict-henselization-limit`

Uses: `def:weil-pbc-pointed-etale-neighbourhoods`

Let

$$\mathcal{I}_{\bar{s}} := (\mathcal{N}_{\bar{s}}^{\mathrm{aff}})^{\mathrm{op}}.$$

For $V = \operatorname{Spec} A_V$, define $D_{\bar{s}} : \mathcal{I}_{\bar{s}} \to \mathbf{CRing}$ by $D_{\bar{s}}(V, \bar{v}) = A_V$. An arrow $(W, \bar{w}) \to (V, \bar{v})$ in $\mathcal{I}_{\bar{s}}$ represented by a pointed map $a : (V, \bar{v}) \to (W, \bar{w})$ is sent to $a^{\#} : A_W \to A_V$. Put

$$A_{S,\bar{s}}^{\mathrm{sh}} := \varinjlim_{\mathcal{I}_{\bar{s}}} D_{\bar{s}}, \qquad S_{(\bar{s})} := \operatorname{Spec} A_{S,\bar{s}}^{\mathrm{sh}}.$$

The affine scheme–ring anti-equivalence carries the colimit cocone to the canonical limit cone

$$S_{(\bar{s})} \xrightarrow{\sim} \varprojlim_{(V,\bar{v}) \in \mathcal{N}_{\bar{s}}^{\mathrm{aff}}} V.$$

Lemma 4.6.4 (Well-definedness certificate for def:weil-pbc-strict-henselization-limit). Lean: `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-strict-henselization-limit`

Uses: `def:weil-pbc-strict-henselization-limit`

The coordinate-ring rule of Definition 4.6.3 is a functor $D_{\bar{s}} : \mathcal{I}_{\bar{s}} \to \mathbf{CRing}$. Applying Spec *to its colimit cocone gives the displayed limit cone, and affine cofinality gives the same strict-local object when nonaffine pointed neighbourhoods are admitted.*

Proof. Uses: `def:weil-pbc-strict-henselization-limit,lemma:weil-pbc-affine-neighbourhood-cofinal`

Coordinate rings carry an identity pointed map to an identity ring map and reverse composition. Passage to $(\mathcal{N}_{\bar{s}}^{\mathrm{aff}})^{\mathrm{op}}$ therefore makes $D_{\bar{s}}$ covariant. The fixed universe of the coefficient convention makes this a small diagram, so its colimit exists in **CRing**.

The anti-equivalence between affine schemes and commutative rings sends this colimit, together with its structure maps, to the inverse-limit cone on the affine pointed neighbourhoods. The initiality statement in Lemma 4.6.2 makes restriction to the affine subcategory limit-cofinal and opposite-colimit-cofinal. Hence the same ring and affine scheme compute the indicated strict-local object. □

Lemma 4.6.5 (Base change commutes with the neighbourhood limit). Lean: `conditional-interface:lemma:weil-pbc-fibre-product-limit`

Uses: `def:weil-pbc-strict-henselization-limit`

For a quasi-compact and quasi-separated $f : X \to S$,

$$X \times_S S_{(\bar{s})} \cong \varprojlim_{(V,\bar{v})} (X \times_S V).$$

Proof. Uses:

On affine charts $X_0 = \operatorname{Spec} B$ over $S_0 = \operatorname{Spec} A$, the left side has coordinate ring $B \otimes_A \varinjlim A_V$. Tensor product commutes with filtered colimits, so this is $\varinjlim (B \otimes_A A_V)$, the coordinate ring of the inverse limit on the right. Quasi-compactness gives finitely many charts and quasi-separatedness gives quasi-compact pairwise intersections; the affine identifications therefore glue. □

Lemma 4.6.6 (Finite etale hypercover skeleta descend). Uses:

Lean: `conditional-interface:lemma:weil-pbc-finite-hypercover-descent`

Let $X_\infty = \varprojlim_i X_i$ be as in the continuity lemma below and fix $N \geq 0$. Every etale hypercover of X_∞ admits a refinement whose N-skeleton is split, has finitely many quasi-compact and quasi-separated components in each degree, and is the pullback of an N-truncated etale hypercover of some X_j. A morphism between two such finite skeleta, and an equality between two such morphisms, also descends after increasing j.

Proof. Uses: `lemma:weil-pbc-fibre-product-limit`

Construct the refinement inductively. In degree zero, quasi-compactness of X_∞ lets us choose a finite subcover by quasi-compact etale objects. Suppose degrees $< r$ have been chosen. The rth matching object is a finite limit of lower-degree objects, because the boundary diagram of the simplex r-simplex has finitely many faces. It is therefore quasi-compact and quasi-separated. Cover it by finitely many quasi-compact etale objects and adjoin the degenerate summands prescribed by the already chosen degeneracy maps. This gives a split hypercover through degree r. Iterating to N gives the required finite skeleton.

Every chosen component is of finite presentation over X_∞, so it is the pullback of an etale component at some stage. There are only finitely many components, face maps, degeneracy maps, and augmentations through degree N; finite presentation descends them all to one common stage. The simplicial identities are finitely many equalities of morphisms between finitely presented schemes and hence hold after a further common stage. For each $r \le N$, the map to the matching object is an etale cover after pullback to X_∞. Etaleness descends by a Jacobian etale algebra presentation $B = A[x_1, \dots, x_m]/(f_1, \dots, f_m)$ with an invertible Jacobian minor. Surjectivity also descends: the complement of the image is constructible and quasi-compact; if it survived at every later stage, its inverse limit would give a point outside the limiting image. Thus, after one more stage, all matching maps are covers. The same finite-presentation argument descends a morphism of skeleta. Equality of two morphisms is the factorization through their equalizer, a finitely presented closed condition, so it too holds at a later stage. □

Lemma 4.6.7 (Etale cohomology continuity)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-etale-cohomology-continuity`

Let (X_i) be a cofiltered inverse system of quasi-compact and quasi-separated schemes with affine transition maps and limit X_∞. For a torsion sheaf F_i descending to one stage, the natural map

$$\varinjlim_{j \ge i} H^q(X_j, F_j) \longrightarrow H^q(X_\infty, F_\infty)$$

is an isomorphism.

Proof. Uses: `lemma:weil-pbc-fibre-product-limit,lemma:weil-pbc-finite-hypercover-descent`

We first recall why finite hypercover data compute the required cohomology. For an etale hypercover $U_\bullet \to Z$, put

$$C^r(U_\bullet, F) = \Gamma(U_r, F), \qquad d = \sum_{a=0}^{r+1} (-1)^a \partial_a^*.$$

Order split hypercovers by common refinement and set $H^q_{\mathrm{hyp}}(Z, F) = \varinjlim_{U_\bullet} H^q(C^\bullet(U_\bullet, F))$. The sheaf equalizer gives $H^0_{\mathrm{hyp}}(Z, F) = \Gamma(Z, F)$. A short exact sequence of sheaves gives connecting maps after refining so that each of the finitely many local lifts of a cocycle exists. The resulting functors form a cohomological delta functor. They are effaceable in positive degree: given a q-cocycle, cover its finitely many degree-q components so that they lift through a chosen monomorphism into an injective sheaf, then fill the matching objects successively through degree $q+1$; on this refinement the cocycle is a coboundary. Hence this delta functor is the universal derived functor of global sections. Therefore

$$H^q_{\mathrm{hyp}}(Z, F) = H^q_{\mathrm{et}}(Z, F).$$

The construction also shows that a degree-q class and a witness that it is zero use only the $(q+1)$-skeleton of one split hypercover.

Let $\alpha \in H^q(X_\infty, F_\infty)$. Represent it by a cocycle $c \in C^q(U_\bullet, F_\infty)$ on a split hypercover with finite qcqs $(q+1)$-skeleton. By the preceding lemma, this skeleton descends to some X_j. Each component of c is, by the definition of the inverse-image sheaf, locally represented by a section at a finite stage. Refine the finite degree-q cover once and choose one stage dominating all those representatives. The equation $dc = 0$ consists of finitely many equalities on the degree-$(q+1)$ components; these equalities descend after one further increase of the stage. We obtain a cocycle c_j and hence a class $\alpha_j \in H^q(X_j, F_j)$ mapping to α. This proves surjectivity.

For injectivity, suppose $\alpha_j \in H^q(X_j, F_j)$ maps to zero on X_∞. Choose a finite split hypercover and cocycle c_j representing it. Vanishing on the limit supplies a refinement $V_\bullet \to U_\bullet \times_{X_j} X_\infty$ and a cochain $b \in C^{q-1}(V_\bullet, F_\infty)$ with $db = c_j|_V$. Only the $(q+1)$-skeleton, the refinement maps, the components of b, and the finite list of equations $db = c_j|_V$ are used. Descend that entire list to one stage X_k by the preceding lemma and the same section argument. It gives a null-homotopy for the image of c_j on X_k, so α_j is zero in the filtered colimit. The comparison map is therefore injective.

All constructions commute with restriction and with the transition maps, so the resulting bijection is the canonical map in the statement. This expands the continuity input of Tag `03Q6`; neither the hypercover nor the common finite stage is suppressed. □

Proposition 4.6.8 (Strict-henselization stalk formula)**.** *Lean:* `conditional-interface:prop:weil-pbc-strict-henselization-stalk-formula`

Uses: `def:weil-pbc-pointed-etale-neighbourhoods, def:weil-pbc-strict-henselization-limit`

For quasi-compact quasi-separated $f : X \to S$, torsion F, and $q \ge 0$,

$$(R^q f_* F)_{\bar s} \cong H^q(X \times_S S_{(\bar s)}, F|).$$

For $K \in D^+(X_{\mathrm{et}})$ this lifts to

$$(Rf_* K)_{\bar s} \xrightarrow{\sim} R\Gamma(X \times_S S_{(\bar s)}, K|)$$

in $D^+(\mathbf{Ab})$.

Proof. Uses: `lemma:weil-pbc-affine-neighbourhood-cofinal,lemma:weil-pbc-fibre-product-limit,lemma:weil-pbc-etale-cohomology-continuity`

The stalk is the colimit over $\mathcal{N}_{\bar s}^{\mathrm{op}}$ of $H^q(X \times_S V, F|)$. Replace the index by affine neighbourhoods, identify their inverse limit by Lemma 4.6.5, and apply cohomology continuity. For the derived statement, choose a bounded-below injective complex $J^\bullet$ representing K. The chain map from the stalk complex $(f_* J^\bullet)_{\bar s}$ to the sections complex on the strict henselization is obtained by restricting each representative section. Its map on every cohomology group is the preceding isomorphism, so it is a quasi-isomorphism. This expands Tag `03Q9`. □

4.6.2 S5.2. Proper curves under separably closed extension

Lemma 4.6.9 (Canonical etale descent across a universal homeomorphism)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-universal-homeomorphism-descent-datum`

Let $u : X_0 \to X$ be a universal homeomorphism. For every etale $V_0 \to X_0$, the two pullbacks to $X_0 \times_X X_0$ have a unique isomorphism whose restriction along the diagonal is the identity. This isomorphism is a descent datum and is compatible with morphisms, identities, and composition.

Proof. Uses:

A universal homeomorphism is integral, surjective, and universally injective. Hence the diagonal

$$\Delta : X_0 \longrightarrow X_0 \times_X X_0$$

is a closed immersion and is surjective on points. On affine charts it is therefore defined by a nil ideal: if $X = \operatorname{Spec} A$ and $X_0 = \operatorname{Spec} B$, the multiplication $B \otimes_A B \to B$ is surjective and its kernel is contained in every prime ideal.

We first record the nil-ideal calculation used below. If $R \to R/J$ has J nil, reduction gives an equivalence between etale R-algebras and etale R/J-algebras. For a finitely presented affine etale algebra this is checked on a Jacobian presentation

$$((R/J)[T_1, \dots, T_m]/(f_1, \dots, f_m))_g$$

with the relevant Jacobian minor invertible. Lift the finitely many coefficients and g to R. The same minor remains invertible after localization because an element invertible modulo a nil ideal is invertible. Existence of a lift of a morphism follows by successively correcting the images of the T_i modulo the powers of the finitely generated nil ideal appearing in those coefficients; the inverse Jacobian gives the correction. The process is finite because that finitely generated subideal is nilpotent. If two lifts agree modulo J, their difference lies in J; the same linearized equations and inverse Jacobian force equality, one nilpotence power at a time. General etale schemes are glued from such affine pieces; the uniqueness just proved makes the overlap maps and their cocycle equations lift uniquely.

Apply this nil-invariance to Δ. The two pullbacks $p_1^* V_0$ and $p_2^* V_0$ restrict along Δ to the same etale X_0-scheme V_0. Thus there is a unique isomorphism

$$\alpha : p_1^* V_0 \xrightarrow{\sim} p_2^* V_0, \qquad \Delta^* \alpha = \mathrm{id}_{V_0}.$$

On $X_0 \times_X X_0 \times_X X_0$, both $p_{23}^* \alpha \circ p_{12}^* \alpha$ and $p_{13}^* \alpha$ restrict to the identity along the small diagonal. That diagonal is again a surjective closed immersion defined by a nil ideal, so uniqueness gives the cocycle equation. If $a : V_0 \to W_0$ is a morphism, the two composites $p_2^* a \circ \alpha_V$ and $\alpha_W \circ p_1^* a$ have the same diagonal restriction and hence are equal. Taking a to be an identity or a composite proves the asserted identity and composition compatibilities. □

Lemma 4.6.10 (Covering compatibility for universal homeomorphisms)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-universal-homeomorphism-coverings`

Let $u : X_0 \to X$ be a universal homeomorphism. Base change gives an equivalence between etale X-schemes and etale X_0-schemes, and a family $\{U_i \to U\}$ is jointly surjective before base change if and only if its base change is jointly surjective.

Proof. Uses: `lemma:weil-pbc-universal-homeomorphism-descent-datum`

We construct a quasi-inverse to base change. First let $X = \operatorname{Spec} A$ and let $V_0 = \operatorname{Spec} C$ be affine etale over an affine inverse image $\operatorname{Spec} B \subset X_0$. The preceding lemma gives an isomorphism between the two $B \otimes_A B$-algebras obtained from C, with its cocycle equation. Define D as the equalizer of the two resulting maps

$$C \rightrightarrows C \otimes_A B,$$

where the second arrow is followed by the descent isomorphism. On a Jacobian etale presentation of C, this equalizer uses only its finitely many generators, relations, and descent equations. The inverse-Jacobian lifting calculation from the preceding proof gives

$$B \otimes_A D \xrightarrow{\sim} C$$

and shows that D is finitely presented and etale over A: surjectivity lifts each generator of C after imposing its descent equation, injectivity uses the cocycle equation on $B \otimes_A B \otimes_A B$, and the lifted Jacobian minor is invertible. This is the affine effective-descent calculation, rather than an appeal to an unnamed descended algebra.

For a general etale $V_0 \to X_0$, choose affine opens $W_a \subset V_0$, each mapping into an affine inverse image in X_0. Their descended affine etale schemes are denoted $V_a \to X$. The intersections $W_a \cap W_b$ are open subspaces. Because $|X_0| \to |X|$ and every one of its base changes are homeomorphisms, each such open is the inverse image of a unique open $V_{ab} \subset V_a$. The two identifications of $W_a \cap W_b$ descend uniquely to $V_{ab} \cong V_{ba}$ by the morphism compatibility in the preceding lemma. On triple intersections the two composites have equal pullback and hence are equal; thus the V_a glue to an etale $V \to X$ with $V \times_X X_0 \cong V_0$. The same uniqueness shows that maps descend and that the unit and counit of the two functors are identities after pullback. Base change is therefore an equivalence.

It remains to check the topology, the omission in Tag `03SI`. For a family $\{U_i \to U\}$ of etale X-schemes, set $Z = |U| \setminus \bigcup_i \operatorname{im}(|U_i| \to |U|)$. Base change gives a homeomorphism $|U \times_X X_0| \to |U|$, and its inverse image of Z is exactly the complement of the union of the base-changed images. Hence $Z = \varnothing$ if and only if that inverse image is empty. This proves joint surjectivity in both directions. The identity and composition compatibilities already proved show that this correspondence of covering families respects refinements and composite coverings. □

Definition 4.6.11 (The eight curve-cohomology assertions)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-eight-curve-assertions`

Let k be algebraically closed, let X be separated of finite type over k with $\dim X \leq 1$, and let F be a torsion sheaf on X_{et}. Write $\mathcal{C}_r(X, F)$, $1 \leq r \leq 8$, for the following assertions:

(1) $H^q(X, F) = 0$ for $q > 2$;

(2) $H^q(X, F) = 0$ for $q > 1$ when X is affine;

(3) $H^q(X, F) = 0$ for $q > 1$ when $\operatorname{char} k = p > 0$ and F is p-power torsion;

(4) every $H^q(X, F)$ is finite when F is constructible and its torsion is prime to $\operatorname{char} k$;

(5) every $H^q(X, F)$ is finite when X is proper and F is constructible;

(6) for every algebraically closed extension k'/k, the restriction $H^q(X, F) \to H^q(X_{k'}, F|_{X_{k'}})$ is an isomorphism when the torsion of F is prime to $\operatorname{char} k$;

(7) the map in (6) is an isomorphism when X is proper, without a restriction on the torsion;

(8) for every open $a : W \hookrightarrow X$, the restriction $H^2(X, F) \to H^2(W, a^{-1}F)$ is surjective.

Lemma 4.6.12 (Filtered cohomology continuity on a curve)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-curve-filtered-cohomology`

Let X be quasi-compact and quasi-separated and let (F_λ) be a filtered system of torsion etale sheaves. For every $q \geq 0$ the canonical map

$$\varinjlim_\lambda H^q(X, F_\lambda) \longrightarrow H^q(X, \varinjlim_\lambda F_\lambda)$$

is an isomorphism, naturally in X, the system, and pullback along a field extension.

Proof. Uses:

For $q = 0$, choose a finite etale cover on which a section is represented. The section and its finitely many equalizer equations on pairwise fibre products occur at one index, proving surjectivity; if two representatives become equal in the colimit, the finite cover and equality witnesses occur at one later index, proving injectivity.

For $q > 0$, represent a class on a split etale hypercover whose $(q+1)$-skeleton has finitely many quasi-compact components. A cocycle uses finitely many sections and face-map equations, so all occur at one index λ. This proves surjectivity. If the class from F_λ dies in the colimit, choose a finite refinement and a degree-$(q-1)$ cochain whose coboundary is the cocycle. The refinement maps, cochain sections, and coboundary equations occur at one $\mu \geq \lambda$, so the class already dies in $H^q(X, F_\mu)$. Pullback of the finite hypercover and all its face maps gives the same construction over an extension field, which proves the stated naturality. □

Lemma 4.6.13 (Short-exact devissage for the eight assertions)**.** *Lean:* `conditional-interface:lemma:weil-pbc-eight-curve-devissage`

Uses: `def:weil-pbc-eight-curve-assertions`

Let

$$0 \longrightarrow F_1 \xrightarrow{a} F \xrightarrow{b} F_2 \longrightarrow 0$$

be exact. If all applicable assertions $\mathcal{C}_r(X, F_i)$ hold for $i = 1, 2$, then all applicable assertions hold for F.

Proof. Uses:

Pullback to an extension field is exact, so the displayed sequence and its base change give natural long exact cohomology sequences.

For (1), (2), and (3), in every degree above the stated bound the exact segment

$$H^q(X, F_1) \xrightarrow{a_*} H^q(X, F) \xrightarrow{b_*} H^q(X, F_2)$$

has zero end terms. Hence its middle term is zero. The hypotheses on affineness and characteristic are unchanged, and in (3) an extension of two p-power torsion sheaves is again killed by a power of p.

For (4) and (5), constructible sheaves form a Serre subcategory: on a common finite stratification the stalk sequence is an exact sequence of finite groups. Prime-to-characteristic torsion is also preserved by extensions. Exactness identifies $H^q(X, F)$ as an extension of $\operatorname{im} a_*$ by $\operatorname{im} b_*$; these are respectively a quotient of the finite group $H^q(X, F_1)$ and a subgroup of the finite group $H^q(X, F_2)$. Thus $H^q(X, F)$ is finite.

For (6) and (7), write the long exact sequence for F_1, F, F_2 above its base-changed sequence. Every square commutes because restriction commutes with a, b, and the boundary map obtained from an injective resolution. For fixed q, the four outer vertical arrows in the five-term portion

$$H^{q-1}(F_2) \to H^q(F_1) \to H^q(F) \to H^q(F_2) \to H^{q+1}(F_1)$$

are isomorphisms; the five lemma makes the middle arrow an isomorphism.

For (8), let $c \in H^2(W, F|_W)$ and put $c_2 = b_*(c)$. Choose $x_2 \in H^2(X, F_2)$ restricting to c_2. Its boundary lies in $H^3(X, F_1) = 0$ by (1), so exactness gives $x \in H^2(X, F)$ with $b_*(x) = x_2$. The difference $c - x|_W$ maps to zero in $H^2(W, F_2|_W)$, hence equals $a_*(d_1)$ for some $d_1 \in H^2(W, F_1|_W)$. Choose $y_1 \in H^2(X, F_1)$ restricting to d_1. Then $x + a_*(y_1)$ restricts to c. This proves the eighth assertion and completes all eight cases compressed in Tag `0A5C`. □

Lemma 4.6.14 (Constructible and primary reduction on curves)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-curve-constructible-primary-reduction`

For a separated finite-type curve X over an algebraically closed field and a torsion sheaf F, extension-invariance of $H^q(X, F)$ reduces to the case where X is reduced and F is a constructible $\mathbf{F}_\ell$-sheaf.

Proof. Uses: `lemma:weil-pbc-universal-homeomorphism-coverings,lemma:weil-pbc-curve-filtered-cohomology,lemma:weil-pbc-eight-curve-devissage`

The reduction map $r : X_{\mathrm{red}} \to X$ is a universal homeomorphism. The site equivalence sends F to $r^{-1}F$, preserves covering families, and therefore identifies the cochain complexes computing all cohomology groups. It also commutes with the corresponding reduction after extending k.

The etale topos of the noetherian scheme X is coherent. Given finitely many local sections of F on quasi-compact etale objects and finitely many relations, their image generates a constructible subsheaf: refine by a finite stratification on which the

finitely generated stalk groups are constant. Ordered by inclusion, these constructible subsheaves F_λ form a filtered system with $F = \varinjlim_\lambda F_\lambda$. The preceding continuity lemma gives the natural square

$$\begin{array}{ccc} \varinjlim_\lambda H^q(X, F_\lambda) & \xrightarrow{\sim} & H^q(X, F) \\ \downarrow & & \downarrow \\ \varinjlim_\lambda H^q(X_{k'}, F_\lambda|) & \xrightarrow{\sim} & H^q(X_{k'}, F|). \end{array}$$

Thus comparison for every constructible F_λ implies comparison for F.

Fix a constructible F. A finite stratification with finite stalks gives one exponent N annihilating every stalk, hence $NF = 0$. Write $N = \prod_{\ell | N} \ell^{r_\ell}$. Bezout supplies integers e_ℓ which are 1 modulo ℓ^{r_ℓ} and 0 modulo the other prime powers. The mutually orthogonal endomorphisms $e_\ell : F \to F$ sum to the identity and give the explicit decomposition

$$F \xrightarrow{\sim} \bigoplus_{\ell | N} F[\ell^{r_\ell}], \qquad s \longmapsto (e_\ell s)_\ell,$$

whose inverse is addition. It remains to treat one summand.

If $\ell^r F = 0$, use the exact kernel–quotient sequence

$$0 \longrightarrow F[\ell] \xrightarrow{\iota} F \xrightarrow{\pi} F/F[\ell] \longrightarrow 0.$$

The quotient is killed by ℓ^{r-1} because $\ell^{r-1} s \in F[\ell]$. Pullback to $X_{k'}$ preserves this kernel, quotient, and both maps. Induction on r, with the base case $\ell F = 0$, and the short-exact devissage lemma prove all applicable eight assertions. In particular assertion (7) gives the required proper field-extension invariance, reduced to constructible $\mathbf{F}_\ell$-module sheaves. □

Lemma 4.6.15 (Dense-open localization reduction)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-curve-dense-open-reduction`

Let X be a reduced curve and F a constructible $\mathbf{F}_\ell$-sheaf. There is a dense open immersion $j : U \hookrightarrow X$ with $U = \coprod_i U_i$ irreducible and $F|_U$ finite locally constant. If $i : Z = X \setminus U \hookrightarrow X$, then

$$0 \longrightarrow j_! j^{-1} F \longrightarrow F \longrightarrow i_* i^{-1} F \longrightarrow 0$$

reduces extension-invariance to $j_! G$ and to sheaves on the zero-dimensional scheme Z.

Proof. Uses: `lemma:weil-pbc-curve-constructible-primary-reduction,lemma:weil-pbc-eight-curve-devissage`

Constructibility gives a finite stratification. Remove the finite closed strata and all intersections of distinct irreducible components. The resulting $U = \coprod_a U_a$ meets every component, each U_a is irreducible, and $G = j^{-1}F$ is finite locally constant.

Let $\epsilon : j_! G \to F$ be extension by zero of the identity of G, and let $\rho : F \to i_* i^{-1} F$ be restriction to Z. At a geometric point of U, the stalks of (ϵ, ρ) are $G_{\bar{x}} \xrightarrow{\mathrm{id}} F_{\bar{x}} \to 0$; at a geometric point of Z they are $0 \to F_{\bar{z}} \xrightarrow{\mathrm{id}} F_{\bar{z}}$. Geometric stalks are exact and conservative, so

$$0 \longrightarrow j_! G \xrightarrow{\epsilon} F \xrightarrow{\rho} i_* i^{-1} F \longrightarrow 0$$

is exact with the displayed maps.

The reduced zero-dimensional scheme Z is a finite disjoint union of k-points. Exact finite direct image and Leray give

$$H^q(X, i_* i^{-1} F) = H^q(Z, i^{-1} F) = 0 \quad (q > 0), \qquad H^0(Z, i^{-1} F) = \bigoplus_{\bar{z} \in Z} F_{\bar{z}}.$$

The last group is finite. The long exact sequence is therefore

$$\begin{aligned} 0 \to H^0(X, j_! G) \to H^0(X, F) \to \bigoplus_{\bar{z} \in Z} F_{\bar{z}} \\ \to H^1(X, j_! G) \to H^1(X, F) \to 0, \end{aligned}$$

and restriction induces $H^q(X, j_! G) \cong H^q(X, F)$ for $q > 1$.

These maps separate the eight assertions as follows. Assertions (1)–(3) and (8) use the isomorphisms in degrees at least two, also after replacing X by an open W and U by $W \cap U$. Assertions (4) and (5) use the displayed five-term sequence and finiteness of the boundary-stalk sum. For (6) and (7), base change preserves Z, its stalk groups, ϵ, ρ, and the boundary map; the two five-term sequences form a commuting diagram, and the five lemma handles degrees $0, 1$, while the higher-degree maps are already identified. Thus the eight assertions hold for F if and only if they hold for $j_! G$.

Finally $j_! G = \bigoplus_a j_{a,!}(G|_{U_a})$. The inclusions and projections split this finite sum, and repeated split exact sequences $0 \to E_1 \to E_1 \oplus E_2 \to E_2 \to 0$ together with short-exact devissage reduce to the individual irreducible components. □

Lemma 4.6.16 (Finite ℓ-group representation filtration)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-finite-l-group-filtration`

Let H be a finite ℓ-group and V a nonzero finite-dimensional $\mathbf{F}_\ell[H]$-module. Then $V^H \neq 0$ and V has a finite filtration by H-stable subspaces whose successive quotients are the trivial one-dimensional representation.

Proof. Uses:

The group algebra augmentation ideal $J \subset \mathbf{F}_\ell[H]$ is nilpotent: induct on $|H|$, choose a central subgroup C of order ℓ, and use $(c-1)^\ell = c^\ell - 1 = 0$ together with nilpotence for H/C. Choose the least r with $J^r V = 0$; then $J^{r-1} V$ is nonzero and fixed by H. Select a one-dimensional fixed subspace and induct on $\dim V$ after quotienting by it. This fills the future-reference gap in Tag `0A3R`. □

Lemma 4.6.17 (Locally constant sheaf reduction)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-curve-locally-constant-reduction`

For irreducible U and finite locally constant G of $\mathbf{F}_\ell$-vector spaces, there is a finite etale $v : V \to U$ of degree prime to ℓ such that $v^{-1}G$ has a filtration by constant $\mathbf{F}_\ell$-sheaves, and G is a direct summand of $v_*v^{-1}G$.

Proof. Uses: `lemma:weil-pbc-finite-l-group-filtration,lemma:weil-pbc-finite-exchange`

Choose a geometric point $\bar{u}$ of U. The locally constant sheaf gives a representation

$$\rho : \pi_1(U, \bar{u}) \longrightarrow \mathrm{GL}_{\mathbf{F}_\ell}(G_{\bar{u}})$$

with finite image Q. If $H \subseteq Q$ is an ℓ-Sylow subgroup, the inverse image $\rho^{-1}(H)$ defines a connected finite etale cover $v : V \to U$ of degree $d = [Q : H]$, which is prime to ℓ. On V, the monodromy factors through H.

The representation filtration gives $0 = V_0 \subset V_1 \subset \cdots \subset V_m = G_{\bar{u}}$ with $V_a/V_{a-1} \cong \mathbf{F}_\ell$ and trivial H-action. Passing from H-modules to locally constant sheaves gives, for every a, the explicit short exact sequence

$$0 \longrightarrow G_{a-1} \longrightarrow G_a \longrightarrow \underline{\mathbf{F}_\ell}_V \longrightarrow 0, \qquad G_m = v^{-1}G.$$

All inclusions, quotient maps, and connecting maps commute with extension of the algebraically closed ground field.

The adjunction unit and the fibrewise sum are

$$\eta : G \longrightarrow v_*v^{-1}G, \qquad \mathrm{tr} : v_*v^{-1}G \longrightarrow G.$$

At a geometric stalk, $\eta(s) = (s, \dots, s)$ in the direct sum indexed by the d points of the fibre, while $\mathrm{tr}(s_1, \dots, s_d) = \sum_a s_a$. Consequently $\mathrm{tr} \circ \eta = d\,\mathrm{id}_G$. Multiplication by $d^{-1} \in \mathbf{F}_\ell$ defines $r = d^{-1}\,\mathrm{tr}$ with $r\eta = \mathrm{id}_G$. The stalk formulas are preserved by base change, so the direct-summand decomposition and every filtration exact sequence are compatible with the cohomology comparison maps. □

Definition 4.6.18 (Finite-open comparison map)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-finite-open-comparison`

Consider a Cartesian square

$$\begin{array}{ccc} U & \xrightarrow{j_X} & X \\ {\scriptstyle v}\downarrow & & \downarrow{\scriptstyle \bar{v}} \\ V & \xrightarrow{j_Y} & Y \end{array}$$

with j_X, j_Y open immersions and $\bar{v}$ finite. For an abelian sheaf G on U_{et}, define

$$\beta_{\bar{v},j_Y,G} : (j_Y)_!v_*G \longrightarrow \bar{v}_*(j_X)_!G$$

as the morphism adjoint to

$$v_*G \xrightarrow{\sim} j_Y^{-1}\bar{v}_*(j_X)_!G.$$

The displayed isomorphism is finite base change $j_Y^{-1}\bar{v}_* \cong v_*j_X^{-1}$ followed by $j_X^{-1}(j_X)_!G \cong G$.

Lemma 4.6.19 (Section and stalk inverse for the finite-open map)**.** *Lean:* `conditional-interface:lemma:weil-pbc-finite-open-comparison-inverse`

Uses: `def:weil-pbc-finite-open-comparison`

The map $\beta_{\bar{v},j_Y,G}$ is an isomorphism. Its inverse is obtained by restricting to U, collecting the finite-fibre tuple on V, and extending that tuple by zero to Y; both inverse equations hold on sections and on geometric stalks.

Proof. Uses:

Let $T \to Y$ be etale and put $T_V = T \times_Y V$, $T_X = T \times_Y X$, and $T_U = T \times_Y U$. A section of $(j_Y)_!v_*G$ over T is etale-locally a finite tuple

$$(s_x)_{x \in T_U/T_V}, \qquad s_x \in G$$

on T_V, extended by zero on $T \setminus T_V$. The section map $\beta(T)$ sends this tuple to the identical tuple on $T_U \subset T_X$ and extends it by zero on $T_X \setminus T_U$. This is a section of $(j_X)_!G$ because the support remains closed after the finite map $T_X \to T$. Restriction along a map $T' \to T$ pulls back every component, so these section maps define a sheaf morphism and agree with the adjunction definition.

Conversely, take $a \in \Gamma(T_X, (j_X)_!G)$. Restrict a to T_U and group its components over the finite fibres of $T_U \to T_V$. Since a vanishes at every geometric point of $T_X \setminus T_U$, the grouped tuple vanishes at every geometric point of $T \setminus T_V$ and therefore defines a section $\gamma(T)(a) \in \Gamma(T, (j_Y)_!v_*G)$. Restriction preserves the grouping, so the $\gamma(T)$ define a sheaf morphism

$$\gamma_{\bar{v},j_Y,G} : \bar{v}_*(j_X)_!G \longrightarrow (j_Y)_!v_*G.$$

At a geometric point $\bar{y} \to Y$, the two stalks are

$$\begin{cases} \displaystyle\bigoplus_{\bar{x} \mapsto \bar{y}} G_{\bar{x}}, & \bar{y} \in V, \\ 0, & \bar{y} \notin V. \end{cases}$$

Here a point $\bar{x}$ above $\bar{y} \in V$ lies in U by Cartesianity. Both $\beta_{\bar{y}}$ and $\gamma_{\bar{y}}$ are the identity of the displayed finite sum; outside V both are the zero map of the zero group. Hence

$$\gamma_{\bar{v},j_Y,G} \circ \beta_{\bar{v},j_Y,G} = \mathrm{id}_{(j_Y)_!v_*G}, \qquad \beta_{\bar{v},j_Y,G} \circ \gamma_{\bar{v},j_Y,G} = \mathrm{id}_{\bar{v}_*(j_X)_!G}$$

on all geometric stalks. Conservativity of geometric stalks proves both equations as sheaf maps. □

Lemma 4.6.20 (Finite pushforward and extension by zero)**.** *Lean:* `conditional-interface:lemma:weil-pbc-finite-pushforward-extension-zero`

Uses: `def:weil-pbc-finite-open-comparison`

In a Cartesian square with j, j' open immersions and $\bar{v}$ finite,

$$j_! v_* G \xrightarrow{\sim} \bar{v}_* j'_! G$$

is the map of Definition 4.6.18. It is functorial in G, compatible with identity and composite finite maps, and commutes with arbitrary base change.

Proof. Uses: `lemma:weil-pbc-curve-locally-constant-reduction,lemma:weil-pbc-finite-open-comparison-inverse`

The preceding lemma proves that the specified map is an isomorphism and gives both inverse equations.

For a morphism $\phi : G \to H$, naturality is the commutative square

$$\begin{array}{ccc} j_! v_* G & \xrightarrow{\beta_G} & \bar{v}_* j'_! G \\ {\scriptstyle j_! v_* \phi}\downarrow & & \downarrow{\scriptstyle \bar{v}_* j'_! \phi} \\ j_! v_* H & \xrightarrow{\beta_H} & \bar{v}_* j'_! H. \end{array}$$

At $\bar{y} \in V$, both routes send $(s_{\bar{x}})_{\bar{x} \mapsto \bar{y}}$ to $(\phi_{\bar{x}}(s_{\bar{x}}))_{\bar{x} \mapsto \bar{y}}$; outside V all four stalks vanish. Thus the square commutes. The same calculation with the inverse maps γ_G, γ_H proves naturality of the inverse.

If $\bar{v} = \mathrm{id}_Y$ and $v = \mathrm{id}_V$, every fibre sum has one summand. Both the unit–base-change construction of β and the section formula are the identity of $j_! G$. This proves identity compatibility.

For composition, let $f_1 : X_1 \to Y$ and $f_2 : X_2 \to X_1$ be finite. Put $U_i = X_i \times_Y V$, let $g_1 : U_1 \to V$, $g_2 : U_2 \to U_1$, and denote the open immersions by $j_0 : V \to Y$, $j_i : U_i \to X_i$. The compatibility to prove is the outer equality in

$$\begin{array}{ccccc} (j_0)_!(g_1 g_2)_* G & \xrightarrow{\beta_{f_1, g_2, * G}} & f_{1,*}(j_1)_! g_{2,*} G & \xrightarrow{f_{1,*}\beta_{f_2, G}} & f_{1,*} f_{2,*} (j_2)_! G \\ {\scriptstyle \beta_{f_1 f_2, G}}\downarrow{\scriptstyle \sim} & & & & \downarrow{=} \\ (f_1 f_2)_* (j_2)_! G & = & (f_1 f_2)_* (j_2)_! G & = & (f_1 f_2)_* (j_2)_! G. \end{array}$$

At $\bar{y} \in V$, the upper route first groups a tuple by $\bar{x}_1 \mapsto \bar{y}$ and then by $\bar{x}_2 \mapsto \bar{x}_1$:

$$\bigoplus_{\bar{x}_1 \mapsto \bar{y}} \left(\bigoplus_{\bar{x}_2 \mapsto \bar{x}_1} G_{\bar{x}_2} \right) \longrightarrow \bigoplus_{\bar{x}_2 \mapsto \bar{y}} G_{\bar{x}_2}.$$

It is the canonical flattening of the doubly indexed finite sum. The lower route is the same flattening for the composite fibre. Outside V every term is zero. This proves composition compatibility; applying the reverse grouping proves the corresponding equation for γ.

Finally let $h : \widetilde{Y} \to Y$ be arbitrary and put tildes on all four schemes and maps after base change. If h_X, h_V, h_U are the projections, open and finite base change give the vertical isomorphisms in

$$\begin{array}{ccc} h^{-1}(j_Y)_! v_* G & \xrightarrow{h^{-1}\beta_{\bar{v}, j_Y, G}} & h^{-1}\bar{v}_*(j_X)_! G \\ {\scriptstyle \sim}\downarrow & & \downarrow{\scriptstyle \sim} \\ (\widetilde{j_Y})_! \widetilde{v}_* h_U^{-1} G & \xrightarrow{\widetilde{\beta}_{\widetilde{v}, j_Y, h_U^{-1} G}} & \widetilde{\bar{v}}_* (\widetilde{j_X})_! h_U^{-1} G. \end{array}$$

The left vertical map is $h^{-1}(j_Y)_! \cong (\widetilde{j_Y})_! h_V^{-1}$ followed by $h_V^{-1} v_* \cong \widetilde{v}_* h_U^{-1}$; the right vertical map uses the same two comparisons in the other order. At a geometric point of $\widetilde{V}$, both routes pull each component $s_{\bar{x}}$ to the corresponding component over the pulled-back geometric fibre. Off $\widetilde{V}$, both routes are zero. The rectangle therefore commutes, and the inverse rectangle follows from the two inverse equations. This supplies the functoriality omitted at Tag `0GJ1`. □

Lemma 4.6.21 (Smooth prime-to-characteristic curve calculation)**.** *Lean:* `conditional-interface:lemma:weil-pbc-smooth-curve-kummer-localization`

Uses: `def:weil-pbc-eight-curve-assertions`

Let k be algebraically closed, let C be a smooth separated finite-type scheme of dimension at most one, and let $\ell \neq \mathrm{char}\, k$. All eight applicable assertions $\mathcal{C}_r(C, \underline{\mathbf{F}_\ell})$ hold, and every field-extension map and open-restriction map is the one induced by pullback of etale cocycles.

Proof. Uses:

Points have no higher etale cohomology, and finite disjoint unions commute with every map under consideration. We may therefore take C to be a smooth connected curve. Let $\overline{C}$ be its smooth proper compactification, of genus g, and let $D = \overline{C} \setminus C$ consist of r points.

Choose a primitive ℓth root in k, identifying $\underline{\mathbf{F}_\ell}$ with μ_ℓ. The Kummer sequence

$$1 \longrightarrow \mu_\ell \longrightarrow \mathbf{G}_m \xrightarrow{(\cdot)^\ell} \mathbf{G}_m \longrightarrow 1$$

gives

$$H^1(\overline{C}, \mu_\ell) = \mathrm{Pic}(\overline{C})[\ell], \qquad 0 \to \mathrm{Pic}(\overline{C})/\ell \to H^2(\overline{C}, \mu_\ell) \to \mathrm{Br}(\overline{C})[\ell] \to 0.$$

The degree sequence $0 \to \mathrm{Pic}^0(\overline{C})(k) \to \mathrm{Pic}(\overline{C}) \to \mathbf{Z} \to 0$ and the finite etale isogeny $[\ell]$ on the g-dimensional Jacobian give

$$\mathrm{Pic}(\overline{C})[\ell] \cong (\mathbf{Z}/\ell)^{2g}, \qquad \mathrm{Pic}(\overline{C})/\ell \cong \mathbf{Z}/\ell.$$

The Brauer group is zero. To expose the curve input, restriction injects it into $\mathrm{Br}(k(\overline{C}))$ by the discrete-valuation residue sequence. The latter group is zero because $k(\overline{C})$ is C_1: writing the unknowns as polynomials of degree at most N turns a homogeneous equation of degree d in more than d variables into fewer coefficient equations than unknown coefficients for N large; a nonzero solution over the algebraically closed field gives a zero of the original form. Applied to a reduced norm form, this splits every central simple algebra. The same residue sequence and ℓ-cohomological dimension one of this C_1 field give $H^q(\overline{C}, \mu_\ell) = 0$ for $q > 2$. Consequently

$$H^0(\overline{C}, \mathbf{F}_\ell) = \mathbf{F}_\ell, \quad H^1(\overline{C}, \mathbf{F}_\ell) = (\mathbf{F}_\ell)^{2g}, \quad H^2(\overline{C}, \mathbf{F}_\ell) = \mathbf{F}_\ell.$$

Purity for the finite set D is the local DVR calculation: a parameter t identifies the boundary class with the valuation of an ℓth-root Kummer torsor. It gives the exact localization segment

$$0 \longrightarrow H^1(\overline{C}, \mathbf{F}_\ell) \longrightarrow H^1(C, \mathbf{F}_\ell) \xrightarrow{\partial} H^0(D, \mathbf{F}_\ell(-1))$$
$$\xrightarrow{\sum \mathrm{tr}_x} H^2(\overline{C}, \mathbf{F}_\ell) \longrightarrow H^2(C, \mathbf{F}_\ell) \longrightarrow 0.$$

The trace sends the basis vector at each $x \in D$ to the fundamental class, so when $r > 0$ it is the summation map $(\mathbf{F}_\ell)^r \to \mathbf{F}_\ell$ and is surjective. Thus a nonproper smooth curve is affine,

$$H^1(C, \mathbf{F}_\ell) \cong (\mathbf{F}_\ell)^{2g+r-1}, \qquad H^q(C, \mathbf{F}_\ell) = 0 \quad (q > 1).$$

Under an algebraically closed extension k'/k, the compactification, genus, boundary points, Kummer sequence, valuation boundary ∂, trace, and the displayed summation map all pull back to their primed counterparts. Hence the vertical maps in the localization diagram are isomorphisms. This proves assertions (1), (2), (4), (5), and (6), and assertion (7) in the proper case. For assertion (8), an open in a proper connected curve is either the whole curve or is affine with zero H^2; opens in an affine curve are affine. The restriction $H^2(C) \to H^2(W)$ is therefore either the identity or the unique map onto zero. All claimed maps are the maps in the Kummer and localization sequences, so their naturality is fixed. □

Lemma 4.6.22 (Smooth characteristic-p Artin–Schreier calculation). *Lean:* `conditional-interface: lemma: weil-pbc-smooth-curve-artin-schreier`

Uses: `def: weil-pbc-eight-curve-assertions`

Let k be algebraically closed of characteristic $p > 0$ and let C be a smooth separated finite-type scheme of dimension at most one. All applicable assertions $\mathcal{C}_r(C, \underline{\mathbf{F}_p})$ hold. For proper C, invariance under algebraically closed extension is induced by the kernel of Frobenius minus the identity on $H^1(C, \mathcal{O}_C)$.

Proof. Uses:

The Artin–Schreier sequence on the etale site is

$$0 \longrightarrow \underline{\mathbf{F}_p} \longrightarrow \mathcal{O}_C \xrightarrow{F-1} \mathcal{O}_C \longrightarrow 0.$$

Exactness on the right is checked on an etale neighbourhood by adjoining a root of $T^p - T - a$; its derivative is -1, so that neighbourhood is etale. Etale cohomology of the quasi-coherent sheaf $\mathcal{O}_C$ equals its Zariski cohomology.

If C is affine, $H^q(C, \mathcal{O}_C) = 0$ for $q > 0$. The long exact sequence gives

$$H^1(C, \mathbf{F}_p) = \mathrm{coker}\big(F - 1 : \Gamma(C, \mathcal{O}_C) \to \Gamma(C, \mathcal{O}_C)\big), \qquad H^q(C, \mathbf{F}_p) = 0 \quad (q > 1).$$

This proves assertions (1)–(3) and (8) in the affine case; no field-extension assertion for affine p-torsion is claimed.

Suppose C is proper and connected. Then $H^0(C, \mathcal{O}_C) = k$, $H^q(C, \mathcal{O}_C) = 0$ for $q > 1$, and $E = H^1(C, \mathcal{O}_C)$ has dimension g. Since $a \mapsto a^p - a$ is surjective on the algebraically closed field k, the long exact sequence reduces to

$$H^1(C, \mathbf{F}_p) = \ker(F - 1 : E \to E), \qquad H^2(C, \mathbf{F}_p) = \mathrm{coker}(F - 1 : E \to E).$$

Choose a basis of E. The additive map $F - 1$ is a morphism $\mathbf{A}^g_k \to \mathbf{A}^g_k$ whose derivative is $-\mathrm{id}$. Its kernel is therefore zero-dimensional etale and finite. The image is an open subgroup of the connected additive group $\mathbf{A}^g_k$, hence is the whole group. Thus the cokernel is zero and the kernel is a finite $\mathbf{F}_p$-vector space.

Proper coherent base change identifies $E \otimes_k k'$ with $H^1(C_{k'}, \mathcal{O}_{C_{k'}})$ and carries Frobenius to Frobenius. The finite etale kernel of $F - 1$ is a finite disjoint union of copies of $\mathrm{Spec}\, k$; after an algebraically closed extension it has the same geometric points. Hence $H^1(C, \mathbf{F}_p) \to H^1(C_{k'}, \mathbf{F}_p)$ is an isomorphism, while degree zero is $\mathbf{F}_p$ and all degrees above one vanish. This proves (1), (3), (5), (7), and (8) in the proper case, with the exact comparison map specified by the Artin–Schreier sequence. □

Lemma 4.6.23 (Normalization comparison exact sequence). *Lean:* `conditional-interface: lemma: weil-pbc-normalization-exact-sequence`

Uses: `def: weil-pbc-finite-open-comparison`

Let X be a reduced curve over an algebraically closed field, $\nu : X^\nu \to X$ its finite normalization, $j : U \hookrightarrow X$ open, and $j^\nu : U^\nu \hookrightarrow X^\nu$ its pullback. For every prime ℓ, the unit and the finite-open comparison define a monomorphism

$$\alpha : j_!\underline{\mathbf{F}_\ell} \longrightarrow \nu_* j^\nu_! \underline{\mathbf{F}_\ell}$$

whose cokernel Q is constructible and supported on the finite nonnormal locus. The sequence

$$0 \longrightarrow j_!\underline{\mathbf{F}_\ell} \xrightarrow{\alpha} \nu_* j^\nu_! \underline{\mathbf{F}_\ell} \longrightarrow Q \longrightarrow 0$$

and all its cohomology boundary maps commute with algebraically closed field extension and restriction to opens.

Proof. Uses: `lemma:weil-pbc-finite-pushforward-extension-zero`, `lemma:weil-pbc-eight-curve-devissage`
Let $v : U^\nu \to U$. Define α as

$$j_!\underline{\mathbf{F}_\ell} \xrightarrow{j_!\eta} j_!v_*\underline{\mathbf{F}_\ell} \xrightarrow{\beta_{\nu,j}} \nu_* j_!^\nu \underline{\mathbf{F}_\ell},$$

where η is the unit of $v^{-1} \dashv v_*$ and β is the comparison map already constructed. At $\bar{x} \in U$, its stalk is the diagonal

$$\mathbf{F}_\ell \longrightarrow \bigoplus_{\bar{z}\in\nu^{-1}(\bar{x})} \mathbf{F}_\ell, \qquad a \longmapsto (a)_{\bar{z}};$$

outside U, both stalks are zero. Normalization is surjective, so the diagonal is injective. Over the dense open where ν is an isomorphism it is an isomorphism. Therefore the cokernel Q is supported on the finite nonnormal locus, with stalk the quotient of the displayed finite sum by its diagonal. This proves exactness and constructibility.

After an algebraically closed extension, X^ν remains smooth and normal and is the normalization of the reduced base-changed curve. The unit stalk map remains the diagonal, while the base-change rectangle for β identifies the two pulled-back comparison maps. Thus the short exact sequence pulls back to the primed sequence. Naturality of the connecting homomorphism follows by applying compatible injective resolutions. For an open $W \subset X$, finite base change replaces ν by $\nu^{-1}W \to W$ and gives the same stalk diagonals, proving compatibility with open restriction.

Finally $H^q(X, Q) = 0$ for $q > 0$, and $H^0(X, Q)$ is the finite direct sum of its stalks; both statements persist after base change. Exact finite direct image and Leray identify $H^q(X, \nu_* j_!^\nu \mathbf{F}_\ell)$ with $H^q(X^\nu, j_!^\nu \mathbf{F}_\ell)$. Hence the displayed long exact sequence transfers vanishing, finiteness, field-extension isomorphisms, and open-restriction surjectivity from the normalized term to $j_!\mathbf{F}_\ell$: in degrees at least two the maps are isomorphisms, and in degrees zero and one the finite sequence

$$0 \to H^0(j_!\mathbf{F}_\ell) \to H^0(\nu_* j_!^\nu \mathbf{F}_\ell) \to H^0(Q) \to H^1(j_!\mathbf{F}_\ell) \to H^1(\nu_* j_!^\nu \mathbf{F}_\ell) \to 0$$

gives the comparison by a five-lemma diagram. □

Lemma 4.6.24 (Passage between separably closed fields)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-curve-separably-closed-passage`

Let $k \subseteq k'$ be separably closed fields, let C be proper of dimension at most one over k, and let F be a torsion etale sheaf. If proper-curve cohomology is invariant for algebraically closed extensions, then the specified restriction $H^q(C, F) \to H^q(C_{k'}, F|)$ is an isomorphism for every q.

Proof. Uses: `lemma:weil-pbc-universal-homeomorphism-coverings`
Choose compatible algebraic closures $\bar{k} \subseteq \bar{k}'$. Since k and k' are separably closed, $\bar{k}/k$ and $\bar{k}'/k'$ are purely inseparable. Their spectrum maps are universal homeomorphisms; the explicit site equivalence and covering compatibility give the vertical isomorphisms in

$$\begin{array}{ccc} H^q(C, F) & \longrightarrow & H^q(C_{k'}, F|) \\ \sim\downarrow & & \downarrow\sim \\ H^q(C_{\bar{k}}, F|) & \longrightarrow & H^q(C_{\bar{k}'}, F|). \end{array}$$

The lower arrow is an isomorphism by the algebraically closed case. The square commutes because each route pulls a representative cocycle through the same composite field map; identity and composition compatibility of the site equivalence identify the two pullbacks. The upper arrow is therefore an isomorphism. In particular spectra of separably closed fields, which are strictly henselian local schemes, introduce no additional stalk ambiguity. □

Lemma 4.6.25 (Normalization and smooth constant case)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-curve-normalization-constant`

For a reduced curve, extension-invariance for $j_!\underline{\mathbf{F}_\ell}$ reduces through the finite normalization to the corresponding calculation on a smooth curve, both for proper and affine opens and also when $\ell = \operatorname{char} k$.

Proof. Uses: `lemma:weil-pbc-smooth-curve-kummer-localization`, `lemma:weil-pbc-smooth-curve-artin-schreier`, `lemma:weil-pbc-normalization-exact-sequence`, `lemma:weil-pbc-curve-dense-open-reduction`
The normalization X^ν is a finite disjoint union of smooth curves and points. On a component disjoint from U^ν the extension-by-zero sheaf is zero and every assertion is immediate; on every other component U^ν is dense. The dense-open localization lemma on X^ν therefore reduces $j_!^\nu \underline{\mathbf{F}_\ell}$ to the constant sheaf on each remaining smooth component, with the five-term sequence and all comparison maps already specified there. If $\ell \neq \operatorname{char} k$, apply the Kummer and localization calculation; if $\ell = \operatorname{char} k$, apply the Artin–Schreier calculation. Thus all applicable eight assertions hold for $j_!^\nu \underline{\mathbf{F}_\ell}$.

Exact finite direct image transfers them to $\nu_* j_!^\nu \underline{\mathbf{F}_\ell}$. The normalization exact sequence then transfers them to $j_!\underline{\mathbf{F}_\ell}$, including the low-degree comparison maps and assertion (8). Properness is preserved by finite normalization; affineness is preserved because a finite morphism over an affine is affine. This completes the normalization, proper/affine localization, prime-to-characteristic, and characteristic-ℓ branches without an unnamed smooth-curve calculation. □

Proposition 4.6.26 (Proper curve invariance over separably closed fields)**.** *Uses:*

Lean: `conditional-interface:prop:weil-pbc-proper-curve-sepclosed-invariance`

Let $k \subseteq k'$ be separably closed fields, C a proper k-scheme of dimension at most one, and F a torsion etale sheaf. Then

$$H^q(C, F) \xrightarrow{\sim} H^q(C_{k'}, F|)$$

for every q.

Proof. Uses: `lemma:weil-pbc-universal-homeomorphism-coverings`, `lemma:weil-pbc-curve-constructible-primary-reduction`, `lemma:weil-pbc-curve-dense-open-reduction`, `lemma:weil-pbc-curve-locally-constant-reduction`, `lemma:weil-pbc-finite-pushforward-extension-zero`, `lemma:weil-pbc-curve-normalization-constant`, `lemma:weil-pbc-curve-separably-closed-passage`

First assume both fields algebraically closed. Apply constructible/primary reduction. The closed immersion $C_{\mathrm{red}} \to C$ is a universal homeomorphism, so the explicit site equivalence first replaces C by its reduction. Then apply dense-open localization, the prime-to-ℓ trace splitting, the finite-pushforward comparison, and the normalized smooth constant calculation in that order. Every step commutes with field extension, giving the result. For separably closed $k \subseteq k'$, apply Lemma 4.6.24 to the algebraically closed result just proved. Its commutative rectangle identifies the resulting isomorphism with the specified restriction map. This expands Tags `03SC` and `0A5E`. □

4.6.3 S5.3. Stalkwise projective-line base change

Lemma 4.6.27 (Exchange on a geometric stalk)**.** *Lean:* `conditional-interface:lemma:weil-pbc-p1-exchange-geometric-stalk`

Uses: `def:weil-pbc-derived-exchange`

For $p : \mathbf{P}^1_S \to S$, a base change $g : T \to S$, and $\bar{t} \to T$ with image $\bar{s} \to S$, the stalk of $\mathrm{Ex}_{\square,F}$ is the restriction map

$$H^q(\mathbf{P}^1_{\kappa(\bar{s})}, F|) \longrightarrow H^q(\mathbf{P}^1_{\kappa(\bar{t})}, F|).$$

Proof. Uses: `prop:weil-pbc-strict-henselization-stalk-formula`, `prop:weil-pbc-p1-strict-henselian-comparison`

Let $\bar{t} \to T$ be represented by an algebraically closed field Ω, and let $\bar{s} \to S$ be its composite. Put $S^{\mathrm{sh}} = \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{S,\bar{s}}$ and $T^{\mathrm{sh}} = \operatorname{Spec} \mathcal{O}^{\mathrm{sh}}_{T,\bar{t}}$. The map $T \to S$ induces a local map $T^{\mathrm{sh}} \to S^{\mathrm{sh}}$ and a separably closed residue-field extension $\kappa(\bar{s}) \hookrightarrow \kappa(\bar{t})$. For an injective resolution $F \to I^\bullet$, the strict-henselization construction gives the following diagram in $D^+(\mathbf{Ab})$:

$$\begin{array}{ccc}
(g^{-1}Rp_*F)_{\bar{t}} & \xrightarrow{(\mathrm{Ex}_{\square,F})_{\bar{t}}} & (Rp'_*g'^{-1}F)_{\bar{t}} \\
{\scriptstyle \Phi_S}\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \Phi_T} \quad \sim V \\
R\Gamma(\mathbf{P}^1_{S^{\mathrm{sh}}}, I|) & \xrightarrow{\lambda} & R\Gamma(\mathbf{P}^1_{T^{\mathrm{sh}}}, I|) \\
{\scriptstyle \rho_S}\downarrow{\scriptstyle \sim} & & \downarrow{\scriptstyle \rho_T} \quad \sim V \\
R\Gamma(\mathbf{P}^1_{\kappa(\bar{s})}, I_{\bar{s}}) & \xrightarrow{\kappa(\bar{s})\subset\kappa(\bar{t})} & R\Gamma(\mathbf{P}^1_{\kappa(\bar{t})}, I_{\bar{t}}).
\end{array}$$

The maps Φ_S and Φ_T are the stalk-to-strict-local maps from the strict-henselization formula. The maps ρ_S and ρ_T are the S4 restriction quasi-isomorphisms, applied after replacing $\mathbf{P}^1$ by its base change to the indicated strict local ring. The bottom arrow is induced by the residue-field map.

To check the upper square, take a representative section of I^r on an affine pointed etale neighbourhood $V \to S$. The left route first applies the unit $I^r \to g'_*g'^{-1}I^r$ and then restricts along $\mathbf{P}^1_T \times_T (V \times_S T)$. The right route first restricts to $\mathbf{P}^1_{S^{\mathrm{sh}}}$ and then along $T^{\mathrm{sh}} \to S^{\mathrm{sh}}$. Both are the same section on $\mathbf{P}^1_{T^{\mathrm{sh}}} \times_{T^{\mathrm{sh}}} V$, so the square commutes degree by degree and hence in the derived category. The lower square is the identical calculation after passing to the closed geometric fibres. Therefore the stalk of the named S0 exchange is exactly the field extension map displayed in the bottom row, not an abstract replacement. □

Lemma 4.6.28 (Conservativity of geometric stalks)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-geometric-stalk-conservative`

A morphism of etale abelian sheaves is an isomorphism if and only if it is an isomorphism at every geometric stalk.

Proof. Uses:

The kernel and cokernel have zero geometric stalks. If a sheaf has zero stalks, every local section vanishes on an etale neighbourhood of each point; quasi-compact refinement on an etale object makes it locally zero on a cover, hence zero by the sheaf axiom. Thus kernel and cokernel vanish. □

Proposition 4.6.29 (Projective-line proper base change)**.** *Uses:*

Lean: `conditional-interface:prop:weil-pbc-projective-line`

For every scheme S, the projection $\mathbf{P}^1_S \to S$ satisfies PBC.

Proof. Uses: `lemma:weil-pbc-p1-exchange-geometric-stalk`, `prop:weil-pbc-proper-curve-sepclosed-invariance`, `lemma:weil-pbc-geometric-stalk-conservative`, `lemma:weil-pbc-property-iff-acyclicity`

At every geometric $\bar{t} \to T$, the exchange map is the field-extension map of the preceding lemma. The two residue fields are separably closed and the fibre is a proper curve, so proper-curve invariance makes this map an isomorphism. Geometric stalk conservativity gives an isomorphism of every R^q sheaf. Specializing to inverse images of injective $\mathbf{F}_\ell$-module sheaves gives IA after every base change; the S0 equivalence gives PBC. □

4.7 S6. Finite-torsion proper base change

Lemma 4.7.1 (Prime-coefficient proper base change)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-prime-proper-base-change`
Every proper morphism satisfies proper base change for sheaves of $\mathbf{F}_\ell$-modules.

Proof. Uses: `prop:weil-pbc-reduction-to-projective-line,prop:weil-pbc-projective-line`
The projective-line proposition supplies the sole geometric input required by the S3 reduction. Finite morphisms, composition, and cancellation in that reduction preserve the named S0 exchange map, so the resulting isomorphism is the exchange map for the original proper morphism. □

Lemma 4.7.2 (Torsion coefficient assembly)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-torsion-assembly`
For every torsion abelian sheaf F and every proper Cartesian square, the specific map $\mathrm{Ex}_{\square,F}$ is an isomorphism.

Proof. Uses: `lemma:weil-pbc-prime-proper-base-change,lemma:weil-pbc-torsion-filtered-union,lemma:weil-pbc-direct-image-filtered-colimit,lemma:weil-pbc-prime-factor-induction,lemma:weil-pbc-five-lemma-transfer`
Write $F = \varinjlim_n F[n]$ and commute both sides of exchange with the filtered colimit. Induct on the number of prime factors of n through $0 \to F[\ell] \to F[n] \to F[n]/F[\ell] \to 0$; the quotient is killed by n/ℓ. Prime coefficients are handled by the preceding lemma, and the five-lemma transfer handles the induction step. Naturality shows at every stage that the comparison is the S0 exchange map. □

Proposition 4.7.3 (Finite-torsion proper base change)**.** *Lean:* `conditional-interface:prop:weil-proet-finite-proper-base-change`

Uses: `def:weil-pbc-derived-exchange`
Let

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be Cartesian with f proper. For every torsion sheaf F on X_{et}, the canonical exchange map

$$g^{-1}Rf_*F \xrightarrow{\sim} Rf'_*g'^{-1}F$$

is an isomorphism. These isomorphisms are natural in F, equal the identity for identity squares, and commute with horizontal and vertical pasting.

Proof. Uses: `lemma:weil-pbc-derived-exchange-coherence,lemma:weil-pbc-torsion-assembly`
The torsion assembly proves the isomorphism in every cohomological degree for the map constructed in S0. A morphism in D^+ is an isomorphism exactly when all its cohomology sheaf maps are isomorphisms. Naturality, identity, and pasting are the corresponding S0 coherence equations and survive each devissage step. This is the expanded endpoint whose source locator is Tag `095T`; that tag is not used as a proof leaf. □

> **Status.** Proposition 4.7.3 is the stable A3 finite-torsion export. It remains conditional until every OPEN in S0–S5 on which its proof depends is discharged.

4.8 S7. Bounded-below finite-torsion complexes

Definition 4.8.1 (Hypercohomology filtrations)**.** *Lean:* `conditional-interface:def:weil-pbc-hypercohomology-filtrations`

Uses: `def:weil-pbc-derived-exchange`
Let $K \in D^+(X_{\mathrm{et}})$ have torsion cohomology sheaves and choose a with $\mathcal{H}^q(K) = 0$ for $q < a$. The canonical truncation tower $\tau_{\le q}K$ defines spectral sequences

$$E_2^{p,q} = R^p f_* \mathcal{H}^q(K) \Longrightarrow \mathcal{H}^{p+q}(Rf_*K),$$

and

$$E'^{p,q}_2 = R^p f'_* g'^{-1} \mathcal{H}^q(K) \Longrightarrow \mathcal{H}^{p+q}(Rf'_* g'^{-1} K).$$

Their filtration in total degree n is the image filtration induced by $\mathcal{H}^n(Rf_*\tau_{\le q}K) \to \mathcal{H}^n(Rf_*K)$; the second filtration is induced by $\mathcal{H}^n(Rf'_*g'^{-1}\tau_{\le q}K) \to \mathcal{H}^n(Rf'_*g'^{-1}K)$.

Lemma 4.8.2 (Half-plane convergence)**.** *Lean:* `conditional-interface:lemma:weil-pbc-hypercohomology-convergence`

Uses: `def:weil-pbc-hypercohomology-filtrations`
The spectral sequences lie in $p \ge 0$, $q \ge a$. For fixed total degree n, only $a \le q \le n$ and $0 \le p \le n - a$ occur; hence the induced filtration is finite, exhaustive, separated, and converges to the displayed abutment.

Proof. Uses:
Right-derived direct image has no negative cohomological degree, giving $p \geq 0$. The lower bound gives $q \geq a$. On $p+q=n$ these inequalities give the finite interval. The truncation triangles show exhaustivity, and finiteness shows separatedness; no inverse-limit convergence issue occurs. □

Definition 4.8.3 (The two truncation exact couples)**.** Lean: `conditional-interface:def:weil-pbc-truncation-exact-couples`

Uses: `def:weil-pbc-hypercohomology-filtrations,def:weil-pbc-derived-exchange`
For every $q \geq a$, set

$$A_q = g^{-1}Rf_*\tau_{\leq q}K, \quad G_q = g^{-1}Rf_*\mathcal{H}^q(K)[-q],$$
$$B_q = Rf'_*g'^{-1}\tau_{\leq q}K, \quad H_q = Rf'_*g'^{-1}\mathcal{H}^q(K)[-q].$$

The canonical truncation triangle and its exchanged image are

$$\begin{array}{ccccccc} A_{q-1} & \xrightarrow{i_q} & A_q & \xrightarrow{j_q} & G_q & \xrightarrow{k_q} & A_{q-1}[1] \\ \downarrow{\scriptstyle e_{q-1}} & & \downarrow{\scriptstyle e_q} & & \downarrow{\scriptstyle \bar{e}_q} & & \downarrow{\scriptstyle e_{q-1}[1]} \\ B_{q-1} & \xrightarrow{i'_q} & B_q & \xrightarrow{j'_q} & H_q & \xrightarrow{k'_q} & B_{q-1}[1]. \end{array}$$

For total degree n, define

$$D_A^{q,n} = \mathcal{H}^n(A_q), \quad \mathcal{E}_A^{p,q} = \mathcal{H}^{p+q}(G_q) = g^{-1}R^pf_*\mathcal{H}^q(K),$$
$$D_B^{q,n} = \mathcal{H}^n(B_q), \quad \mathcal{E}_B^{p,q} = \mathcal{H}^{p+q}(H_q) = R^pf'_*g'^{-1}\mathcal{H}^q(K).$$

The exact-couple differential is

$$d_1 = (j_{q-1})_* \circ (k_q)_* : \mathcal{E}_A^{p,q} \longrightarrow \mathcal{E}_A^{p+2,q-1},$$

and likewise $d'_1 = (j'_{q-1})_*(k'_q)_*$. This first exact-couple page is the page conventionally denoted E_2 in the hypercohomology indexing.

Lemma 4.8.4 (Hypercover model and E_1-map)**.** Lean: `conditional-interface:lemma:weil-pbc-hypercover-exact-couple-map`

Uses: `def:weil-pbc-truncation-exact-couples`
On every split etale hypercover, the exchange maps define a morphism between the two exact couples. Its first exact-couple page is

$$\bar{e}_1^{p,q} : g^{-1}R^pf_*\mathcal{H}^q(K) \longrightarrow R^pf'_*g'^{-1}\mathcal{H}^q(K),$$

the S6 finite-torsion map, and $d'_1\bar{e}_1^{p,q} = \bar{e}_1^{p+2,q-1}d_1$ *with no residual sign.*

Proof. Uses: `def:weil-pbc-truncation-exact-couples,prop:weil-proet-finite-proper-base-change`
Evaluate the two rows of truncation triangles on an etale object and choose a split etale hypercover $U_\bullet$. A Cartan–Eilenberg representative gives on each side a double complex $C^{m,r}$, where m is the cosimplicial degree and r the complex degree. Write

$$d_{\mathrm{simp}} = \sum_{t=0}^{m+1}(-1)^t d_t^*, \qquad d_{\mathrm{tot}} = d_{\mathrm{simp}} + (-1)^m d_{\mathrm{int}}.$$

The simplicial identities give $d_{\mathrm{simp}}^2 = 0$; naturality of the face maps gives $d_{\mathrm{simp}}d_{\mathrm{int}} = d_{\mathrm{int}}d_{\mathrm{simp}}$, and the factor $(-1)^m$ makes $d_{\mathrm{tot}}^2 = 0$. The left and right totalizations use this same sign convention.

The sectionwise exchange map pulls a section on every U_m through the Cartesian square. It commutes with every face and degeneracy map and with d_{int}, hence is a chain map of the two total complexes. Filtering by canonical truncation gives exactly the objects A_q, G_q and B_q, H_q of Definition 4.8.3. Naturality with respect to $\tau_{\leq q-1}K \to \tau_{\leq q}K$ gives the first three squares of the displayed triangle.

For the last square, use the cohomological cone convention

$$\mathrm{Cone}(u)^n = Y^n \oplus X^{n+1}, \qquad d(y,x) = (d_Y y + u(x), -d_X x).$$

The connecting map is induced by $(y,x) \mapsto x$. Applying exchange before or after this projection gives $e_{q-1}[1](x)$ in both routes; the minus sign belongs to the shifted differential on both sides and does not change the map. Thus $k'_q\bar{e}_q = e_{q-1}[1]k_q$, with no extra sign.

Taking cohomology yields a map of exact couples. On the graded object $\mathcal{H}^q(K)[-q]$, degree $p+q$ cohomology is $g^{-1}R^pf_*\mathcal{H}^q(K)$ on the left and $R^pf'_*g'^{-1}\mathcal{H}^q(K)$ on the right. The induced map is the S0 exchange for that torsion sheaf and therefore the specified S6 map. Finally

$$\begin{aligned} d'_1\bar{e}_1 &= (j'_{q-1})_*(k'_q)_*\bar{e}_1 \\ &= (j'_{q-1})_*(e_{q-1}[1])_*(k_q)_* = \bar{e}_1(j_{q-1})_*(k_q)_* = \bar{e}_1 d_1, \end{aligned}$$

which proves differential compatibility. □

Lemma 4.8.5 (Finite rows and the abutment comparison)**.** Lean: `conditional-interface:lemma:weil-pbc-exact-couple-abutment`

Uses: `def:weil-pbc-hypercohomology-filtrations, def:weil-pbc-truncation-exact-couples`
In total degree n, the two exact couples have only the finite row range $a \leq q \leq n$. Their induced abutment map is the cohomology map

$$\mathcal{H}^n(g^{-1}Rf_*K) \longrightarrow \mathcal{H}^n(Rf'_*g'^{-1}K)$$

of the derived exchange morphism, with the image filtrations of Definition 4.8.1.

Proof. Uses: lemma:weil-pbc-hypercohomology-convergence
On $\mathcal{E}_1^{p,q}$ one has $p \geq 0$ and $q \geq a$. The equality $p+q=n$ therefore forces $a \leq q \leq n$. On a hypercover total complex, only these finitely many filtered quotients contribute to degree n. Consequently direct-sum and product totalization agree in that degree, the filtration is finite and separated, and passage from a finite skeleton to the full hypercover changes none of its graded terms.

The maps $e_q : A_q \to B_q$ commute with $\tau_{\leq q}K \to K$. For $q \geq n$, the omitted cohomology sheaves of K have degrees greater than n; after a nonnegative right-derived functor they cannot contribute to total degree n. Hence

$$\mathcal{H}^n(A_q) \xrightarrow{\sim} \mathcal{H}^n(g^{-1}Rf_*K), \qquad \mathcal{H}^n(B_q) \xrightarrow{\sim} \mathcal{H}^n(Rf'_*g'^{-1}K).$$

Under these identifications, $\mathcal{H}^n(e_q)$ is $\mathcal{H}^n(\mathrm{Ex}_{\square,K})$ by naturality of exchange in the truncation map. Its filtration is the image of $\mathcal{H}^n(A_q)$ and maps to the image of $\mathcal{H}^n(B_q)$, exactly the two filtrations defined earlier. Thus the exact-couple abutment map is the specified derived exchange map, rather than an unnamed filtered isomorphism. □

Lemma 4.8.6 (Morphism of hypercohomology spectral sequences). Lean: conditional-interface:lemma:weil-pbc-hypercohomology-map

Uses: def:weil-pbc-hypercohomology-filtrations, def:weil-pbc-truncation-exact-couples

The S0 exchange map induces a morphism from the inverse image of the first spectral sequence to the second, whose E_2 component is

$$g^{-1}R^pf_*\mathcal{H}^q(K) \longrightarrow R^pf'_*g'^{-1}\mathcal{H}^q(K),$$

the S6 finite-torsion proper-base-change map.

Proof. Uses: def:weil-pbc-truncation-exact-couples, lemma:weil-pbc-hypercohomology-convergence, lemma:weil-pbc-hypercover-exact-couple-map, lemma:weil-pbc-exact-couple-abutment, prop:weil-proet-finite-proper-base-change
Definition 4.8.3 displays both exact couples and every structure map. Lemma 4.8.4 constructs their morphism, identifies its first page with the S6 map, and checks the differential and connecting-map sign. Lemma 4.8.5 proves finite-row convergence and identifies the abutment map with $\mathcal{H}^n(\mathrm{Ex}_{\square,K})$. Deriving the exact couples therefore gives the asserted morphism on every page. □

Proposition 4.8.7 (Bounded-below finite derived proper base change). Uses:

Lean: conditional-interface:prop:weil-proet-finite-derived-proper-base-change

For a proper Cartesian square and $K \in D^+(X_{\mathrm{et}})$ with torsion cohomology, the exchange map

$$g^{-1}Rf_*K \xrightarrow{\sim} Rf'_*g'^{-1}K$$

is an isomorphism.

Proof. Uses: lemma:weil-pbc-hypercohomology-convergence, lemma:weil-pbc-hypercohomology-map, prop:weil-proet-finite-proper-base-change
S6 makes every E_2 component of the spectral-sequence morphism an isomorphism. Induction on the page gives isomorphisms on E_r and E_∞. In each total degree the finite filtrations have isomorphic associated graded pieces, so induction on the filtration length gives an isomorphism of abutments. A morphism in the derived category is an isomorphism when every cohomology sheaf map is an isomorphism. This expands Tag ODDE, including its exact-couple comparison. □

4.9 S8. Etale-to-pro-etale finite-level transport

Definition 4.9.1 (Etale objects inside the pro-etale site). Lean: conditional-interface:def:weil-pbc-etale-proetale-inclusion

Uses: def:weil-small-etale-topology, def:weil-small-etale-pretopology, def:weil-small-proetale-category, def:weil-small-proetale-precoverage, def:weil-small-proetale-topology

For a scheme X, define the assignment

$$\iota_X : X_{\mathrm{et}} \longrightarrow X_{\mathrm{proet}}$$

as follows. An object $p : U \to X$, etale in the A2 small etale category, is sent to the same structural morphism p. It is an object of the A2 small pro-etale category because an etale morphism is weakly etale. An X-morphism $h : U \to V$ is sent to the same scheme morphism h. Thus both forgetful functors to $\mathrm{Over}(X)$ agree on this assignment.

Lemma 4.9.2 (Identity law for the etale-to-pro-etale assignment). Lean: conditional-interface:lemma:weil-efi-site-inclusion-identity

Uses: def:weil-pbc-etale-proetale-inclusion

For every etale object $U \to X$,

$$\iota_X(1_U) = 1_{\iota_X U}.$$

Proof. Uses: def:weil-pbc-etale-proetale-inclusion
Both arrows have underlying scheme morphism 1_U, and morphisms in the selected over-category are determined by their underlying scheme maps. □

Lemma 4.9.3 (Composition law for the etale-to-pro-etale assignment). Lean: conditional-interface:lemma:weil-efi-site-inclusion-composition

Uses: def:weil-pbc-etale-proetale-inclusion

For composable etale-site arrows $U \xrightarrow{h} V \xrightarrow{k} W$,

$$\iota_X(kh) = \iota_X(k)\iota_X(h).$$

Proof. Uses: def:weil-pbc-etale-proetale-inclusion
Each side has underlying scheme morphism kh. Equality follows from the faithful forgetful functor to $\mathrm{Over}(X)$. □

Definition 4.9.4 (Functor-law certificate for the site assignment). Lean: conditional-interface:def:weil-efi-site-inclusion-functor-certificate

Uses: def:weil-pbc-etale-proetale-inclusion
Let $\mathsf{FunctorLaw}(\iota_X)$ be the conjunction of the identity equation for every etale object and the composition equation for every composable pair of etale-site arrows.

Proposition 4.9.5 (The site assignment satisfies its functor laws). Lean: conditional-interface:prop:weil-efi-site-inclusion-functor

Uses: def:weil-efi-site-inclusion-functor-certificate, def:weil-pbc-etale-proetale-inclusion
The predicate $\mathsf{FunctorLaw}(\iota_X)$ *holds. Hence the assignment of Definition 4.9.1 is a functor.*

Proof. Uses: lemma:weil-efi-site-inclusion-identity,lemma:weil-efi-site-inclusion-composition
The first lemma supplies the identity field and the second supplies the composition field of the displayed predicate. □

Definition 4.9.6 (Image of an etale sieve in the pro-etale site). Lean: conditional-interface:def:weil-efi-site-covering-sieve-image

Uses: def:weil-pbc-etale-proetale-inclusion
For an etale object $U \to X$ and a sieve $R \subset h_U$, let $\iota^{\mathrm{sieve}}_{X,!}R$ be the sieve on $\iota_X U$ generated by the arrows $\iota_X V \to \iota_X U$ with $V \to U$ in R.

Lemma 4.9.7 (Etale covering families remain pro-etale covering). Lean: conditional-interface:lemma:weil-efi-site-covering-family-image

Uses: def:weil-pbc-etale-proetale-inclusion
The image under ι_X *of an etale covering family is a pro-etale covering family.*

Proof. Uses: def:weil-pbc-etale-proetale-inclusion,def:weil-small-etale-pretopology,def:weil-small-proetale-precoverage
Every arrow in the family is etale and hence weakly etale. The underlying scheme maps are unchanged, so their joint surjectivity is unchanged. These are exactly the membership conditions in the selected pro-etale precoverage. □

Lemma 4.9.8 (Etale covering sieves remain pro-etale covering). Lean: conditional-interface:lemma:weil-efi-site-covering-sieve-image

Uses: def:weil-efi-site-covering-sieve-image
If R *is covering on* U *for the selected etale topology, then* $\iota^{\mathrm{sieve}}_{X,!}R$ *is covering for the selected pro-etale topology.*

Proof. Uses: lemma:weil-efi-site-covering-family-image,def:weil-small-etale-topology,def:weil-small-proetale-topology
The etale topology is generated by its pretopology, so R contains the sieve generated by an etale covering family. The preceding lemma sends that family to a pro-etale covering family. Its generated sieve is contained in $\iota^{\mathrm{sieve}}_{X,!}R$, and covering sieves are upward closed. □

Lemma 4.9.9 (Pullback equation for the image-sieve operation). Lean: conditional-interface:lemma:weil-efi-site-covering-sieve-pullback

Uses: def:weil-efi-site-covering-sieve-image
For an etale arrow $f : V \to U$ *and a sieve* R *on* U,

$$(\iota_X f)^*(\iota^{\mathrm{sieve}}_{X,!}R) = \iota^{\mathrm{sieve}}_{X,!}(f^*R).$$

Proof. Uses: def:weil-efi-site-covering-sieve-image,def:weil-pbc-etale-proetale-inclusion
Both sieves are generated by the base changes $V \times_U V' \to V$ as $V' \to U$ ranges through R. The selected etale and pro-etale over-categories compute these fiber products by the same scheme $V \times_U V'$, and ι_X fixes its projections. Hence their generating arrows, and therefore the generated sieves, agree. □

Definition 4.9.10 (Continuity predicate for the site inclusion). Lean: conditional-interface:def:weil-efi-site-continuity-certificate

Uses: def:weil-efi-site-covering-sieve-image
Let $\mathsf{Continuous}(\iota_X)$ mean that the image-sieve operation sends a covering sieve to a covering sieve and satisfies the displayed pullback equation.

Proposition 4.9.11 (Continuity certificate for the site inclusion). Lean: conditional-interface:prop:weil-efi-site-continuity-certificate

Uses: def:weil-efi-site-continuity-certificate
The predicate $\mathsf{Continuous}(\iota_X)$ *holds.*

Proof. Uses: lemma:weil-efi-site-covering-sieve-image,lemma:weil-efi-site-covering-sieve-pullback
The first lemma supplies covering preservation and the second supplies the pullback field. □

Lemma 4.9.12 (The etale/pro-etale assignment is a functor)**.** Lean: `conditional-interface:lemma:weil-pbc-etale-proetale-functor`

Uses: `def:weil-pbc-etale-proetale-inclusion`

The assignment ι_X preserves identity morphisms and composition.

Proof. Uses:

For an etale object $U \to X$, both $\iota_X(1_U)$ and $1_{\iota_X(U)}$ have underlying scheme morphism 1_U. For composable X-morphisms $U \xrightarrow{h} V \xrightarrow{k} W$, both $\iota_X(kh)$ and $\iota_X(k)\iota_X(h)$ have underlying scheme morphism kh. Morphisms in either A2 over-category are determined by this underlying map, which proves both equations. □

Lemma 4.9.13 (Continuity of the etale/pro-etale inclusion)**.** Lean: `conditional-interface:lemma:weil-pbc-etale-proetale-continuous`

Uses: `def:weil-pbc-etale-proetale-inclusion`

The functor $\iota_X : X_{\mathrm{et}} \to X_{\mathrm{proet}}$ is continuous for the final A2 topologies.

Proof. Uses: `lemma:weil-pbc-etale-proetale-functor`

Let $\{U_a \to U\}_{a\in A}$ be a covering family in the A2 small etale pretopology. Each $U_a \to U$ is etale, hence weakly etale, and the underlying family of scheme maps is unchanged by ι_X. Its images are jointly surjective, so the same family belongs to the pulled-back pro-etale precoverage. Consequently the sieve that it generates is covering for the A2 pro-etale topology.

More generally, if $R \subset h_U$ is an etale covering sieve, send it to the sieve on $\iota_X U$ generated by the arrows $\iota_X V \to \iota_X U$ for $V \to U$ in R. The preceding argument shows that this image sieve is pro-etale covering. Formation of the image sieve commutes with pullback, because the underlying scheme fiber products are unchanged. Since both A2 topologies are generated by the displayed precoverages, this is the continuity criterion. □

Lemma 4.9.14 (Finite-limit preservation of the inclusion)**.** Lean: `conditional-interface:lemma:weil-pbc-etale-proetale-left-exact`

Uses: `def:weil-pbc-etale-proetale-inclusion`

The functor ι_X preserves the terminal object and fiber products, hence is left exact.

Proof. Uses: `lemma:weil-pbc-etale-proetale-functor`

The terminal object in both A2 sites is $1_X : X \to X$, and ι_X fixes it. For a diagram $U \to W \leftarrow V$ of etale X-objects, its fiber product in both over-categories has underlying scheme $U \times_W V$ with the same two projections and structural map to X. Etale morphisms are stable under base change and composition, so this is again an etale object. Thus the canonical map

$$\iota_X(U \times_W V) \longrightarrow \iota_X U \times_{\iota_X W} \iota_X V$$

is the identity on $U \times_W V$, and is an isomorphism. Terminal objects and fiber products generate finite limits. □

Definition 4.9.15 (Etale/pro-etale comparison morphism)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-epsilon-comparison`

The continuous left-exact functor ι_X induces the geometric morphism

$$\epsilon_X : X_{\mathrm{proet}} \longrightarrow X_{\mathrm{et}}.$$

For every commutative coefficient ring A, its inverse- and direct-image functors have the typed source and target

$$\epsilon_X^{-1} : \mathrm{Sh}_A(X_{\mathrm{et}}) \longrightarrow \mathrm{Sh}_A(X_{\mathrm{proet}}), \qquad \epsilon_{X,*} : \mathrm{Sh}_A(X_{\mathrm{proet}}) \longrightarrow \mathrm{Sh}_A(X_{\mathrm{et}}).$$

For $G \in \mathrm{Sh}_A(X_{\mathrm{proet}})$ one has $(\epsilon_{X,*}G)(U) = G(\iota_X U)$. For $F \in \mathrm{Sh}_A(X_{\mathrm{et}})$, the inverse image is the pro-etale sheafification of the left Kan extension along ι_X^{op}; before sheafification its value at $W \to X$ is

$$\operatorname*{colim}_{(W \to \iota_X U)} F(U).$$

Definition 4.9.16 (Etale factorization index for a pro-etale object)**.** Lean: `conditional-interface:def:weil-pbc-epsilon-factorization-index-v1`

Uses: `def:weil-pbc-epsilon-comparison`

For a pro-etale object $W \to X$, let I_W be the category whose objects are factorizations $W \xrightarrow{a} \iota_X U \to X$ with $U \to X$ etale. A morphism from (a, U) to (a', U') is an etale-site arrow $h : U' \to U$ such that $a = \iota_X(h)a'$. The transition map on a sheaf F is restriction $F(U) \to F(U')$.

Lemma 4.9.17 (The etale factorization index is filtered)**.** Lean: `conditional-interface:lemma:weil-pbc-epsilon-factorization-index-filtered-v1`

Uses: `def:weil-pbc-epsilon-factorization-index-v1`

For every W, the category I_W is nonempty and filtered.

Proof. Uses: `def:weil-pbc-epsilon-factorization-index-v1,lemma:weil-pbc-etale-proetale-left-exact`

The structural map $W \to X = \iota_X X$ gives an object. Two factorizations through U_1 and U_2 are dominated by the factorization through $U_1 \times_X U_2$, since ι_X preserves this fibre product. For two parallel arrows $h_0, h_1 : U' \rightrightarrows U$ that agree after precomposition with $W \to \iota_X U'$, their equalizer is the pullback of the diagonal of the etale morphism $U \to X$. This diagonal is an open immersion, so the equalizer is an etale open subobject of U' through which W factors. The resulting object receives a morphism from (a', U') and equalizes the two arrows in I_W. These are the three filteredness axioms. □

Lemma 4.9.18 (Construction of the comparison geometric morphism). Lean: `conditional-interface:lemma:weil-pbc-epsilon-geometric-morphism-construction-v1`

Uses: `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-epsilon-factorization-index-v1`

Restriction along ι_X^{op}, pro-etale sheafification of left Kan extension, and the displayed evaluation maps define an adjoint pair $\epsilon_X^{-1} \dashv \epsilon_{X,*}$. The adjunction bijection is natural in both variables, and ϵ_X^{-1} preserves finite limits.

Proof. Uses: `lemma:weil-pbc-etale-proetale-continuous`, `lemma:weil-pbc-etale-proetale-left-exact`, `lemma:weil-pbc-epsilon-factorization-index-filtered-v1`, `def:weil-pbc-epsilon-comparison`

Continuity makes restriction of a pro-etale sheaf an etale sheaf. Presheaf left Kan extension is left adjoint to presheaf restriction, and sheafification is left adjoint to inclusion of sheaves. Composing these two adjunctions gives

$$\mathrm{Hom}(\epsilon_X^{-1}F, G) \cong \mathrm{Hom}(F, \epsilon_{X,*}G).$$

The two universal properties make this bijection natural in F and G and make its identity and composition equations automatic.

Before sheafification, evaluation at W is $\varinjlim_{I_W} F(U)$. The preceding lemma makes I_W filtered, so this colimit of A-modules commutes with finite limits. Sheafification of A-module presheaves is left exact, and the left-exactness of ι_X identifies the transition diagrams for terminal objects and fibre products. Thus ϵ_X^{-1} preserves finite limits. All maps used are induced by the same restriction maps, so they are compatible with restriction in W and with morphisms of coefficient sheaves. □

Definition 4.9.19 (Forward Hom map for the comparison). Lean: `conditional-interface:def:weil-efi-site-adjunction-hom-forward`

Uses: `def:weil-pbc-epsilon-comparison`

For A-module sheaves F on X_{et} and G on X_{proet}, send $\alpha : \epsilon_X^{-1}F \to G$ to the map $\Phi(\alpha) : F \to \epsilon_{X,*}G$ whose component at U is

$$s \longmapsto \alpha_{\iota_X U}\left[(1_{\iota_X U}, s)\right].$$

Definition 4.9.20 (Representative formula for the reverse Hom map). Lean: `conditional-interface:def:weil-efi-site-adjunction-hom-backward-presheaf`

Uses: `def:weil-pbc-epsilon-comparison`

For $\beta : F \to \epsilon_{X,*}G$, assign to a left-Kan-extension representative $(W \xrightarrow{a} \iota_X U, s)$ the section

$$G(a)(\beta_U(s)) \in G(W).$$

Lemma 4.9.21 (Descent of the reverse Hom formula). Lean: `conditional-interface:lemma:weil-efi-site-adjunction-hom-backward-descent`

Uses: `def:weil-efi-site-adjunction-hom-backward-presheaf`

The representative formula factors uniquely through pro-etale sheafification.

Proof. Uses: `def:weil-efi-site-adjunction-hom-backward-presheaf`, `def:weil-pbc-epsilon-factorization-index-v1`

For a transition $h : U' \to U$ with $a = \iota_X(h)a'$, naturality of β and functoriality of G identify the values of (a, s) and $(a', F(h)s)$. Restriction along $b : W' \to W$ changes each formula to $G(ab)(\beta_U(s))$. Local equality descends because G is a sheaf. The universal property of sheafification gives the unique factorization. □

Definition 4.9.22 (Reverse Hom map for the comparison). Lean: `conditional-interface:def:weil-efi-site-adjunction-hom-backward`

Uses: `def:weil-efi-site-adjunction-hom-backward-presheaf`

Let $\Psi(\beta) : \epsilon_X^{-1}F \to G$ be the sheaf map induced from the representative formula by the universal property of sheafification.

Lemma 4.9.23 (Left inverse equation for the comparison Hom maps). Lean: `conditional-interface:lemma:weil-efi-site-adjunction-hom-left-inverse`

Uses: `def:weil-efi-site-adjunction-hom-forward`, `def:weil-efi-site-adjunction-hom-backward`

For every $\alpha : \epsilon_X^{-1}F \to G$,

$$\Psi(\Phi(\alpha)) = \alpha.$$

Proof. Uses: `def:weil-efi-site-adjunction-hom-forward`, `def:weil-efi-site-adjunction-hom-backward`

On a representative $(W \xrightarrow{a} \iota_X U, s)$, the left side is $G(a)(\alpha_{\iota_X U}[(1, s)])$. Naturality of α identifies this with $\alpha_W[(a, s)]$. Representatives are locally exhaustive after sheafification, so the sheaf maps agree. □

Lemma 4.9.24 (Right inverse equation for the comparison Hom maps). Lean: `conditional-interface:lemma:weil-efi-site-adjunction-hom-right-inverse`

Uses: `def:weil-efi-site-adjunction-hom-forward`, `def:weil-efi-site-adjunction-hom-backward`

For every $\beta : F \to \epsilon_{X,*}G$,

$$\Phi(\Psi(\beta)) = \beta.$$

Proof. Uses: `def:weil-efi-site-adjunction-hom-forward`, `def:weil-efi-site-adjunction-hom-backward`

At an etale object U, the left side sends s to the evaluation of $(1_{\iota_X U}, s)$, namely $G(1_{\iota_X U})(\beta_U(s)) = \beta_U(s)$. □

Lemma 4.9.25 (Source naturality of the comparison Hom correspondence). Lean: `conditional-interface:lemma:weil-efi-site-adjunction-hom-source-naturality`

Uses: `def:weil-efi-site-adjunction-hom-forward`

For $u : F' \to F$ and $\alpha : \epsilon_X^{-1} F \to G$,

$$\Phi(\alpha\, \epsilon_X^{-1} u) = \Phi(\alpha) u.$$

Proof. Uses: `def:weil-efi-site-adjunction-hom-forward`

At U, each side sends $s \in F'(U)$ to $\alpha_{\iota_X U}[(1_{\iota_X U}, u_U(s))]$. □

Lemma 4.9.26 (Target naturality of the comparison Hom correspondence). Lean: `conditional-interface:lemma:weil-efi-site-adjunction-hom-target-naturality`

Uses: `def:weil-efi-site-adjunction-hom-forward`

For $v : G \to G'$ and $\alpha : \epsilon_X^{-1} F \to G$,

$$\Phi(v\alpha) = \epsilon_{X,*}(v)\Phi(\alpha).$$

Proof. Uses: `def:weil-efi-site-adjunction-hom-forward`

At U, each side sends $s \in F(U)$ to $v_{\iota_X U}(\alpha_{\iota_X U}[(1_{\iota_X U}, s)])$. □

Definition 4.9.27 (Adjunction-certificate predicate for the comparison functors). Lean: `conditional-interface:def:weil-efi-site-sheaf-adjunction-certificate`

Uses: `def:weil-efi-site-adjunction-hom-forward`, `def:weil-efi-site-adjunction-hom-backward`

Let $\mathsf{AdjunctionCert}(\Phi, \Psi)$ be the conjunction of the left inverse equation, the right inverse equation, source naturality, and target naturality for the displayed Hom-map pair.

Proposition 4.9.28 (Adjunction certificate for the comparison functors). Lean: `conditional-interface:prop:weil-efi-site-sheaf-adjunction-certificate`

Uses: `def:weil-efi-site-sheaf-adjunction-certificate`

The predicate $\mathsf{AdjunctionCert}(\Phi, \Psi)$ holds.

Proof. Uses: `lemma:weil-efi-site-adjunction-hom-left-inverse`, `lemma:weil-efi-site-adjunction-hom-right-inverse`, `lemma:weil-efi-site-adjunction-hom-source-naturality`, `lemma:weil-efi-site-adjunction-hom-target-naturality`

The four cited lemmas supply the four fields in their displayed order. □

Lemma 4.9.29 (Finite-limit preservation by comparison inverse image). Lean: `conditional-interface:lemma:weil-efi-site-inverse-image-finite-limits`

Uses: `def:weil-pbc-epsilon-comparison`

The functor ϵ_X^{-1} preserves finite limits of A-module sheaves.

Proof. Uses: `def:weil-pbc-epsilon-factorization-index-v1`, `lemma:weil-pbc-epsilon-factorization-index-filtered-v1`, `prop:weil-efi-site-continuity-certificate`

Before sheafification, evaluation at W is the filtered colimit $\varinjlim_{I_W} F(U)$. Filtered colimits of A-modules commute with finite limits. Sheafification of A-module presheaves is left exact, and the pullback equation in the continuity certificate identifies the transition diagrams. The resulting sheaf functor therefore preserves finite limits. □

Definition 4.9.30 (Geometric-morphism certificate predicate). Lean: `conditional-interface:def:weil-efi-site-geometric-morphism-certificate`

Uses: `def:weil-pbc-epsilon-comparison`, `def:weil-efi-site-sheaf-adjunction-certificate`

Let $\mathsf{GeometricMorphismCert}(\epsilon_X^{-1}, \epsilon_{X,*})$ mean that the Hom correspondence satisfies the adjunction-certificate predicate and that ϵ_X^{-1} preserves finite limits.

Proposition 4.9.31 (Geometric-morphism package for the comparison). Lean: `conditional-interface:prop:weil-efi-site-geometric-morphism-package`

Uses: `def:weil-efi-site-geometric-morphism-certificate`

The predicate $\mathsf{GeometricMorphismCert}(\epsilon_X^{-1}, \epsilon_{X,*})$ holds.

Proof. Uses: `prop:weil-efi-site-sheaf-adjunction-certificate`, `lemma:weil-efi-site-inverse-image-finite-limits`

The proposition supplies the adjunction field and the lemma supplies the finite-limit field. □

Lemma 4.9.32 (Exactness of the packaged comparison inverse image). Lean: `conditional-interface:lemma:weil-efi-site-inverse-image-exact`

Uses: `def:weil-pbc-epsilon-comparison`

The functor ϵ_X^{-1} is exact on A-module sheaves.

Proof. Uses: `prop:weil-efi-site-sheaf-adjunction-certificate`, `lemma:weil-efi-site-inverse-image-finite-limits`

Finite-limit preservation gives preservation of kernels. The adjunction certificate makes ϵ_X^{-1} a left adjoint, so it preserves cokernels. Kernel and cokernel preservation gives exactness in the abelian category of A-module sheaves. □

Lemma 4.9.33 (Well-definedness certificate for def:weil-pbc-epsilon-comparison)**.** *Lean:* `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-epsilon-comparison`

Uses: `def:weil-pbc-epsilon-comparison`

The construction in Definition 4.9.15 is a typed raw datum. Its theorem-valued naturality, comparison, descent, or coherence fields are isolated in this property lemma.

Proof. Uses: `lemma:weil-pbc-etale-proetale-continuous,lemma:weil-pbc-etale-proetale-left-exact,lemma:weil-pbc-epsilon-factorization-index-filtered-v1,lemma:weil-pbc-epsilon-geometric-morphism-construction-v1`

The inclusion functor is continuous and left exact. The factorization-index lemma makes the displayed Kan-extension formula a filtered colimit, and the construction lemma supplies its sheafification, the adjunction, naturality, and finite-limit preservation. The formulas for direct image and inverse image are therefore independent of representatives and restrictions and define the stated geometric morphism with all comparison fields. □

Lemma 4.9.34 (The comparison functors form a geometric morphism)**.** *Lean:* `conditional-interface:lemma:weil-pbc-epsilon-geometric-morphism`

Uses: `def:weil-pbc-epsilon-comparison`

For A-module sheaves F on X_{et} and G on X_{proet} there is a natural bijection

$$\mathrm{Hom}_{X_{\mathrm{proet}}}(\epsilon_X^{-1}F, G) \cong \mathrm{Hom}_{X_{\mathrm{et}}}(F, \epsilon_{X,*}G),$$

and ϵ_X^{-1} preserves finite limits.

Proof. Uses: `lemma:weil-pbc-etale-proetale-continuous,lemma:weil-pbc-etale-proetale-left-exact`

Restriction along ι_X^{op} sends pro-etale sheaves to etale sheaves by Lemma 4.9.13. The left-Kan extension adjunction, followed by the sheafification adjunction, gives the displayed bijection.

For a pro-etale object W, let I_W be the category of factorizations $W \to \iota_X U$ with U etale over X. It is filtered: the map to X gives an object, two factorizations are dominated by their fiber product over X, and two parallel arrows are equalized by their scheme-theoretic equalizer, an open etale subobject through which W factors. Filtered colimits of A-modules commute with finite limits. The pointwise Kan extension therefore preserves finite limits, and A-module sheafification does as well. This proves left exactness of ϵ_X^{-1} and hence the geometric-morphism assertion. □

Lemma 4.9.35 (Exactness of comparison inverse image)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-epsilon-inverse-image-exact`

For every commutative ring A, the functor $\epsilon_X^{-1} : \mathrm{Sh}_A(X_{\mathrm{et}}) \to \mathrm{Sh}_A(X_{\mathrm{proet}})$ is exact.

Proof. Uses: `lemma:weil-pbc-epsilon-geometric-morphism`

It preserves kernels by the finite-limit assertion in Lemma 4.9.34. It preserves cokernels because it is a left adjoint. Applying these two facts to a short exact sequence

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

shows that its inverse-image sequence is exact at all three terms. □

Definition 4.9.36 (Unit and counit of the comparison)**.** Lean: `conditional-interface:def:weil-pbc-epsilon-unit-counit`

Uses: `def:weil-pbc-epsilon-comparison`

Write the unit and counit of the preceding adjunction as

$$\eta^X : \mathrm{id}_{\mathrm{Sh}_A(X_{\mathrm{et}})} \longrightarrow \epsilon_{X,*}\epsilon_X^{-1}, \qquad \varepsilon^X : \epsilon_X^{-1}\epsilon_{X,*} \longrightarrow \mathrm{id}_{\mathrm{Sh}_A(X_{\mathrm{proet}})}.$$

For $F \in \mathrm{Sh}_A(X_{\mathrm{et}})$, the component $\eta^X_{F,U}$ sends $s \in F(U)$ to the class represented by $(1_{\iota_X U} : \iota_X U \to \iota_X U, s)$. For $G \in \mathrm{Sh}_A(X_{\mathrm{proet}})$, the component $\varepsilon^X_{G,W}$ sends a representative $(W \xrightarrow{a} \iota_X U, s)$ to the restriction $G(a)(s) \in G(W)$. This map on the Kan-extension presheaf factors uniquely through pro-etale sheafification.

Lemma 4.9.37 (Representative descent for the comparison counit)**.** *Lean:* `conditional-interface:lemma:weil-pbc-epsilon-counit-representative-descent-v1`

Uses: `def:weil-pbc-epsilon-unit-counit`

The rule $(W \xrightarrow{a} \iota_X U, s) \mapsto G(a)(s)$ respects the left-Kan-extension relations, restriction in W, and local equality, hence factors uniquely through pro-etale sheafification.

Proof. Uses: `def:weil-pbc-epsilon-unit-counit,def:weil-pbc-epsilon-factorization-index-v1,lemma:weil-pbc-epsilon-geometric-morphism-construction-v1`

If $h : U' \to U$ is a transition and $a = \iota_X(h)a'$, then functoriality of G gives

$$G(a')G(\iota_X h)(s) = G(a)(s),$$

so the rule is constant on the colimit relation. For $b : W' \to W$, both restriction before evaluation and evaluation before restriction send (a, s) to $G(ab)(s)$. If two representatives agree after a pro-etale cover, their images agree on that cover; the sheaf equalizer for G makes the images equal. Thus the presheaf map is local and has a unique factor through sheafification. This factor is precisely the counit attached to the adjunction constructed above. □

Lemma 4.9.38 (Naturality equation for the comparison unit). *Lean:* `conditional-interface: lemma: weil-efi-site-unit-naturality`

Uses: `def: weil-pbc-epsilon-unit-counit`

For $\alpha : F \to F'$ on X_{et},

$$(\epsilon_{X,*}\epsilon_X^{-1}\alpha)\eta_F^X = \eta_{F'}^X\alpha.$$

Proof. Uses: `def:weil-pbc-epsilon-unit-counit`

At an etale object U, each route sends $s \in F(U)$ to $(1_{\iota_X U}, \alpha_U(s))$. □

Lemma 4.9.39 (Naturality equation for the comparison counit). *Lean:* `conditional-interface: lemma: weil-efi-site-counit-naturality`

Uses: `def: weil-pbc-epsilon-unit-counit`

For $\beta : G \to G'$ on X_{proet},

$$\beta\varepsilon_G^X = \varepsilon_{G'}^X(\epsilon_X^{-1}\epsilon_{X,*}\beta).$$

Proof. Uses: `def:weil-pbc-epsilon-unit-counit,lemma:weil-pbc-epsilon-counit-representative-descent-v1`

On a representative $(W \xrightarrow{a} \iota_X U, s)$, each route gives $\beta_W(G(a)(s)) = G'(a)(\beta_{\iota_X U}(s))$. The representative-descent lemma carries this equality through sheafification. □

Lemma 4.9.40 (Restriction equation for the comparison unit). *Lean:* `conditional-interface: lemma: weil-efi-site-unit-restriction`

Uses: `def: weil-pbc-epsilon-unit-counit`

For an etale arrow $b : U' \to U$, restriction carries $\eta_{F,U}^X(s)$ to $\eta_{F,U'}^X(F(b)s)$.

Proof. Uses: `def:weil-pbc-epsilon-unit-counit`

Restriction carries $(1_{\iota_X U}, s)$ to $(\iota_X b, s)$. The left-Kan-extension relation identifies this representative with $(1_{\iota_X U'}, F(b)s)$. □

Lemma 4.9.41 (Restriction equation for the comparison counit). *Lean:* `conditional-interface: lemma: weil-efi-site-counit-restriction`

Uses: `def: weil-pbc-epsilon-unit-counit`

For a pro-etale arrow $b : W' \to W$, restriction carries the counit value of $(W \xrightarrow{a} \iota_X U, s)$ to the counit value of $(W' \xrightarrow{ab} \iota_X U, s)$.

Proof. Uses: `def:weil-pbc-epsilon-unit-counit,lemma:weil-pbc-epsilon-counit-representative-descent-v1`

The first route gives $G(b)G(a)(s)$ and the second gives $G(ab)(s)$; these are equal by functoriality of G and descend through sheafification. □

Definition 4.9.42 (Naturality predicate for the comparison unit and counit). Lean: `conditional-interface: def:weil-efi-site-unit-counit-certificate`

Uses: `def:weil-pbc-epsilon-unit-counit`

Let NaturalUnitCounit(η^X, ε^X) be the conjunction of unit naturality, counit naturality, the unit restriction equation, and the counit restriction equation.

Proposition 4.9.43 (Natural unit-counit package for the comparison). *Lean:* `conditional-interface: prop: weil-efi-site-unit-counit-package`

Uses: `def: weil-efi-site-unit-counit-certificate`

The predicate NaturalUnitCounit(η^X, ε^X) *holds.*

Proof. Uses: `lemma:weil-pbc-epsilon-counit-representative-descent-v1,lemma:weil-efi-site-unit-naturality,lemma:weil-efi-site-counit-naturality,lemma:weil-efi-site-unit-restriction,lemma:weil-efi-site-counit-restriction`

Representative descent types the counit, and the four atomic equation lemmas supply the fields of the displayed predicate. □

Lemma 4.9.44 (Left triangle equation for the comparison adjunction). *Lean:* `conditional-interface: lemma: weil-efi-site-left-triangle`

Uses: `def: weil-pbc-epsilon-unit-counit`

For every etale A-module sheaf F,

$$\varepsilon^X_{\epsilon_X^{-1}F}\epsilon_X^{-1}(\eta_F^X) = 1_{\epsilon_X^{-1}F}.$$

Proof. Uses: `def:weil-pbc-epsilon-unit-counit`

A local section is represented by $(W \xrightarrow{a} \iota_X U, s)$. The composite replaces s by its identity representative and evaluates along a, returning the class (a, s). Local representatives detect sheaf maps. □

Lemma 4.9.45 (Right triangle equation for the comparison adjunction). Lean: `conditional-interface:lemma:weil-efi-site-right-triangle`

Uses: `def:weil-pbc-epsilon-unit-counit`

For every pro-etale A-module sheaf G,

$$\epsilon_{X,*}(\varepsilon^X_G)\,\eta^X_{\epsilon_{X,*}G} = 1_{\epsilon_{X,*}G}.$$

Proof. Uses: `def:weil-pbc-epsilon-unit-counit`

At an etale object U, a section $s \in G(\iota_X U)$ first becomes $(1_{\iota_X U}, s)$ and is then evaluated along the identity, returning s. □

Definition 4.9.46 (Triangle-certificate predicate for the comparison adjunction). Lean: `conditional-interface:def:weil-efi-site-triangle-certificate`

Uses: `def:weil-pbc-epsilon-unit-counit`

Let $\mathsf{TriangleCert}(\eta^X, \varepsilon^X)$ be the conjunction of the left-triangle equation and the right-triangle equation.

Proposition 4.9.47 (Triangle certificate for the comparison adjunction). Lean: `conditional-interface:prop:weil-efi-site-triangle-certificate`

Uses: `def:weil-efi-site-triangle-certificate`

The predicate $\mathsf{TriangleCert}(\eta^X, \varepsilon^X)$ holds.

Proof. Uses: `lemma:weil-efi-site-left-triangle,lemma:weil-efi-site-right-triangle`

The cited lemmas supply the left and right fields, respectively. □

Lemma 4.9.48 (Naturality and triangle equations for the comparison adjunction). Lean: `conditional-interface:lemma:weil-pbc-epsilon-adjunction-equations-v1`

Uses: `def:weil-pbc-epsilon-unit-counit`

The displayed unit and counit are natural in coefficient sheaves and satisfy the two triangle identities. Their components commute with restriction on both sites.

Proof. Uses: `def:weil-pbc-epsilon-unit-counit,lemma:weil-pbc-epsilon-counit-representative-descent-v1,lemma:weil-pbc-epsilon-geometric-morphism-construction-v1`

For $\alpha : F \to F'$, both routes around the unit naturality square send $s \in F(U)$ to $(1_{\iota_X U}, \alpha_U(s))$. For $\beta : G \to G'$, both routes around the counit square send (a, s) to $\beta_W(G(a)(s)) = G'(a)(\beta_{\iota_X U}(s))$. These equalities commute with restriction and descend to sheaves by the preceding lemma.

A local section of $\epsilon_X^{-1}F$ is represented by (a, s). The first triangle composite replaces s by its identity representative and evaluates along a, returning (a, s). A section $s \in (\epsilon_{X,*}G)(U) = G(\iota_X U)$ is sent by the second composite to $(1_{\iota_X U}, s)$ and then evaluated along the identity, returning s. Local representatives detect sheaf maps, so both composites are identities. □

Lemma 4.9.49 (Well-definedness certificate for def:weil-pbc-epsilon-unit-counit). Lean: `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-epsilon-unit-counit`

Uses: `def:weil-pbc-epsilon-unit-counit`

The construction in Definition 4.9.36 is a typed raw datum. Its theorem-valued naturality, comparison, descent, or coherence fields are isolated in this property lemma.

Proof. Uses: `lemma:weil-pbc-epsilon-geometric-morphism-construction-v1,lemma:weil-pbc-epsilon-counit-representative-descent-v1,lemma:weil-pbc-epsilon-adjunction-equations-v1`

The geometric-morphism construction supplies the adjunction. Representative descent makes the displayed counit formula a sheaf morphism, while the unit formula is the identity-factorization component of the same adjunction. The final lemma proves naturality, restriction compatibility, and both triangle identities, so every theorem-valued field of the raw unit–counit datum is well-defined. □

Lemma 4.9.50 (Naturality of the comparison unit and counit). Lean: `conditional-interface:lemma:weil-pbc-epsilon-unit-counit-naturality`

Uses: `def:weil-pbc-epsilon-unit-counit`

For $\alpha : F \to F'$ on X_{et} and $\beta : G \to G'$ on X_{proet}, one has

$$\begin{aligned}(\epsilon_{X,*}\epsilon_X^{-1}\alpha)\eta^X_F &= \eta^X_{F'}\alpha,\\ \beta\varepsilon^X_G &= \varepsilon^X_{G'}(\epsilon_X^{-1}\epsilon_{X,*}\beta).\end{aligned}$$

Proof. Uses: `def:weil-pbc-epsilon-comparison,def:weil-pbc-epsilon-unit-counit`

At an etale object U, both sides of the first equation send $s \in F(U)$ to the class $(1_{\iota_X U}, \alpha_U(s))$. At a pro-etale object W, apply the second equation to a representative $(W \xrightarrow{a} \iota_X U, s)$. Both sides give $\beta_W(G(a)(s)) = G'(a)(\beta_{\iota_X U}(s))$ by naturality of β. These equations survive sheafification and prove the asserted naturality. □

Lemma 4.9.51 (Triangle identities for the comparison adjunction). Lean: `conditional-interface:lemma:weil-pbc-epsilon-triangle-identities`

Uses: `def:weil-pbc-epsilon-unit-counit`

The unit and counit satisfy

$$\begin{aligned}\varepsilon^X_{\epsilon_X^{-1}F}\,\epsilon_X^{-1}(\eta^X_F) &= 1_{\epsilon_X^{-1}F},\\ \epsilon_{X,*}(\varepsilon^X_G)\,\eta^X_{\epsilon_{X,*}G} &= 1_{\epsilon_{X,*}G}.\end{aligned}$$

Proof. Uses: `lemma:weil-pbc-epsilon-unit-counit-naturality`

A section of $\epsilon_X^{-1}F$ is locally represented by $(W \xrightarrow{a} \iota_X U, s)$. The first composite replaces s by its identity representative at $\iota_X U$ and then restricts it along a, which returns the original class. For $s \in (\epsilon_{X,*}G)(U) = G(\iota_X U)$, the second composite forms $(1_{\iota_X U}, s)$ and evaluates it along the identity, returning s. Local representatives determine maps of sheaves, so both displayed composites are identities. □

Lemma 4.9.52 (Naturality of the comparison in the scheme). Lean: `conditional-interface:lemma:weil-pbc-epsilon-scheme-naturality`

Uses: `def:weil-pbc-etale-proetale-inclusion`, `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-epsilon-unit-counit`

For a scheme morphism $u : X \to Y$, base change on the two sites gives a canonical natural isomorphism

$$u^*_{\mathrm{proet}}\iota_Y \xrightarrow{\sim} \iota_X u^*_{\mathrm{et}}.$$

It is compatible with identities and composition in u, and the induced square of geometric morphisms carries η^Y, ε^Y to η^X, ε^X.

Proof. Uses: `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-epsilon-unit-counit`

On an etale object $V \to Y$, both composites have underlying X-scheme $X \times_Y V \to X$; on a morphism they both use its base change. The identity map of this fiber product gives the component of the natural isomorphism. The associativity and unit isomorphisms for fiber products give its composition and identity coherence. Under this identification, the unit is still represented by an identity arrow and the counit is still restriction along $W \to \iota_X U$. The formulas in Definition 4.9.36 therefore give the two compatibility squares. □

Definition 4.9.53 (Witness for coming from the etale site). Lean: `conditional-interface:def:weil-pbc-comes-from-etale`

Uses: `def:weil-pbc-epsilon-comparison`

For a coefficient ring A and $K \in D^+(X_{\mathrm{proet}}, A)$, a witness that K comes from the etale site is a pair

$$(L, \theta), \qquad L \in D^+(X_{\mathrm{et}}, A), \quad \theta : \epsilon_X^{-1}L \xrightarrow{\sim} K,$$

including the two inverse equations for θ.

Lemma 4.9.54 (Comparison inverse image commutes with pullback). Lean: `conditional-interface:lemma:weil-pbc-epsilon-pullback`

Uses: `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-comes-from-etale`

For $u : X \to Y$ and $L \in D^+(Y_{\mathrm{et}}, A)$ there is a natural isomorphism

$$Lu^*\epsilon_Y^{-1}L \xrightarrow{\sim} \epsilon_X^{-1}Lu^*L.$$

It is compatible with identities, composition, and both horizontal maps in a Cartesian square.

Proof. Uses: `lemma:weil-pbc-epsilon-scheme-naturality`, `lemma:weil-pbc-epsilon-inverse-image-exact`

Lemma 4.9.52 identifies the two composite site functors. Their inverse images are therefore sheafifications of the same presheaf left Kan extension, and the identity on that presheaf induces the displayed isomorphism. The identity and composition coherence proved there gives the corresponding equations and the two horizontal maps in a Cartesian square. Exactness of ϵ^{-1} and K-flat replacement extend these equations to derived pullback. □

Lemma 4.9.55 (The comparison unit is an isomorphism). Lean: `conditional-interface:lemma:weil-pbc-epsilon-unit-isomorphism`

Uses: `def:weil-pbc-epsilon-unit-counit`, `def:weil-pbc-epsilon-comparison`

For every A-module sheaf F on X_{et}, the unit

$$\eta_F^X : F \longrightarrow \epsilon_{X,*}\epsilon_X^{-1}F$$

is an isomorphism.

Proof. Uses: `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-epsilon-unit-counit`

Evaluate at an etale object $U \to X$. Restriction of the left-Kan-extension presheaf in Definition 4.9.15 back to $\iota_X U$ has the identity factorization $\iota_X U \to \iota_X U$. Refining any other factorization by its fiber product with U shows that the associated sheaf value is $F(U)$. Under this identification, the formula in Definition 4.9.36 sends s to the same section s. Thus every component of η_F^X is the identity. Etale objects detect maps of etale sheaves, so the unit is an isomorphism. □

Definition 4.9.56 (Derived etale-to-pro-etale cohomology comparison). Lean: `conditional-interface:def:weil-efi-site-derived-cohomology-comparison`

Uses: `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-epsilon-unit-counit`

For a commutative ring A and $L \in D^+(X_{\mathrm{et}}, A)$, let

$$u^R_{X,L} : L \longrightarrow R\epsilon_{X,*}\mathbf{L}\epsilon_X^{-1}L$$

be the derived adjunction unit. Define

$$c_{X,L} : R\Gamma(X_{\mathrm{et}}, L) \longrightarrow R\Gamma(X_{\mathrm{proet}}, \mathbf{L}\epsilon_X^{-1}L)$$

as $R\Gamma(X_{\mathrm{et}}, u^R_{X,L})$ followed by the canonical identity $R\Gamma(X_{\mathrm{et}}, R\epsilon_{X,*}(-)) = R\Gamma(X_{\mathrm{proet}}, -)$. Write $c^n_{X,L}$ for its map on nth cohomology.

Lemma 4.9.57 (Acyclicity of pulled-back injectives)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-epsilon-injective-acyclicity`

Let U be affine and etale over X, and let I be an injective A-module sheaf on U_{et}. Then

$$H^r(U_{\mathrm{proet}}, \epsilon_U^{-1} I) = 0 \qquad (r > 0).$$

Consequently $R^r \epsilon_{X,*}(\epsilon_X^{-1} J) = 0$ for every injective J on X_{et} and every $r > 0$.

Proof. Uses: `lemma:weil-pbc-epsilon-unit-isomorphism,lemma:weil-pbc-epsilon-scheme-naturality`

On U_{proet}, take the basis $\mathcal{B}$ of affine objects $V = \varprojlim_j V_j$ for which every $V_j \to U$ is affine etale and the transition maps are affine. Use finite pro-etale covers $\{V_a \to V\}_{a=1}^m$ inside this basis. Write each V_a as an inverse limit of affine etale $V_{a,j} \to V$. Fiber products commute with these limits, so every term of the Cech complex of the cover is the filtered colimit of the corresponding terms for the finite-stage families $\{V_{a,j} \to V_j\}$.

After increasing j, all finitely many arrows and their joint surjectivity descend, so $\{V_{a,j} \to V_j\}$ is an etale covering. The comparison unit identifies the restriction of $\epsilon_U^{-1} I$ on every finite-stage etale object with I. Hence

$$\check{H}^r(\{V_a \to V\}, \epsilon_U^{-1} I) = \varinjlim_j \check{H}^r(\{V_{a,j} \to V_j\}, I) = 0 \qquad (r > 0).$$

The last equality uses exactness of filtered colimits and Cech acyclicity of an injective sheaf. The basis is closed under fiber products, and every pro-etale object is covered by its members; the basis Cech criterion therefore gives the displayed cohomology vanishing.

For an affine etale $U \to X$, scheme naturality identifies the restriction of $\epsilon_X^{-1} J$ with $\epsilon_U^{-1}(J|_U)$. The restriction of an injective to the localization site is injective because extension by zero is its exact left adjoint. Thus the first part applies. Affine etale objects cover every etale object, so the sheaf associated to $U \mapsto H^r(U_{\mathrm{proet}}, \epsilon_U^{-1}(J|_U))$ is zero. This sheaf is $R^r \epsilon_{X,*}(\epsilon_X^{-1} J)$. □

Lemma 4.9.58 (Derived comparison for the epsilon unit)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-epsilon-derived-unit`

For $L \in D^+(X_{\mathrm{et}}, A)$, the derived unit

$$L \longrightarrow R\epsilon_{X,*} \epsilon_X^{-1} L$$

is an isomorphism, naturally in L.

Proof. Uses: `lemma:weil-pbc-epsilon-injective-acyclicity,lemma:weil-pbc-epsilon-unit-isomorphism,lemma:weil-pbc-epsilon-inverse-image-exact`

Choose a bounded-below injective resolution $L \to I^\bullet$. Exactness of ϵ_X^{-1} represents the inverse image by $\epsilon_X^{-1} I^\bullet$. Lemma 4.9.57 makes every term $\epsilon_{X,*}$-acyclic. The bounded-below acyclic-resolution lemma then computes the derived direct image termwise:

$$R\epsilon_{X,*} \epsilon_X^{-1} L \simeq \epsilon_{X,*} \epsilon_X^{-1} I^\bullet.$$

The unit isomorphism identifies the last complex with $I^\bullet$, including its differentials. A morphism of bounded-below complexes lifts to the chosen injective resolutions, and two lifts are homotopic; termwise naturality of the unit makes the resulting derived isomorphism independent of the lift. □

Proposition 4.9.59 (Finite-level etale/pro-etale cohomology comparison)**.** *Lean:* `conditional-interface:prop:weil-efi-site-derived-comparison-isomorphism`

Uses: `def:weil-efi-site-derived-cohomology-comparison`

Let ℓ be prime, let $r \geq 1$, and let $\mathcal{A}$ be a constructible $\mathbf{Z}/\ell^r$-module sheaf on X_{et}. For $n \geq 0$, the map

$$c^n_{X,\mathcal{A}} : H^n(X_{\mathrm{et}}, \mathcal{A}) \longrightarrow H^n(X_{\mathrm{proet}}, \epsilon_X^{-1} \mathcal{A})$$

is an isomorphism.

Proof. Uses: `lemma:weil-pbc-epsilon-derived-unit,lemma:weil-efi-site-inverse-image-exact`

Exactness identifies $\mathbf{L}\epsilon_X^{-1}(\mathcal{A}[0])$ with $\epsilon_X^{-1}\mathcal{A}[0]$. The derived-unit lemma makes

$$\mathcal{A}[0] \longrightarrow R\epsilon_{X,*} \epsilon_X^{-1} \mathcal{A}[0]$$

an isomorphism. Derived global sections preserve this isomorphism, and the canonical identity $R\Gamma(X_{\mathrm{et}}, R\epsilon_{X,*}(-)) = R\Gamma(X_{\mathrm{proet}}, -)$ identifies the resulting arrow with $c_{X,\mathcal{A}[0]}$. Applying H^n gives the displayed isomorphism. □

Definition 4.9.60 (Canonical direct-image comparison)**.** *Lean:* `conditional-interface:def:weil-pbc-epsilon-direct-image-comparison`

Uses: `def:weil-pbc-epsilon-unit-counit`

Let $f : X \to Y$ be quasi-compact and quasi-separated. For an A-module sheaf F on X_{et}, define

$$\delta_{f,F} : \epsilon_Y^{-1} f_{\mathrm{et},*} F \longrightarrow f_{\mathrm{proet},*} \epsilon_X^{-1} F$$

to be adjoint to

$$f_{\mathrm{et},*} F \xrightarrow{f_{\mathrm{et},*} \eta_F^X} f_{\mathrm{et},*} \epsilon_{X,*} \epsilon_X^{-1} F = \epsilon_{Y,*} f_{\mathrm{proet},*} \epsilon_X^{-1} F.$$

The equality follows by evaluating both restrictions on an etale object of Y. Right deriving this natural transformation and using exactness of ϵ_Y^{-1} defines

$$D_f(L) : \epsilon_Y^{-1} R f_{\mathrm{et},*} L \longrightarrow R f_{\mathrm{proet},*} \epsilon_X^{-1} L$$

for $L \in D^+(X_{\mathrm{et}}, A)$.

Lemma 4.9.61 (Well-definedness certificate for def:weil-pbc-epsilon-direct-image-comparison)**.** *Lean:* `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-epsilon-direct-image-comparison`

Uses: `def:weil-pbc-epsilon-direct-image-comparison`

The construction in Definition 4.9.60 is a typed raw datum. Its theorem-valued naturality, comparison, descent, or coherence fields are isolated in this property lemma.

Proof. Uses: `lemma:weil-pbc-epsilon-scheme-naturality,lemma:weil-pbc-epsilon-adjunction-equations-v1,lemma:weil-pbc-epsilon-inverse-image-exact`

The underived transformation is the adjoint of $f_*\eta^X$. Naturality of η^X proves naturality in F; the identity formula and the triangle identities prove the identity and composition equations. Scheme naturality identifies the two restrictions in the displayed equality $f_{\mathrm{et},*}\epsilon_{X,*} = \epsilon_{Y,*}f_{\mathrm{proet},*}$ and makes the construction compatible with Cartesian pullback.

For the derived map choose a bounded-below injective resolution $L \to I^\bullet$ on X_{et}, and then a bounded-below injective resolution $\epsilon_X^{-1}I^\bullet \to J^\bullet$ on X_{proet}. Exactness of ϵ_X^{-1} and ϵ_Y^{-1} identifies the two derived source representatives, and the complex map

$$\epsilon_Y^{-1}f_{\mathrm{et},*}I^\bullet \xrightarrow{\delta_{f,I^\bullet}} f_{\mathrm{proet},*}\epsilon_X^{-1}I^\bullet \longrightarrow f_{\mathrm{proet},*}J^\bullet$$

represents $D_f(L)$. Comparisons between two choices of either resolution are unique up to homotopy; naturality of δ_f carries both comparison homotopies to the target, so the localized map is independent of every choice. The underived identity, composition, and base-change equations persist after localization. Thus $D_f(L)$ and all its stated coherence fields are well-defined. □

Lemma 4.9.62 (Site-local components of direct-image comparison)**.** *Lean:* `conditional-interface:lemma:weil-pbc-epsilon-direct-image-site-comparison`

Uses: `def:weil-pbc-epsilon-direct-image-comparison`

Suppose Y is affine and let $W = \varprojlim_i W_i$ be an affine pro-etale object, with $W_i \to Y$ affine etale and affine transition maps. Put $X_i = X \times_Y W_i$ and $X_W = \varprojlim_i X \times_Y W$. Then

$$(\epsilon_Y^{-1}f_{\mathrm{et},*}F)(W) = \varinjlim_i \Gamma(X_i, F),$$

$$(f_{\mathrm{proet},*}\epsilon_X^{-1}F)(W) = \Gamma(X_W, \epsilon_{X_W}^{-1}g_{\mathrm{et}}^{-1}F),$$

where $g : X_W \to X$. Under these identifications, $\delta_{f,F}(W)$ is the map induced by pullback from the finite stages. Such W form a covering basis of Y_{proet}.

Proof. Uses: `lemma:weil-pbc-epsilon-scheme-naturality`

The first equality is the left-Kan-extension formula for ϵ_Y^{-1}: the subsystem of the displayed affine etale stages is cofinal among etale objects receiving a map from W. The second equality is the definition of pro-etale direct image, followed by the scheme-naturality identification of the restricted inverse image. Expanding the adjoint of $f_{\mathrm{et},*}\eta_F^X$ shows that a section over X_i is pulled back to X_W; hence its filtered-colimit map is exactly $\delta_{f,F}(W)$. Every pro-etale object over affine Y admits a cover by inverse limits of this form, so these evaluations detect sheaf isomorphisms. □

Lemma 4.9.63 (Higher direct-image comparison on finite sheaves)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-epsilon-direct-image-higher-sheaves`

Let A be finite, let $f : X \to Y$ be quasi-compact and quasi-separated, and let F be a constructible A-module sheaf on X_{et}. For every $r \geq 0$, the canonical map

$$\epsilon_Y^{-1}R^r f_{\mathrm{et},*}F \longrightarrow R^r f_{\mathrm{proet},*}\epsilon_X^{-1}F$$

is an isomorphism.

Proof. Uses: `lemma:weil-pbc-epsilon-direct-image-site-comparison,lemma:weil-pbc-epsilon-derived-unit`

The assertion is Zariski local on Y, so take Y affine and use the basis $W = \varprojlim_i W_i$ from Lemma 4.9.62. The left side at W is

$$\varinjlim_i H^r_{\mathrm{et}}(X_i, F).$$

For the right side, apply Lemma 4.9.58 on X_W to $g_{\mathrm{et}}^{-1}F$. It identifies its value at W with

$$H^r_{\mathrm{proet}}(X_W, \epsilon_{X_W}^{-1}g_{\mathrm{et}}^{-1}F) = H^r_{\mathrm{et}}(X_W, g_{\mathrm{et}}^{-1}F).$$

Because f is quasi-compact and quasi-separated, X_W is the inverse limit of the X_i with affine transition maps and etale cohomology is continuous for this system. Hence

$$\varinjlim_i H^r_{\mathrm{et}}(X_i, F) \xrightarrow{\sim} H^r_{\mathrm{et}}(X_W, g_{\mathrm{et}}^{-1}F).$$

The component description in the preceding lemma identifies this isomorphism with the canonical comparison map. The basis evaluations are conservative, proving the sheaf assertion. □

Lemma 4.9.64 (Finite constructibility and lower bounds)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-epsilon-direct-image-constructible-bounds`

Let A be finite and let $L \in D_c^+(X_{\mathrm{et}}, A)$ vanish in degrees $< a$. Then $\epsilon_X^{-1}L$ has constructible cohomology and also vanishes in degrees $< a$. Both $Rf_{\mathrm{et},*}L$ and $Rf_{\mathrm{proet},*}\epsilon_X^{-1}L$ vanish in degrees $< a$.

Proof. Uses: `lemma:weil-pbc-epsilon-inverse-image-exact,lemma:weil-pbc-epsilon-scheme-naturality`
Exactness gives

$$\mathcal{H}^q(\epsilon_X^{-1}L) = \epsilon_X^{-1}\mathcal{H}^q(L).$$

For each q, choose a finite stratification on which $\mathcal{H}^q(L)$ is locally constant with finite stalks. Scheme naturality identifies its inverse image on each pro-etale stratum with the same finite locally constant sheaf, so that stratification witnesses constructibility of the corresponding cohomology sheaf. The equality also preserves the lower vanishing bound. Right-derived direct image has no negative derived degree; its hypercohomology terms have $p \geq 0$ and $q \geq a$, which gives the same lower bound for both direct images. □

Lemma 4.9.65 (Naturality of direct-image comparison). Lean: `conditional-interface:lemma:weil-pbc-epsilon-direct-image-naturality`

Uses: `def:weil-pbc-epsilon-direct-image-comparison`
The maps $D_f(L)$ are natural in L, equal the identity for $f = 1_X$, and respect composition of quasi-compact quasi-separated morphisms. For a Cartesian square

$$\begin{array}{ccc} X' & \xrightarrow{v} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{u} & Y, \end{array}$$

the square formed by $u^*D_f(L)$, $D_{f'}(v^*L)$, and the two canonical base-change transformations commutes.

Proof. Uses: `lemma:weil-pbc-epsilon-unit-counit-naturality,lemma:weil-pbc-epsilon-scheme-naturality`
The underived map is adjoint to $f_*\eta^X$. Naturality of η^X gives naturality in F, its identity formula gives the identity case, and the triangle identity shows that the adjoint for a composite is the ordered composite of the two adjoints. Deriving preserves these equations.

For the Cartesian square, expand both routes before taking adjoints. Scheme naturality identifies $v^*\eta_L^X$ with $\eta_{v^*L}^{X'}$, and both routes then consist of this unit followed by restriction along the same projection $X' \to X$ and the same base-change map. They are equal before derivation, hence the derived base-change square commutes. □

Lemma 4.9.66 (Comparison direct-image transport). Lean: `conditional-interface:lemma:weil-pbc-epsilon-direct-image`

Uses: `def:weil-pbc-epsilon-direct-image-comparison`
Let A be finite, let $f : X \to Y$ be quasi-compact and quasi-separated, and let $L \in D_c^+(X_{\mathrm{et}}, A)$. The canonical map

$$D_f(L) : \epsilon_Y^{-1}Rf_{\mathrm{et},*}L \longrightarrow Rf_{\mathrm{proet},*}\epsilon_X^{-1}L$$

is an isomorphism, natural in L, in f, and in Cartesian base change.

Proof. Uses: `lemma:weil-pbc-epsilon-direct-image-higher-sheaves,lemma:weil-pbc-epsilon-direct-image-constructible-bounds,lemma:weil-pbc-epsilon-direct-image-naturality,lemma:weil-pbc-hypercohomology-convergence`
Choose a below the cohomology of L. The truncation towers on both sides give morphisms of hypercohomology spectral sequences whose second pages are

$$\epsilon_Y^{-1}R^p f_{\mathrm{et},*}\mathcal{H}^q(L) \longrightarrow R^p f_{\mathrm{proet},*}\epsilon_X^{-1}\mathcal{H}^q(L).$$

Every component is an isomorphism by Lemma 4.9.63. The bounds lemma puts both spectral sequences in $p \geq 0$, $q \geq a$. Thus each total degree has a finite exhaustive separated filtration, by the numerical argument of Lemma 4.8.2. Page induction and finite-filtration induction make the abutment map an isomorphism in every degree. Hence $D_f(L)$ is an isomorphism in the derived category. Its naturality statements are Lemma 4.9.65. □

Proposition 4.9.67 (Finite-level pro-etale proper base change). Lean: `conditional-interface:prop:weil-pbc-proet-finite-level-transport`

Uses: `def:weil-pbc-comes-from-etale`
Let A be a finite torsion ring and let (L, θ) witness that $K \in D^+(X_{\mathrm{proet}}, A)$ comes from the etale site, with $L \in D_c^+(X_{\mathrm{et}}, A)$. For a proper Cartesian square, the pro-etale exchange map

$$Lg^*Rf_*K \longrightarrow Rf'_*L(g')^*K$$

is an isomorphism.

Proof. Uses: `lemma:weil-pbc-epsilon-pullback,lemma:weil-pbc-epsilon-direct-image,prop:weil-proet-finite-derived-proper-base-change`
Use θ to replace K by $\epsilon_X^{-1}L$. Commute ϵ^{-1} past the two pullbacks and two direct images using the preceding lemmas. The resulting middle arrow is $\epsilon_{Y'}^{-1}$ applied to the S7 etale exchange map and is therefore an isomorphism. Expanding all four comparison isomorphisms shows the transported arrow is the pro-etale unit–counit exchange map; naturality follows from the two comparison squares. □

4.10 S9. Derived completion and inverse limits

Definition 4.10.1 (Adic coefficient datum and reduction tower)**.** Uses:

Lean: `conditional-interface:def:weil-pbc-complete-coefficient-datum`

Let Λ be Noetherian, let $I = (f_1, \dots, f_r) \subseteq \Lambda$ have Λ/I torsion, and suppose every $P_n := \Lambda/I^n$ is perfect over Λ. For $K \in D(X_{\mathrm{proet}}, \Lambda)$ put

$$K_n := K \otimes^L_\Lambda P_n.$$

For $b \geq a$, let $q_{b,a} : K_b \to K_a$ be induced by $P_b \to P_a$. The quotient identities give $q_{a,a} = 1$ and $q_{c,a} = q_{b,a} q_{c,b}$. Associativity also gives the normalization isomorphisms

$$K_b \otimes^L_{P_b} P_a \xrightarrow{\sim} K_a,$$

and these obey the same cocycle.

Definition 4.10.2 (Explicit adic derived inverse limit)**.** Lean: `conditional-interface:def:weil-pbc-adic-derived-inverse-limit`

Uses: `def:weil-pbc-complete-coefficient-datum`

For a tower $M_\bullet = (M_n, t_{n+1,n})$, put

$$s((x_n)_n) = (t_{n+1,n}(x_{n+1}))_n, \qquad R\varprojlim_n M_n := \mathrm{Cone}\left(1 - s : \prod_n M_n \longrightarrow \prod_n M_n\right)[-1].$$

Thus there is a functorial distinguished triangle

$$R\varprojlim_n M_n \longrightarrow \prod_n M_n \xrightarrow{1-s} \prod_n M_n \longrightarrow .$$

Maps of towers act componentwise on the two products and hence on the cone.

Lemma 4.10.3 (Milnor sequence on pro-etale test objects)**.** *Lean:* `conditional-interface:lemma:weil-pbc-adic-milnor-sequence`

Uses: `def:weil-pbc-adic-derived-inverse-limit`

Let U be a weakly contractible quasi-compact object of the pro-etale site. For every tower $M_\bullet$ and every i there is a functorial exact sequence

$$0 \longrightarrow \varprojlim{}^1_n H^{i-1}(U, M_n) \longrightarrow H^i\left(U, R\varprojlim_n M_n\right) \longrightarrow \varprojlim_n H^i(U, M_n) \longrightarrow 0.$$

Proof. Uses:

Weak contractibility makes $\Gamma(U, -)$ exact, and evaluation on U commutes with products. Apply it to the product-cone triangle. The long exact cohomology sequence identifies the kernel of $1 - s$ on $\prod_n H^i(U, M_n)$ with $\varprojlim H^i(U, M_n)$ and its cokernel in degree $i-1$ with $\varprojlim{}^1 H^{i-1}(U, M_n)$. Kernel–cokernel exactness gives the displayed sequence, and every construction is natural in a map of towers. □

Lemma 4.10.4 (Finite towers satisfy Mittag–Leffler)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-adic-mittag-leffler`

For an inverse system of finite abelian groups (A_n), the images $\mathrm{Im}(A_m \to A_n)$ are constant for all sufficiently large $m \geq n$; hence (A_n) is Mittag–Leffler.

Proof. Uses:

For fixed n these images form a descending chain of subgroups of the finite group A_n. Such a chain stabilizes. Doing this for every n is exactly the Mittag–Leffler condition. □

Lemma 4.10.5 (Mittag–Leffler kills the Milnor obstruction)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-adic-lim-one-zero`

For a countable Mittag–Leffler system (A_n) one has $\varprojlim{}^1_n A_n = 0$. In particular this holds for a tower of finite abelian groups.

Proof. Uses: `lemma:weil-pbc-adic-mittag-leffler`

Replace A_n by the stable image in A_n; this does not change either $\varprojlim$ or the cokernel description of $\varprojlim{}^1$, and the new transition maps are surjective. Given $(y_n) \in \prod_n A_n$, choose x_1 arbitrarily and then choose x_{n+1} recursively so that $x_n - t_{n+1,n}(x_{n+1}) = y_n$. Thus $1 - s$ is surjective on the product and its cokernel, namely $\varprojlim{}^1 A_n$, is zero. □

Definition 4.10.6 (Intrinsic Koszul completion)**.** Lean: `conditional-interface:def:weil-pbc-intrinsic-koszul-completion`

Uses: `def:weil-pbc-adic-derived-inverse-limit`

Let

$$Q_m := K(\Lambda; f_1^m, \dots, f_r^m)$$

be the Koszul complex in degrees $[-r, 0]$. Its map $Q_{m+1} \to Q_m$ is the tensor product, over j, of the two-term chain maps whose degree-zero component is the identity and whose degree-minus-one component is multiplication by f_j. Define

$$K^{\wedge,\mathrm{Kos}} := R\varprojlim_m (K \otimes^L_\Lambda Q_m).$$

The augmentations $\Lambda \to Q_m$ give $\kappa_K : K \to K^{\wedge,\mathrm{Kos}}$. Call K intrinsically derived I-complete when κ_K is an isomorphism; below, "derived I-complete" always means this intrinsic condition.

Definition 4.10.7 (Completion map). Lean: `conditional-interface:def:weil-pbc-completion-map`

Uses: `def:weil-pbc-adic-derived-inverse-limit`

Define

$$\mathrm{comp}_K : K \longrightarrow R\varprojlim_n K_n$$

as the map induced by the compatible quotient arrows $K \to K_n$.

Lemma 4.10.8 (Koszul and quotient towers have the same limit). Lean: `conditional-interface:lemma:weil-pbc-koszul-quotient-limit`

Uses: `def:weil-pbc-intrinsic-koszul-completion`, `def:weil-pbc-completion-map`

There is a functorial isomorphism

$$c_K : K^{\wedge,\mathrm{Kos}} \xrightarrow{\sim} R\varprojlim_n K_n$$

such that $c_K \kappa_K = \mathrm{comp}_K$.

Proof. Uses:

Put $J_m = (f_1^m, \dots, f_r^m)$. The augmentation $Q_m \to \Lambda/J_m$ has cone $\tau_{<0}Q_m$. For $p < 0$ the modules $H^p(Q_m)$ are finite and killed by J_m. The transition in degree p has entries in $(f_1^{b-a}, \dots, f_r^{b-a})$ from level b to level a. Artin–Rees, applied to the cycles inside the finite free term Q_a^p, therefore makes $H^p(Q_b) \to H^p(Q_a)$ zero for $b \gg a$. Hence $(\tau_{<0}Q_m)_m$ is pro-zero and $(Q_m)_m \to (\Lambda/J_m)_m$ is a pro-isomorphism.

The inclusions

$$I^{r(m-1)+1} \subseteq J_m \subseteq I^m$$

show that $(\Lambda/J_m)_m$ and $(P_n)_n$ are cofinal pro-systems. Tensor the comparison maps with K and apply the product-cone limit. Pro-zero cones have zero product-cone limit after a cofinal reindexing, so the result is c_K. Both units are induced by the same augmentations $\Lambda \to Q_m \to \Lambda/J_m \to P_n$; this proves the triangle equation. □

Lemma 4.10.9 (Derived-complete reconstruction). Lean: `conditional-interface:lemma:weil-pbc-derived-complete-reconstruction`

Uses: `def:weil-pbc-intrinsic-koszul-completion`

If K is intrinsically derived I-complete, then

$$\mathrm{comp}_K : K \xrightarrow{\sim} R\varprojlim_n K_n$$

is an isomorphism, functorially in K.

Proof. Uses: `lemma:weil-pbc-koszul-quotient-limit`

By intrinsic completeness, κ_K is an isomorphism. The preceding lemma identifies comp_K with the composite $c_K\kappa_K$, so it too is an isomorphism. For $u : K \to L$, naturality of the Koszul augmentations and of the product-cone construction gives

$$(R\varprojlim u_n)\,\mathrm{comp}_K = \mathrm{comp}_L\, u.$$

Conjugating by the two reconstruction isomorphisms proves functoriality, including functoriality of their inverses. This is the explicit Koszul-to-adic argument behind Tag 099M. □

Lemma 4.10.10 (Finite reduction of adic completion). Uses:

Lean: `conditional-interface:lemma:weil-pbc-adic-completion-reduction`

For every $a \geq 1$, projection to level a induces a functorial isomorphism

$$\left(R\varprojlim_n K_n\right) \otimes^L_\Lambda P_a \xrightarrow{\sim} K_a.$$

Proof. Uses: `lemma:weil-pbc-koszul-quotient-limit`

Use the preceding comparison to replace the quotient tower by $K \otimes^L Q_m$. Since P_a is perfect, tensoring with P_a commutes with both products in the product-cone model. For $m \geq a$, the differential of $Q_m \otimes^L P_a$ is zero. Its degree-zero transition is the identity, whereas every composite of a successive transitions in a negative degree contains a factor $f_j^a \in I^a$ and is zero over P_a. Thus the negative-degree towers are pro-zero and the degree-zero tower is the constant tower P_a. After tensoring with K, its product-cone limit is $K \otimes^L P_a = K_a$. The surviving degree-zero projection is precisely the displayed map. This also exposes the finite Tor-amplitude input: the bounded finite-projective complex representing P_a is what permits it to pass through the two countable products. □

Definition 4.10.11 (Uniformly bounded constructible adic tower). Lean: `conditional-interface:def:weil-pbc-bounded-constructible-adic-tower`

Uses: `def:weil-pbc-complete-coefficient-datum`

Say that the reduction tower $K_\bullet$ attached to K is uniformly bounded constructible if there are integers $d_0 \leq d_1$ and one finite stratification of X, each of whose strata has finitely many connected components, such that, for every n, $H^i(K_n) = 0$ outside $[d_0, d_1]$ and each $H^i(K_n)$ is finite lisse on every stratum. The normalization isomorphisms and their cocycle are part of this datum.

Lemma 4.10.12 (Bounded constructible towers have no Milnor obstruction)**.** *Lean:* `conditional-interface:lemma:weil-pbc-constructible-adic-lim-one-zero`

Uses: `def:weil-pbc-bounded-constructible-adic-tower`

If the reduction tower $K_\bullet$ attached to K is uniformly bounded constructible, then on every weakly contractible quasi-compact U lying over one of the displayed strata,

$$H^i\left(U, R\varprojlim_n K_n\right) \xrightarrow{\sim} \varprojlim_n H^i(U, K_n),$$

and both sides vanish outside $[d_0, d_1]$.

Proof. Uses: `lemma:weil-pbc-adic-milnor-sequence,lemma:weil-pbc-adic-lim-one-zero`

Fix n and i. As m increases, the image sheaves of $H^i(K_m) \to H^i(K_n)$ form a descending chain of finite lisse subsheaves on each displayed stratum. On each connected component their fibers are descending chains of subgroups of a finite group, so they stabilize. There are only finitely many strata and, by hypothesis, finitely many connected components on each stratum; hence one index works globally. Exactness of $\Gamma(U, -)$ identifies the image of $H^i(U, K_m) \to H^i(U, K_n)$ with the sections of that image sheaf. Consequently $(H^i(U, K_n))_n$ is Mittag–Leffler even when U has infinitely many connected components. The vanishing lemma now kills its $\varprojlim^1$. The Milnor sequence is therefore the displayed isomorphism, and uniform finite-level vanishing gives the last assertion. □

Lemma 4.10.13 (Bounded constructibility survives reconstruction)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-bounded-constructible-reconstruction`

If the reduction tower $K_\bullet$ attached to K is uniformly bounded constructible, then $\widehat{K} := R\varprojlim_n K_n$ is intrinsically derived I-complete, has finite reduction $\widehat{K} \otimes^L P_n \simeq K_n$, and retains the common cohomological interval $[d_0, d_1]$ on the conservative weakly contractible basis subordinate to the displayed stratification.

Proof. Uses: `lemma:weil-pbc-adic-completion-reduction,lemma:weil-pbc-constructible-adic-lim-one-zero,lemma:weil-pbc-koszul-quotient-limit`

Finite reduction is the preceding reduction lemma. Hence the quotient completion of $\widehat{K}$ is the limit of the same tower and its completion map is the identity under the product-cone model. The Koszul–quotient comparison then makes its intrinsic Koszul unit an isomorphism. The common cohomological bounds and the absence of a hidden degree from $\varprojlim^1$ are exactly the conclusion of the preceding lemma. Weakly contractible objects subordinate to a finite stratification form a conservative basis, so these bounds detect the bounds of $\widehat{K}$. □

Definition 4.10.14 (Completed pullback)**.** *Lean:* `conditional-interface:def:weil-pbc-completed-pullback`

Uses: `def:weil-pbc-complete-coefficient-datum`

For $g : Y' \to Y$ define

$$Lg^*_{\mathrm{comp}} K := R\varprojlim_n \left(Lg^* K \otimes^L_\Lambda \Lambda/I^n\right).$$

The comparison from ordinary pullback is the completion map of Lg^*K.

Lemma 4.10.15 (Reduction commutes with completed pullback)**.** *Lean:* `conditional-interface:lemma:weil-pbc-completed-pullback-reduction`

Uses: `def:weil-pbc-completed-pullback`

For every n,

$$(Lg^*_{\mathrm{comp}} K) \otimes^L_\Lambda \Lambda/I^n \xrightarrow{\sim} Lg^* K_n.$$

Proof. Uses: `lemma:weil-pbc-adic-completion-reduction`

Derived pullback commutes with derived tensor, so its defining tower is the reduction tower of Lg^*K. Apply finite reduction of adic completion to this object. Its level n is $Lg^*K \otimes^L_\Lambda P_n \simeq Lg^*(K \otimes^L_\Lambda P_n) = Lg^* K_n$. □

Definition 4.10.16 (Adic operator and base-change datum)**.** *Lean:* `conditional-interface:def:weil-pbc-adic-operator-base-change-datum`

Uses: `def:weil-pbc-completed-pullback`

Let $g : Y' \to Y$, let K be on Y_{proet}, and let L be on Y'_{proet}. An adic operator and base-change datum consists of compatible tower endomorphisms

$$\phi_n : K_n \to K_n, \qquad \psi_n : L_n \to L_n,$$

and compatible maps $b_n : Lg^* K_n \to L_n$ such that

$$b_n \, Lg^*(\phi_n) = \psi_n b_n$$

for every n. Compatibility includes the transition squares, the normalization squares, and their cocycles. This is a datum of ordinary endomorphisms. A genuine Frobenius or Weil structure has the different type $\mathrm{Fr}^* K_n \to K_n$ and is not identified with this datum.

Lemma 4.10.17 (Reconstruction preserves operators and base change)**.** *Lean:* `conditional-interface:lemma:weil-pbc-adic-reconstruction-compatibility`

Uses: `def:weil-pbc-adic-operator-base-change-datum`

An adic operator and base-change datum induces maps

$$\widehat{\phi} = R\varprojlim_n \phi_n, \quad \widehat{\psi} = R\varprojlim_n \psi_n, \quad \widehat{Lg^*\phi} = R\varprojlim_n Lg^*\phi_n, \quad \widehat{b} = R\varprojlim_n b_n : Lg^*_{\mathrm{comp}} K \longrightarrow \widehat{L}$$

satisfying $\widehat{b}\,\widehat{Lg^*\phi} = \widehat{\psi}\widehat{b}$. Under the canonical finite-reduction isomorphisms, reduction modulo I^n gives exactly ϕ_n, ψ_n, b_n. If K and L are intrinsically derived complete, the reconstruction isomorphisms identify these with operators and a base-change map on K and L themselves.

Proof. Uses: `lemma:weil-pbc-adic-completion-reduction`
Compatibility with transitions makes the three families maps of towers. They therefore give maps between the two product-cone diagrams. The levelwise equations give an equality on both product terms, hence on their fibres, proving the displayed completed equation. The finite-reduction lemma identifies projection to the nth factor with reduction modulo I^n, so the reduced maps are precisely the named level maps. Finally, conjugate by the functorial reconstruction isomorphisms for complete K and L; the same equality survives conjugation. □

Definition 4.10.18 (Local strict-perfect coefficient package)**.** Uses:
Lean: `conditional-interface:def:weil-pbc-strict-perfect-coefficient-package`
On a localization of a ringed topos, let Q be represented by a bounded complex $F^\bullet$ of finite projective Λ-modules. A strict-perfect package for this representation consists, for every r, of an integer N_r and maps

$$F^r \xrightarrow{i_r} \Lambda^{N_r} \xrightarrow{p_r} F^r, \qquad p_r i_r = 1.$$

Write $e_r = i_r p_r$; then $e_r^2 = e_r$, $p_r e_r = p_r$, and $e_r i_r = i_r$. Perfectness means that such packages exist after a covering. When $Q \in D(\Lambda)$ is a perfect coefficient complex, one global package and its associated constant sheaf may be used.

Definition 4.10.19 (Projection-formula arrows)**.** Lean: `conditional-interface:def:weil-pbc-projection-formula-arrows`
Uses: `def:weil-pbc-strict-perfect-coefficient-package`
For a morphism of ringed topoi $h : \mathcal{T} \to \mathcal{S}$, an object $E \in D(\mathcal{T}, \Lambda)$, and a strict-perfect package $F^\bullet$ on $\mathcal{S}$, define

$$\pi_{h,E,F} : Rh_*E \otimes^L_\Lambda F^\bullet \longrightarrow Rh_*(E \otimes^L_\Lambda Lh^*F^\bullet)$$

to be adjoint to

$$Lh^*(Rh_*E \otimes^L F^\bullet) \simeq Lh^*Rh_*E \otimes^L Lh^*F^\bullet \xrightarrow{\epsilon_E \otimes 1} E \otimes^L Lh^*F^\bullet.$$

For $F^r = \Lambda^{N_r}$, let ρ_r be the inverse of the finite-sum identification

$$(Rh_*E)^{N_r} \xrightarrow{\pi_r} Rh_*(E^{N_r}).$$

For a projective term with the displayed splitting, define

$$\rho_{F^r} := (Rh_*E \otimes p_r)\, \rho_r\, Rh_*(1_E \otimes h^*i_r).$$

The bounded totalization of the ρ_{F^r} is denoted $\rho_{h,E,F}$.

Lemma 4.10.20 (Finite-free projection formula)**.** *Lean:* `conditional-interface:lemma:weil-pbc-projection-formula-finite-free`
Uses: `def:weil-pbc-projection-formula-arrows`
For $F = \Lambda^N[a]$, the maps $\pi_{h,E,F}$ and $\rho_{h,E,F}$ are inverse:

$$\rho_{h,E,F}\pi_{h,E,F} = 1, \qquad \pi_{h,E,F}\rho_{h,E,F} = 1.$$

Proof. Uses:
Both source and target are the shift by a of a finite direct sum of N copies of Rh_*E. A right adjoint preserves finite products, and in an additive category finite products are finite coproducts. Under these two identifications, the counit defining π is the identity in every summand. The map ρ is the reverse finite-sum identification, so the two displayed composites are componentwise identities. □

Lemma 4.10.21 (Finite-projective summand equations)**.** *Lean:* `conditional-interface:lemma:weil-pbc-projection-formula-projective-summand`
Uses: `def:weil-pbc-projection-formula-arrows`
For a finite projective F^r and a displayed splitting (i_r, p_r), the arrows π_{h,E,F^r} and ρ_{F^r} are inverse and are independent of the chosen finite-free summand presentation.

Proof. Uses: `lemma:weil-pbc-projection-formula-finite-free`
Let $A(u) = Rh_*E \otimes u$ and $B(u) = Rh_*(E \otimes h^*u)$. Naturality of the counit gives

$$B(i_r)\pi_{F^r} = \pi_r A(i_r), \qquad B(p_r)\pi_r = \pi_{F^r} A(p_r).$$

Using $p_r i_r = 1$ and the finite-free inverse equations, compute

$$\rho_{F^r}\pi_{F^r} = A(p_r)\rho_r B(i_r)\pi_{F^r} = A(p_r)\rho_r\pi_r A(i_r) = 1,$$

and

$$\pi_{F^r}\rho_{F^r} = \pi_{F^r} A(p_r)\rho_r B(i_r) = B(p_r)\pi_r\rho_r B(i_r) = 1.$$

Thus ρ_{F^r} is the unique inverse of the intrinsic map π_{F^r}; uniqueness removes the dependence on (i_r, p_r). □

Lemma 4.10.22 (Signed totalization of the perfect formula)**.** *Lean:* `conditional-interface:lemma:weil-pbc-projection-formula-totalization`
Uses: `def:weil-pbc-projection-formula-arrows`
For a bounded finite-projective complex $F^\bullet$, the total arrows $\pi_{h,E,F}$ and $\rho_{h,E,F}$ are inverse isomorphisms.

Proof. Uses: `lemma:weil-pbc-projection-formula-projective-summand`
Choose a complex representing Rh_*E. On a homogeneous tensor $x \otimes y$ with x of degree a, both total complexes use

$$d(x \otimes y) = d(x) \otimes y + (-1)^a x \otimes d_F(y).$$

Naturality of π_{F^r} in a map of finite projective modules and uniqueness of its inverse give naturality of ρ_{F^r}. Apply these two naturality statements to every differential $d_F^r : F^r \to F^{r+1}$. They say the term maps commute with the horizontal differentials; their degree zero says they commute with the signed vertical differentials. Hence their finite totalizations are chain maps. On every bidegree the two composites are the identities proved in the summand lemma. Finite totalization preserves these equalities, so

$$\rho_{h,E,F}\pi_{h,E,F} = 1, \qquad \pi_{h,E,F}\rho_{h,E,F} = 1.$$

□

Definition 4.10.23 (Chosen compactification and compact-support image)**.** Uses:
Lean: `conditional-interface:def:weil-pbc-projection-compactification`
Let $f : X \to Y$ admit a chosen compactification

$$X \xrightarrow{j} \overline{X} \xrightarrow{p} Y, \qquad f = pj,$$

with j a quasi-compact open immersion and p proper. For this datum set

$$Rf_!^{(j,p)}E := Rp_*j_!E.$$

For proper f, its identity compactification is $(j,p) = (1_X, f)$.

Lemma 4.10.24 (Open-immersion tensor piece)**.** *Lean:* `conditional-interface:lemma:weil-pbc-projection-formula-open-piece`

Uses: `def:weil-pbc-projection-compactification`
For a bounded finite-projective complex $F^\bullet$ on $\overline{X}$, the canonical map

$$(j_!E) \otimes^L F^\bullet \xrightarrow{\sim} j_!(E \otimes^L Lj^*F^\bullet)$$

is an isomorphism, natural in E and $F^\bullet$.

Proof. Uses:
Extension by zero along an open immersion is exact. Resolve only E; tensoring with each finite projective term remains exact. At a point of the pro-etale topos lying over X, both sides are $E_x \otimes F_x^\bullet$ and the map is the identity. At a point lying over $\overline{X} \setminus X$, both sides are zero. These points form a conservative family, so the term map is an isomorphism. It commutes with d_F and with the signed total differential, hence gives the displayed derived isomorphism. □

Lemma 4.10.25 (Proper compactification piece)**.** *Lean:* `conditional-interface:lemma:weil-pbc-projection-formula-proper-piece`

Uses: `def:weil-pbc-projection-compactification`
For the proper piece p and every strict-perfect $F^\bullet$ on Y,

$$(Rp_*j_!E) \otimes^L F^\bullet \xrightarrow[\pi_{p,j_!E,F}]{\sim} Rp_*(j_!E \otimes^L Lp^*F^\bullet),$$

with inverse $\rho_{p,j_!E,F}$.

Proof. Uses: `lemma:weil-pbc-projection-formula-totalization`
Apply the signed-totalization lemma to the morphism of ringed pro-etale topoi induced by p and the object $j_!E$. □

Lemma 4.10.26 (Compactified finite-level projection formula)**.** *Uses:*
Lean: `conditional-interface:lemma:weil-pbc-projection-formula-compactified`
For a chosen compactification and a perfect Q on Y, the ordered composite

$$\begin{aligned}(Rf_!^{(j,p)}E) \otimes^L Q &= (Rp_*j_!E) \otimes^L Q \\ &\xrightarrow{\pi_{p,j_!E,Q}} Rp_*(j_!E \otimes^L Lp^*Q) \\ &\xrightarrow{Rp_*\theta_j} Rp_*j_!(E \otimes^L Lf^*Q) = Rf_!^{(j,p)}(E \otimes^L Lf^*Q)\end{aligned}$$

is an isomorphism. Here θ_j is the open-piece isomorphism and the inverse is $\rho_{p,j_!E,Q}\, Rp_*(\theta_j^{-1})$ in reverse order.

Proof. Uses: `lemma:weil-pbc-projection-formula-open-piece,lemma:weil-pbc-projection-formula-proper-piece`
The assertion is local on Y. On a cover choose a strict-perfect package for Q. The first arrow and its displayed inverse are the proper-piece inverse pair. The second arrow and its inverse come from the open-piece inverse pair, using $Lj^*Lp^*Q = Lf^*Q$. Associativity gives

$$\rho_{p,j_!E,Q}(Rp_*\theta_j^{-1})(Rp_*\theta_j)\pi_{p,j_!E,Q} = 1$$

and

$$(Rp_*\theta_j)\pi_{p,j_!E,Q}\rho_{p,j_!E,Q}(Rp_*\theta_j^{-1}) = 1.$$

These are precisely the two inverse equations for the ordered composite. They prove that the globally defined forward map is an isomorphism; uniqueness of an inverse glues the local reverse maps. □

Lemma 4.10.27 (Identity compactification recovers direct image)**.** *Uses:*

Lean: `conditional-interface: lemma: weil-pbc-projection-formula-identity-compactification`

If f is proper, the identity compactification identifies $Rf_!^{(1,f)}$ with Rf_*, makes θ_1 the identity, and identifies the compactified formula with $\pi_{f,E,Q}$ and its inverse with $\rho_{f,E,Q}$.

Proof. Uses: `lemma:weil-pbc-projection-formula-compactified`

For $j = 1_X$, extension by zero, inverse image, and both unit constraints are identities. Substitution into the ordered composite leaves $\pi_{f,E,Q}$; substitution into its inverse leaves $\rho_{f,E,Q}$. □

Lemma 4.10.28 (Projection formulas form a coefficient tower)**.** *Lean:* `conditional-interface: lemma: weil-pbc-projection-formula-tower-compatibility`

Uses: `def: weil-pbc-complete-coefficient-datum, def: weil-pbc-projection-formula-arrows`

For every n, let $q_n : P_{n+1} \to P_n$ be the quotient. The square

$$\begin{array}{ccc} (Rf_*K) \otimes^L P_{n+1} & \xrightarrow{\pi_{n+1}} & Rf_*(K \otimes^L P_{n+1}) \\ {\scriptstyle 1\otimes q_n}\downarrow & & \downarrow{\scriptstyle Rf_*(1\otimes q_n)} \\ (Rf_*K) \otimes^L P_n & \xrightarrow{\pi_n} & Rf_*(K \otimes^L P_n) \end{array}$$

commutes, and so does the square with every π replaced by ρ and horizontal arrows reversed; its vertical arrows remain the two quotient maps.

Proof. Uses:

The two adjoints of the forward composites are both

$$Lf^*Rf_*K \otimes^L P_{n+1} \xrightarrow{\epsilon_K\otimes 1} K \otimes^L P_{n+1} \xrightarrow{1\otimes q_n} K \otimes^L P_n.$$

Thus the forward square commutes. Invert the two horizontal isomorphisms in the equality $Rf_*(1 \otimes q_n)\pi_{n+1} = \pi_n(1 \otimes q_n)$ to obtain the asserted compatibility for ρ. □

Lemma 4.10.29 (Operator and base-change naturality of the formula)**.** *Lean:* `conditional-interface: lemma: weil-pbc-projection-formula-base-change-naturality`

Uses: `def: weil-pbc-cartesian-square, def: weil-pbc-projection-formula-arrows`

The projection-formula arrows are natural in K; hence for every endomorphism $\phi : K \to K$ with reductions $\phi_n : K_n \to K_n$,

$$Rf_*\phi_n\, \pi_n = \pi_n((Rf_*\phi) \otimes 1), \qquad ((Rf_*\phi) \otimes 1)\rho_n = \rho_n Rf_*\phi_n.$$

For a Cartesian square, the following two routes are equal:

$$\begin{aligned} Lg^*((Rf_*K) \otimes^L Q) &\xrightarrow{Lg^*\pi_f} Lg^*Rf_*(K \otimes^L Lf^*Q) \xrightarrow{\mathrm{Ex}_{K\otimes Lf^*Q}} Rf'_*L(g')^*(K \otimes^L Lf^*Q), \\ Lg^*((Rf_*K) \otimes^L Q) &\xrightarrow{\sim} (Lg^*Rf_*K) \otimes^L Lg^*Q \xrightarrow{\mathrm{Ex}_K\otimes 1} (Rf'_*L(g')^*K) \otimes^L Lg^*Q \\ &\xrightarrow{\pi_{f'}} Rf'_*(L(g')^*K \otimes^L L(f')^*Lg^*Q). \end{aligned}$$

The targets are identified by the chosen Cartesian isomorphism.

Proof. Uses:

For a map $u : K \to L$, both naturality composites are adjoint to

$$Lf^*Rf_*K \otimes Lf^*Q \xrightarrow{\epsilon_K\otimes 1} K \otimes Lf^*Q \xrightarrow{u\otimes 1} L \otimes Lf^*Q.$$

This proves the forward operator equation; conjugation by $\rho_n = \pi_n^{-1}$ gives the second. For base change, expand both exchange maps as unit followed by counit. After adjunction to X', both displayed routes become

$$L(g')^*Lf^*Rf_*K \otimes L(g')^*Lf^*Q \xrightarrow{L(g')^*(\epsilon_K)\otimes 1} L(g')^*K \otimes L(g')^*Lf^*Q.$$

The Cartesian identification $L(g')^*Lf^* = L(f')^*Lg^*$ identifies the coefficient factors. Hence the adjoints, and therefore the two routes, are equal. □

Lemma 4.10.30 (Perfect projection formula for reductions)**.** *Lean:* `conditional-interface: lemma: weil-pbc-perfect-projection-formula`

Uses: `def: weil-pbc-complete-coefficient-datum`

For every n there is a natural isomorphism

$$(Rf_*K) \otimes^L_\Lambda \Lambda/I^n \xrightarrow[\pi_n]{\sim} Rf_*K_n$$

whose inverse is ρ_n. The families (π_n) and (ρ_n) commute with the reduction tower, compatible coefficient endomorphisms, and base change.

Proof. Uses: lemma:weil-pbc-projection-formula-totalization,lemma:weil-pbc-projection-formula-identity-compactification,lemma:weil-pbc-projection-formula-tower-compatibility,lemma:weil-pbc-projection-formula-base-change-naturality

Choose a strict-perfect package $F_n^\bullet \to P_n$. Signed totalization gives inverse arrows

$$(Rf_*K) \otimes^L F_n^\bullet \underset{\rho_n}{\overset{\pi_n}{\rightleftarrows}} Rf_*(K \otimes^L Lf^*F_n^\bullet).$$

Because P_n is constant, $Lf^*P_n = P_n$, and the target is Rf_*K_n. The identity-compactification lemma shows that this is also the open/proper compactification composite for proper f. The totalization lemma gives $\rho_n\pi_n = 1$ and $\pi_n\rho_n = 1$. Uniqueness of an inverse makes the map independent of the strict-perfect package. Tower, operator, and base-change compatibility are exactly the preceding two lemmas. This expands the finite-projective argument located at Tag 0944. □

Lemma 4.10.31 (Direct image commutes with sequential homotopy limits)**.** *Uses:*

Lean: conditional-interface:lemma:weil-pbc-direct-image-r-lim

For an inverse tower M_n on X_{proet},

$$Rf_*R\lim_{\leftarrow n} M_n \xrightarrow{\sim} R\lim_{\leftarrow n} Rf_*M_n.$$

Proof. Uses:

Use the product/equalizer model

$$R\lim_{\leftarrow} M_n \longrightarrow \prod_n M_n \xrightarrow{1-\mathrm{shift}} \prod_n M_n$$

as a distinguished triangle. Since Rf_* is a right adjoint, it commutes with products. Applying it to the triangle gives the product/equalizer model for the right-hand limit, and the identity maps on the two products give the isomorphism of fibres. □

Definition 4.10.32 (Completed tensor product)**.** Lean: conditional-interface:def:weil-pbc-completed-tensor-product

Uses: def:weil-pbc-complete-coefficient-datum,def:weil-pbc-adic-derived-inverse-limit

For A, B on the same ringed pro-etale topos, define

$$A\widehat{\otimes}^L_\Lambda B := R\lim_{\leftarrow n}(A \otimes^L_\Lambda B \otimes^L_\Lambda P_n).$$

Thus $A\widehat{\otimes}^L\Lambda$ is the derived I-completion of A.

Lemma 4.10.33 (Finite reduction of completed tensor)**.** Lean: conditional-interface:lemma:weil-pbc-completed-tensor-reduction

Uses: def:weil-pbc-completed-tensor-product

For every a there is a functorial isomorphism

$$(A\widehat{\otimes}^L B) \otimes^L P_a \xrightarrow{\sim} A \otimes^L B \otimes^L P_a.$$

Proof. Uses: lemma:weil-pbc-adic-completion-reduction

The completed tensor is the adic completion of $A \otimes^L B$. Apply finite reduction of adic completion to that object and use associativity of derived tensor product. □

Lemma 4.10.34 (Completed projection-formula composite)**.** Lean: conditional-interface:lemma:weil-pbc-completed-projection-formula-composite

Uses: def:weil-pbc-completed-tensor-product

Let B be an object on Y_{proet} such that $B_n := B \otimes^L P_n$ is perfect for every n. Locally choose a strict-perfect package for each B_n. Then the ordered composite

$$\begin{aligned}(Rf_*E)\widehat{\otimes}^L B &= R\lim_{\leftarrow n}((Rf_*E) \otimes^L B_n)\\ &\xrightarrow{R\lim \pi_{E,B_n}} R\lim_{\leftarrow n} Rf_*(E \otimes^L Lf^*B_n)\\ &\xrightarrow{(Rf_*\text{-}R\lim)^{-1}} Rf_*R\lim_{\leftarrow n}(E \otimes^L Lf^*B \otimes^L P_n)\\ &= Rf_*(E\widehat{\otimes}^L Lf^*B)\end{aligned}$$

is an isomorphism, natural in E, B, compatible endomorphisms, and base-change squares. Its inverse is the reverse composite using direct-image/limit commutation and $R\lim \rho_{E,B_n}$.

Proof. Uses: lemma:weil-pbc-projection-formula-totalization,lemma:weil-pbc-projection-formula-tower-compatibility,lemma:weil-pbc-direct-image-r-lim

Naturality in the coefficient maps $B_{n+1} \to B_n$ makes the levelwise projection formulas a map of towers. Every π_{E,B_n} is an isomorphism locally by signed totalization, with inverse ρ_{E,B_n}; uniqueness of an inverse glues the local inverses. The second displayed arrow is an isomorphism because direct image commutes with the product-cone limit. The forward composite followed by the stated reverse composite is

$$R\lim(\rho_{E,B_n}\pi_{E,B_n}) = R\lim 1 = 1;$$

the reverse order uses $R\lim(\pi_{E,B_n}\rho_{E,B_n}) = 1$ and the two inverse equations for direct-image/limit commutation. Levelwise operator and base-change naturality passes through products and the product-cone fibre, proving the last assertion. □

Lemma 4.10.35 (Direct image commutes with derived completion)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-direct-image-derived-completion`

There is a functorial isomorphism

$$\widehat{Rf_*E} \xrightarrow{\sim} Rf_*\widehat{E}.$$

If E is intrinsically derived I-complete, then Rf_*E is intrinsically derived I-complete. Both statements commute with compatible endomorphisms and base change.

Proof. Uses: `lemma:weil-pbc-completed-projection-formula-composite,lemma:weil-pbc-derived-complete-reconstruction`

Apply the completed projection-formula composite with $B = \Lambda$; its finite reductions are P_n, which are perfect by the coefficient datum. This gives the displayed map and its explicit inverse. If E is complete, reconstruction identifies $\widehat{E}$ with E. Naturality of the composite identifies the resulting inverse $Rf_*E \to \widehat{Rf_*E}$ with the completion unit, so that unit is an isomorphism. The naturality assertion is inherited from the completed composite. □

Lemma 4.10.36 (Limits preserve compatible levelwise isomorphisms)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-R-lim-levelwise-isomorphism`

If $u_n : M_n \to N_n$ is a compatible map of towers and every u_n is an isomorphism, then $R\varprojlim u_n$ is an isomorphism.

Proof. Uses:

The maps $\prod_n u_n$ are isomorphisms and commute with $1 -$ shift. Taking fibres in the homotopy-limit triangles preserves the isomorphism. □

Lemma 4.10.37 (Complete cone detected by reductions)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-complete-cone-zero`

If C is derived complete and $C \otimes^L \Lambda/I^n = 0$ for every n, then $C = 0$.

Proof. Uses: `lemma:weil-pbc-derived-complete-reconstruction`

Reconstruction identifies C with the derived limit of its zero reductions; the product/equalizer model of the zero tower is zero. □

Lemma 4.10.38 (Perfectness of $\mathbf{Z}/\ell^n$)**.** *Uses:*

Lean: `conditional-interface:lemma:weil-pbc-zl-reduction-perfect`

For $\Lambda = \mathbf{Z}_\ell$ and $I = (\ell)$, every Λ/I^n is perfect over Λ.

Proof. Uses:

The two-term complex

$$[\mathbf{Z}_\ell \xrightarrow{\ell^n} \mathbf{Z}_\ell]$$

in degrees $-1, 0$ consists of finite free modules. Multiplication by ℓ^n is injective and its cokernel is $\mathbf{Z}/\ell^n$, so the complex is a finite projective resolution. □

Definition 4.10.39 (Canonical finite reduction tower over $\mathbf{Z}_\ell$)**.** Lean: `conditional-interface:def:weil-efi-coeff-zl-reduction-tower`

Uses: `def:weil-a2-derived-category,def:weil-pbc-complete-coefficient-datum`

Constructed by: `lemma:weil-pbc-zl-reduction-perfect`

Specialize the adic coefficient construction to $\Lambda = \mathbf{Z}_\ell$, $I = (\ell)$, and $P_a = \mathbf{Z}_\ell/\ell^a$. For $K \in \mathcal{D}(X, \Lambda)$ set

$$K_a := K \otimes^L_\Lambda P_a, \qquad q_{b,a} : K_b \longrightarrow K_a \quad (b \geq a)$$

using the quotient map $P_b \to P_a$.

Lemma 4.10.40 (Identity equation for the canonical reduction tower)**.** Lean: `conditional-interface:lemma:weil-efi-coeff-zl-reduction-identity`

Uses: `def:weil-efi-coeff-zl-reduction-tower`

For every a, $q_{a,a} = \mathrm{id}_{K_a}$.

Proof. Uses: `def:weil-efi-coeff-zl-reduction-tower`

The coefficient map $P_a \to P_a$ is the identity, and derived tensor sends it to the identity of $K \otimes^L_\Lambda P_a$. □

Lemma 4.10.41 (Composition equation for the canonical reduction tower)**.** Lean: `conditional-interface:lemma:weil-efi-coeff-zl-reduction-composition`

Uses: `def:weil-efi-coeff-zl-reduction-tower`

For $c \geq b \geq a$, $q_{c,a} = q_{b,a} q_{c,b}$.

Proof. Uses: `def:weil-efi-coeff-zl-reduction-tower`

The two coefficient maps are induced by the equal quotient homomorphisms $P_c \to P_a$. Functoriality of derived tensor preserves that equality. □

Definition 4.10.42 (Normalization comparison for the canonical reduction tower). Lean: `conditional-interface:def:weil-efi-coeff-zl-reduction-normalization`

Uses: `def:weil-efi-coeff-zl-reduction-tower`

For $b \geq a$, let

$$\nu_{b,a,K} : K_b \otimes^L_{P_b} P_a \longrightarrow K_a$$

be the derived-tensor associator followed by the multiplication isomorphism $P_b \otimes^L_{P_b} P_a \simeq P_a$.

Lemma 4.10.43 (The normalization comparison is invertible). Lean: `conditional-interface:lemma:weil-efi-coeff-zl-reduction-normalization-iso`

Uses: `def:weil-efi-coeff-zl-reduction-normalization`

The map $\nu_{b,a,K}$ is an isomorphism.

Proof. Uses: `lemma:weil-pbc-zl-reduction-perfect`

The derived associator is an isomorphism. The second factor is the standard derived unitor $P_b \otimes^L_{P_b} P_a \simeq P_a$; composing the two isomorphisms gives the claim. □

Lemma 4.10.44 (Cocycle equation for reduction normalization). Lean: `conditional-interface:lemma:weil-efi-coeff-zl-reduction-normalization-cocycle`

Uses: `def:weil-efi-coeff-zl-reduction-normalization`

For $c \geq b \geq a$, the direct normalization from level c to level a equals the composite through level b.

Proof. Uses: `lemma:weil-efi-coeff-zl-reduction-composition`

Expand the two maps into derived-tensor associators and quotient maps. The associator pentagon makes the reassociations equal, and the cited quotient composition equation identifies the remaining coefficient map. □

Definition 4.10.45 (Intrinsic constructible integral category candidate). Lean: `conditional-interface:def:weil-efi-coeff-intrinsic-constructible-zl-category`

Uses: `def:weil-a2-derived-category`, `def:weil-pbc-intrinsic-koszul-completion`, `def:weil-pbc-bounded-constructible-adic-tower`, `def:weil-efi-coeff-zl-reduction-tower`

Define $\mathscr{A}^{\mathrm{intr}}_{\mathbf{Z}_\ell}(X)$ to be the full subcategory of $\mathcal{D}(X, \mathbf{Z}_\ell)$ on those K for which the intrinsic Koszul completion unit is an isomorphism and the canonical finite reduction tower is uniformly bounded constructible. This definition has no K-flat or K-injective package parameter.

Lemma 4.10.46 (Adic completion lands in the intrinsic candidate). Lean: `conditional-interface:lemma:weil-efi-coeff-completion-in-intrinsic-category`

Uses: `def:weil-efi-coeff-intrinsic-constructible-zl-category`

If the canonical reduction tower of K is uniformly bounded constructible, then its explicit adic completion is an object of $\mathscr{A}^{\mathrm{intr}}_{\mathbf{Z}_\ell}(X)$.

Proof. Uses: `lemma:weil-pbc-bounded-constructible-reconstruction`, `lemma:weil-pbc-adic-completion-reduction`

The reconstruction lemma proves intrinsic completeness and preserves the common constructible bounds. The finite-reduction lemma identifies the reductions of the completion with the original tower, so the same constructibility witness applies. □

4.11 S10. Complete pro-etale proper base change

Definition 4.11.1 (Four finite-level rectangle corners). Lean: `conditional-interface:def:weil-pbc-level-map-corners`

Uses: `def:weil-pbc-cartesian-square`, `def:weil-pbc-complete-coefficient-datum`

For the proper Cartesian square □, put $E = L(g')^*K$ and, for every $n \geq 1$, define

$$A_n := Lg^*((Rf_*K) \otimes^L P_n), \quad B_n := Lg^*Rf_*K_n,$$
$$C_n := (Rf'_*E) \otimes^L P_n, \quad D_n := Rf'_*E_n,$$

where $E_n = L(g')^*K_n = E \otimes^L P_n$. The chosen Cartesian identification supplies $L(g')^*Lf^* = L(f')^*Lg^*$ throughout.

Definition 4.11.2 (The four finite-level arrows). Lean: `conditional-interface:def:weil-pbc-level-map-arrows`

Uses: `def:weil-pbc-level-map-corners`

Let $\xi_K : Lg^*Rf_*K \to Rf'_*E$ be the unit–counit exchange map

$$Lg^*Rf_*K \longrightarrow Rf'_*L(f')^*Lg^*Rf_*K \simeq Rf'_*L(g')^*Lf^*Rf_*K \xrightarrow{Rf'_*L(g')^*\epsilon_K} Rf'_*E.$$

Define the four edges

$$h_n := Lg^*(\pi_{f,K,n}) : A_n \longrightarrow B_n,$$
$$v_n := \xi_K \otimes^L 1_{P_n} : A_n \longrightarrow C_n,$$
$$h'_n := \pi_{f',E,n} : C_n \longrightarrow D_n,$$
$$v'_n := \xi_{K_n} : B_n \longrightarrow D_n.$$

For v_n use $Lg^*((Rf_*K) \otimes P_n) \simeq (Lg^*Rf_*K) \otimes P_n$. The transported S8 map is v'_n, because its construction was identified there with this unit–counit exchange map.

Lemma 4.11.3 (Well-definedness certificate for def:weil-pbc-level-map-arrows). *Lean:* `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-level-map-arrows`

Uses: `def:weil-pbc-level-map-arrows`

For $n \geq 1$, the tuple (h_n, v_n, h'_n, v'_n) belongs to

$$\mathrm{Hom}(A_n, B_n) \times \mathrm{Hom}(A_n, C_n) \times \mathrm{Hom}(C_n, D_n) \times \mathrm{Hom}(B_n, D_n),$$

and h_n, h'_n are isomorphisms. Under finite-level pro-etale transport, the fourth component is the unit–counit exchange map ξ_{K_n}.

Proof. Uses: `lemma:weil-pbc-perfect-projection-formula,prop:weil-pbc-proet-finite-level-transport`

Perfectness of P_n makes $\pi_{f,K,n} : (Rf_*K) \otimes^L P_n \to Rf_*K_n$ and $\pi_{f',E,n} : (Rf'_*E) \otimes^L P_n \to Rf'_*E_n$ isomorphisms. Derived pullback preserves the first isomorphism, giving the stated type and invertibility of h_n; the second gives the same facts for h'_n. Symmetric monoidality of derived pullback gives

$$Lg^*((Rf_*K) \otimes^L P_n) \simeq (Lg^*Rf_*K) \otimes^L P_n,$$

so tensoring ξ_K with 1_{P_n} has type $A_n \to C_n$. Proposition 4.9.67 transports the finite etale-origin exchange to a map $B_n \to D_n$ and identifies that map with the unit–counit formula ξ_{K_n}. These are exactly the four components asserted above; no rectangle or tower equation is used here. □

Lemma 4.11.4 (The finite-level exchange rectangle). *Lean:* `conditional-interface:lemma:weil-pbc-level-map-rectangle`

Uses: `def:weil-pbc-level-map-arrows`

For every n, the diagram

$$\begin{array}{ccc} A_n & \xrightarrow{h_n} & B_n \\ {\scriptstyle v_n}\downarrow & & \downarrow{\scriptstyle v'_n} \\ C_n & \xrightarrow{h'_n} & D_n \end{array}$$

commutes; equivalently, its two composites satisfy

$$v'_n h_n = h'_n v_n : A_n \longrightarrow D_n.$$

Proof. Uses: `lemma:weil-pbc-projection-formula-base-change-naturality`

This is the base-change naturality rectangle for the projection-formula map with coefficient P_n. To expose the equality, adjoint both composites to X'. Their common adjoint is

$$L(g')^*Lf^*Rf_*K \otimes^L P_n \xrightarrow{L(g')^*(\epsilon_K)\otimes 1} L(g')^*K \otimes^L P_n = E_n.$$

Thus every unit, counit, tensor associator, and Cartesian comparison occurs in the same order in the two adjoints. Equality after adjunction proves the displayed component equality. □

Lemma 4.11.5 (Transition compatibility of all four edges). *Lean:* `conditional-interface:lemma:weil-pbc-level-map-tower-compatibility`

Uses: `def:weil-pbc-level-map-arrows`

The four corner transitions from level $n+1$ to level n are

$$\begin{aligned} a_{n+1,n} &= Lg^*(1 \otimes q_n) : A_{n+1} \to A_n, \\ b_{n+1,n} &= Lg^*Rf_*(K_{n+1} \to K_n) : B_{n+1} \to B_n, \\ c_{n+1,n} &= 1 \otimes q_n : C_{n+1} \to C_n, \\ d_{n+1,n} &= Rf'_*L(g')^*(K_{n+1} \to K_n) : D_{n+1} \to D_n. \end{aligned}$$

Every edge is a map of towers:

$$\begin{aligned} b_{n+1,n}h_{n+1} &= h_n a_{n+1,n}, \quad & c_{n+1,n}v_{n+1} &= v_n a_{n+1,n}, \\ d_{n+1,n}h'_{n+1} &= h'_n c_{n+1,n}, \quad & d_{n+1,n}v'_{n+1} &= v'_n b_{n+1,n}. \end{aligned}$$

Proof. Uses: `lemma:weil-pbc-projection-formula-tower-compatibility`

The first and third equations are the coefficient-tower equations for the two projection formulas. The second follows by tensor naturality of ξ_K with respect to q_n. The last is naturality of the unit–counit exchange map with respect to $K_{n+1} \to K_n$. Expanded after adjunction, both sides of the last equation are the counit of K followed by the same coefficient quotient. Hence all four equations hold componentwise. □

Lemma 4.11.6 (Operator naturality of the finite rectangle). *Lean:* `conditional-interface:lemma:weil-pbc-level-map-operator-naturality`

Uses: `def:weil-pbc-level-map-arrows`

Let $\phi : K \to K$ be a tower-compatible endomorphism. Give A_n, B_n, C_n, D_n the endomorphisms induced by $Rf_*\phi$, $Rf_*\phi_n$, $Rf'_*L(g')^*\phi$, and $Rf'_*L(g')^*\phi_n$, respectively. Then h_n, v_n, h'_n, v'_n intertwine these endomorphisms, and the equality $v'_n h_n = h'_n v_n$ is an equality of operator-equivariant maps.

Proof. Uses: `lemma:weil-pbc-projection-formula-base-change-naturality`

The horizontal statements are operator naturality of the projection formula. The vertical statements follow by naturality of the unit and counit defining exchange. Composing either pair of equivariant edges gives an equivariant map; the preceding rectangle identifies those two maps. □

Definition 4.11.7 (Frobenius-equivariant square and Weil tower)**.** Lean: `conditional-interface:def:weil-pbc-frobenius-tower-datum`

Uses: `def:weil-pbc-level-map-corners,def:weil-pbc-level-map-arrows`

A Frobenius-equivariant datum for the proper Cartesian square $\square$ and the tower $K_\bullet$ includes displayed witnesses

$$\epsilon_X^{-1} L_n \xrightarrow{\sim} K_n, \qquad L_n \in D_c^+(X_{\mathrm{et}}, P_n),$$

compatible with the tower transitions. Pullback supplies the corresponding etale-origin witnesses for $E_n = L(g')^* K_n$. The datum then consists of automorphisms

$$\mathrm{Fr}_X, \quad \mathrm{Fr}_Y, \quad \mathrm{Fr}_{X'}, \quad \mathrm{Fr}_{Y'}$$

such that

$$f \,\mathrm{Fr}_X = \mathrm{Fr}_Y \, f, \qquad g \,\mathrm{Fr}_{Y'} = \mathrm{Fr}_Y \, g,$$
$$g' \,\mathrm{Fr}_{X'} = \mathrm{Fr}_X \, g', \quad f' \,\mathrm{Fr}_{X'} = \mathrm{Fr}_{Y'} \, f'.$$

The four resulting Frobenius faces are equipped with their Cartesian identifications, and these identifications make the resulting cube commute with the chosen Cartesian identification for $\square$. Put $\Phi_Z = L \,\mathrm{Fr}_Z^*$. Since every Fr_Z is an automorphism, every Φ_Z is an exact equivalence and preserves products and fibres.

The pullback faces give canonical isomorphisms

$$\kappa_g : \Phi_{Y'} L g^* \xrightarrow{\sim} L g^* \Phi_Y, \qquad \kappa_{g'} : \Phi_{X'} L(g')^* \xrightarrow{\sim} L(g')^* \Phi_X.$$

The proper Frobenius faces give finite-level exchange isomorphisms

$$\beta_{f,n} : \Phi_Y R f_* K_n \xrightarrow{\sim} R f_* \Phi_X K_n, \qquad \beta_{f',M} : \Phi_{Y'} R f'_* M \xrightarrow{\sim} R f'_* \Phi_{X'} M.$$

Finally the datum contains isomorphisms

$$\vartheta_n : \Phi_X K_n \xrightarrow{\sim} K_n$$

such that, for $r_{n+1,n} : K_{n+1} \to K_n$,

$$r_{n+1,n} \vartheta_{n+1} = \vartheta_n \Phi_X(r_{n+1,n}).$$

The constant coefficient quotients carry their identity Frobenius structures, and all displayed comparisons are required to respect their normalizations and cocycles.

Lemma 4.11.8 (Well-definedness certificate for def:weil-pbc-frobenius-tower-datum)**.** Lean: `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-frobenius-tower-datum`

Uses: `def:weil-pbc-frobenius-tower-datum`

The displayed Frobenius record has fields

$$\kappa_g : \Phi_{Y'} L g^* \xrightarrow{\sim} L g^* \Phi_Y, \qquad \kappa_{g'} : \Phi_{X'} L(g')^* \xrightarrow{\sim} L(g')^* \Phi_X,$$
$$\beta_{f,n} : \Phi_Y R f_* K_n \xrightarrow{\sim} R f_* \Phi_X K_n, \qquad \beta_{f',E_n} : \Phi_{Y'} R f'_* E_n \xrightarrow{\sim} R f'_* \Phi_{X'} E_n,$$
$$\vartheta_n : \Phi_X K_n \xrightarrow{\sim} K_n.$$

The two sides of the tower relation belong to $\mathrm{Hom}(\Phi_X K_{n+1}, K_n)$*, so the relation is a typed field of the record.*

Proof. Uses: `lemma:weil-pbc-derived-exchange-coherence,prop:weil-pbc-proet-finite-level-transport`

Pullback by the inverse of Fr_Z is a quasi-inverse to $\Phi_Z = L \,\mathrm{Fr}_Z^*$; hence Φ_Z is an exact equivalence. The two pullback faces in the Frobenius cube give κ_g and $\kappa_{g'}$ with the displayed endpoints. Applied to the two proper faces, finite-level pro-etale transport gives the displayed exchange isomorphisms because K_n and E_n carry their etale-origin witnesses. The identity and pasting equations required of these exchanges are those of Lemma 4.1.24.

Finally, $\Phi_X(r_{n+1,n})$ has source $\Phi_X K_{n+1}$ and target $\Phi_X K_n$. Thus $r_{n+1,n}\vartheta_{n+1}$ and $\vartheta_n \Phi_X(r_{n+1,n})$ have the common source and target stated in the lemma. This verifies the record typing without asserting a new Frobenius-equivariance theorem for the finite rectangle. $\square$

Definition 4.11.9 (Weil structures on the four finite corners)**.** Lean: `conditional-interface:def:weil-pbc-frobenius-corner-structures`

Uses: `def:weil-pbc-frobenius-tower-datum`

For $E_n = L(g')^* K_n$, define

$$\vartheta_{E,n} := L(g')^*(\vartheta_n)\, \kappa_{g',K_n} : \Phi_{X'} E_n \xrightarrow{\sim} E_n.$$

The Weil structures on B_n and D_n are the typed composites

$$\Theta_{B,n} : \Phi_{Y'} B_n \xrightarrow{\kappa_g} L g^* \Phi_Y R f_* K_n \xrightarrow{L g^* \beta_{f,n}} L g^* R f_* \Phi_X K_n \xrightarrow{L g^* R f_* \vartheta_n} B_n,$$
$$\Theta_{D,n} : \Phi_{Y'} D_n \xrightarrow{\beta_{f',E_n}} R f'_* \Phi_{X'} E_n \xrightarrow{R f'_* \vartheta_{E,n}} D_n.$$

Transport these structures across the two projection-formula isomorphisms:

$$\Theta_{A,n} := h_n^{-1} \Theta_{B,n} \Phi_{Y'}(h_n), \qquad \Theta_{C,n} := (h'_n)^{-1} \Theta_{D,n} \Phi_{Y'}(h'_n).$$

Thus every $\Theta_{Z,n}$ has the genuine Weil type $\Phi_{Y'} Z_n \xrightarrow{\sim} Z_n$ for $Z = A, B, C, D$.

Lemma 4.11.10 (Well-definedness certificate for def:weil-pbc-frobenius-corner-structures)**.** Lean: `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-frobenius-corner-structures`

Uses: `def:weil-pbc-frobenius-corner-structures`

The tuple $(\Theta_{A,n}, \Theta_{B,n}, \Theta_{C,n}, \Theta_{D,n})$ *belongs to*

$$\mathrm{Iso}(\Phi_{Y'} A_n, A_n) \times \mathrm{Iso}(\Phi_{Y'} B_n, B_n) \times \mathrm{Iso}(\Phi_{Y'} C_n, C_n) \times \mathrm{Iso}(\Phi_{Y'} D_n, D_n).$$

This assertion records the four corner types and does not assert edge equivariance.

Proof. Uses: `lemma:weil-sf-well-defined-def-weil-pbc-level-map-arrows,lemma:weil-sf-well-defined-def-weil-pbc-frobenius-tower-datum,lemma:weil-pbc-perfect-projection-formula`

The certified tower datum makes $\kappa_g, \kappa_{g'}, \beta_{f,n}, \beta_{f',E_n}$ and ϑ_n isomorphisms with the displayed endpoints. Their ordered composites therefore give $\vartheta_{E,n}$, $\Theta_{B,n}$ and $\Theta_{D,n}$ with the asserted types. The level-arrow certificate and the perfect projection formula make h_n and h'_n invertible. Conjugating $\Theta_{B,n}$ by h_n and $\Theta_{D,n}$ by h'_n gives the asserted types for $\Theta_{A,n}$ and $\Theta_{C,n}$. Edge equivariance remains the next lemma. □

Lemma 4.11.11 (The finite rectangle is a Weil rectangle). *Lean:* `conditional-interface:lemma:weil-pbc-frobenius-level-rectangle`

Uses: `def:weil-pbc-frobenius-corner-structures`

All four finite-level edges are Weil-equivariant:

$$h_n\Theta_{A,n} = \Theta_{B,n}\Phi_{Y'}(h_n), \quad v_n\Theta_{A,n} = \Theta_{C,n}\Phi_{Y'}(v_n),$$
$$h'_n\Theta_{C,n} = \Theta_{D,n}\Phi_{Y'}(h'_n), \quad v'_n\Theta_{B,n} = \Theta_{D,n}\Phi_{Y'}(v'_n).$$

In particular the two explicitly typed composites

$$\mathcal{P}_n := \Phi_{Y'}A_n \xrightarrow{\Theta_{A,n}} A_n \xrightarrow{v'_n h_n} D_n,$$
$$\mathcal{Q}_n := \Phi_{Y'}A_n \xrightarrow{\Phi_{Y'}(v'_n h_n)} \Phi_{Y'}D_n \xrightarrow{\Theta_{D,n}} D_n$$

are equal. The same common map is obtained after replacing $v'_n h_n$ by $h'_n v_n$.

Proof. Uses: `lemma:weil-pbc-level-map-rectangle,lemma:weil-pbc-derived-exchange-coherence`

The equations for h_n and h'_n are the two transport definitions. For v'_n, expand the upper-right and lower-left routes in the Frobenius cube. Pasting coherence for exchange and naturality in ϑ_n identify both with

$$\Phi_{Y'}Lg^*Rf_*K_n \xrightarrow{\kappa_g} Lg^*\Phi_Y Rf_*K_n \xrightarrow{Lg^*\beta_{f,n}} Lg^*Rf_*\Phi_X K_n$$
$$\xrightarrow{\xi_{\Phi_X K_n}} Rf'_*L(g')^*\Phi_X K_n \xrightarrow{Rf'_*L(g')^*\vartheta_n} D_n.$$

This proves the fourth equation with every pullback and exchange map typed. For the remaining vertical edge, use the ordinary rectangle and the three equations already proved:

$$h'_n v_n \Theta_{A,n} = v'_n h_n \Theta_{A,n} = v'_n \Theta_{B,n}\Phi_{Y'}(h_n)$$
$$= \Theta_{D,n}\Phi_{Y'}(v'_n h_n) = h'_n \Theta_{C,n}\Phi_{Y'}(v_n).$$

Cancel the isomorphism h'_n. Composing either pair of edge equations and using $v'_n h_n = h'_n v_n$ gives $\mathcal{P}_n = \mathcal{Q}_n$ and the final assertion. □

Lemma 4.11.12 (Frobenius compatibility with tower transitions). *Lean:* `conditional-interface:lemma:weil-pbc-frobenius-tower-compatibility`

Uses: `def:weil-pbc-frobenius-corner-structures`

For the four transitions $a_{n+1,n}, b_{n+1,n}, c_{n+1,n}, d_{n+1,n}$,

$$a_{n+1,n}\Theta_{A,n+1} = \Theta_{A,n}\Phi_{Y'}(a_{n+1,n}),$$
$$b_{n+1,n}\Theta_{B,n+1} = \Theta_{B,n}\Phi_{Y'}(b_{n+1,n}),$$
$$c_{n+1,n}\Theta_{C,n+1} = \Theta_{C,n}\Phi_{Y'}(c_{n+1,n}),$$
$$d_{n+1,n}\Theta_{D,n+1} = \Theta_{D,n}\Phi_{Y'}(d_{n+1,n}).$$

Hence the four corner Weil structures and all four edges are maps of towers.

Proof. Uses: `lemma:weil-pbc-frobenius-level-rectangle,lemma:weil-pbc-level-map-tower-compatibility`

The equation for B follows by naturality of κ_g and β_f and the defining equation $r_{n+1,n}\vartheta_{n+1} = \vartheta_n\Phi_X(r_{n+1,n})$. The same argument for $\kappa_{g'}$, $\beta_{f'}$, and $\vartheta_{E,n}$ proves the equation for D. For A, combine $b_{n+1,n}h_{n+1} = h_n a_{n+1,n}$ with the transport formula for Θ_A and the equation for B; cancelling h_n gives the first displayed equation. The equation for C follows identically from $d_{n+1,n}h'_{n+1} = h'_n c_{n+1,n}$ and the equation for D. The finite Weil rectangle and the four transition equations now say precisely that every edge is a morphism of Weil towers. □

Lemma 4.11.13 (The rectangle passes to the derived inverse limit). *Lean:* `conditional-interface:lemma:weil-pbc-level-map-derived-limit`

Uses: `def:weil-pbc-adic-derived-inverse-limit`

Write $A_\infty = R\varprojlim A_n$ and likewise for the other three towers, and write H, V, H', V' for the limits of h, v, h', v'. Then

$$V'H = H'V : A_\infty \longrightarrow D_\infty.$$

For every n, both maps have the same nth component

$$\mathrm{pr}_{D,n}\, V'H = v'_n h_n \,\mathrm{pr}_{A,n} = h'_n v_n \,\mathrm{pr}_{A,n} = \mathrm{pr}_{D,n}\, H'V.$$

The equality is natural for compatible tower endomorphisms.

Proof. Uses: `lemma:weil-pbc-level-map-rectangle,lemma:weil-pbc-level-map-tower-compatibility,lemma:weil-pbc-level-map-operator-naturality`

Transition compatibility makes each family a map between the product-cone diagrams. On both product terms, the nth coordinate of $V'H$ is $v'_n h_n$, while that of $H'V$ is $h'_n v_n$; these are equal by the finite-level rectangle. Hence the product maps, their $1 -$ shift squares, and the induced maps on the fibres are equal. This proves both the limit equality and the three displayed component equations. Operator naturality holds on every product coordinate, so it also holds on the fibre. □

Lemma 4.11.14 (Frobenius structures pass through the product cone)**.** Lean: `conditional-interface:lemma:weil-pbc-frobenius-product-cone-limit`

Uses: `def:weil-pbc-adic-derived-inverse-limit`

For $Z = A, B, C, D$, the product-cone limit carries the Weil structure

$$\Theta_{Z,\infty} : \Phi_{Y'} Z_\infty \xrightarrow{\sim} R\varprojlim_n \Phi_{Y'} Z_n \xrightarrow{R\varprojlim_n \Theta_{Z,n}} Z_\infty.$$

The first arrow is the canonical comparison, which is an isomorphism because $\Phi_{Y'}$ is an equivalence and therefore preserves the two products and the fibre in the product-cone model. Each of H, V, H', V' is Weil-equivariant. For the common limit map $W := V'H = H'V : A_\infty \to D_\infty$, the two typed composites

$$\Phi_{Y'} A_\infty \xrightarrow{\Theta_{A,\infty}} A_\infty \xrightarrow{W} D_\infty,$$
$$\Phi_{Y'} A_\infty \xrightarrow{\Phi_{Y'}(W)} \Phi_{Y'} D_\infty \xrightarrow{\Theta_{D,\infty}} D_\infty$$

are equal.

Proof. Uses: `lemma:weil-pbc-frobenius-level-rectangle`, `lemma:weil-pbc-frobenius-tower-compatibility`, `lemma:weil-pbc-level-map-derived-limit`

Apply $\Phi_{Y'}$ to the product-cone triangle for each tower. Preservation of products identifies the two product terms with the products of $\Phi_{Y'} Z_n$, and transition compatibility identifies the two $1-$shift arrows. Thus the displayed comparison is an isomorphism of fibres. On its nth product coordinate, equivariance of H, V, H', V' is exactly one of the four finite edge equations. Hence the product maps and their maps on fibres are equivariant. Composing the equations for H, V' or for V, H' gives

$$W\Theta_{A,\infty} = \Theta_{D,\infty}\Phi_{Y'}(W).$$

The equality $V'H = H'V$ is already an equality of maps of product-cone diagrams, so both factorizations give the same typed Weil morphism. □

Definition 4.11.15 (Completed base-change map)**.** Lean: `conditional-interface:def:weil-pbc-completed-base-change-map`

Uses: `def:weil-pbc-completed-pullback`

Let K be derived complete and let every K_n have a displayed witness $\epsilon_X^{-1} L_n \simeq K_n$ with $L_n \in D_c^+(X_{\mathrm{et}}, \Lambda/I^n)$. Define

$$\mathrm{Ex}^{\mathrm{comp}}_{\square,K} : Lg^*_{\mathrm{comp}} Rf_* K \longrightarrow Rf'_* L(g')^*_{\mathrm{comp}} K$$

to be the ordered composite

$$\begin{aligned} Lg^*_{\mathrm{comp}} Rf_* K &\xrightarrow{\sim} R\varprojlim_n Lg^* Rf_* K_n \\ &\xrightarrow{R\varprojlim_n \mathrm{Ex}_{\square,K_n}} R\varprojlim_n Rf'_* L(g')^* K_n \\ &\xrightarrow{\sim} Rf'_* R\varprojlim_n L(g')^* K_n = Rf'_* L(g')^*_{\mathrm{comp}} K. \end{aligned}$$

In the notation above, the first two arrows are H followed by V'. The first isomorphism uses reconstruction and the perfect projection formula; the second displayed isomorphism is the inverse of direct-image/limit commutation. This definition fills the map-construction omission in Tag `09C9`.

Lemma 4.11.16 (Well-definedness certificate for def:weil-pbc-completed-base-change-map)**.** Lean: `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-completed-base-change-map`

Uses: `def:weil-pbc-completed-base-change-map`

With $S = Lg^*_{\mathrm{comp}} Rf_* K$ and $T = Rf'_* L(g')^*_{\mathrm{comp}} K$, the ordered expression

$$S \xrightarrow{s} A_\infty \xrightarrow{H} B_\infty \xrightarrow{V'} D_\infty \xrightarrow{t} T$$

is a morphism $S \to T$ and is the map $\mathrm{Ex}^{\mathrm{comp}}_{\square,K}$ of Definition 4.11.15. The endpoint comparisons s, t and the limit projection-formula map H are isomorphisms.

Proof. Uses: `lemma:weil-pbc-perfect-projection-formula`, `lemma:weil-pbc-level-map-derived-limit`, `lemma:weil-pbc-direct-image-r-lim`

Symmetric monoidality of Lg^* identifies the defining completed-pullback tower with the A_n tower. The perfect projection formula then gives the map $H : A_\infty \to B_\infty$, while finite exchange gives the compatible tower map $V' : B_\infty \to D_\infty$ recorded by Lemma 4.11.13. Direct image commutation with the product-cone limit gives $t : D_\infty \xrightarrow{\sim} T$. The target of each factor is the source of the next one.

At level n, h_n is invertible by the perfect projection formula. From $b_{n+1,n} h_{n+1} = h_n a_{n+1,n}$, multiplication by the two inverses gives $a_{n+1,n} h_{n+1}^{-1} = h_n^{-1} b_{n+1,n}$. Hence the inverse family is a map of towers, and its derived limit is inverse to H. Thus s, H, t are isomorphisms. Expanding the ordered definition gives the displayed composite. No invertibility or Frobenius assertion for the whole composite is made in this certificate. □

Definition 4.11.17 (Weil structures on the completed source and target)**.** Lean: `conditional-interface:def:weil-pbc-completed-frobenius-structures`

Uses: `def:weil-pbc-completed-base-change-map`

Write

$$S = Lg^*_{\mathrm{comp}} Rf_* K, \qquad T = Rf'_* L(g')^*_{\mathrm{comp}} K.$$

Let $s : S \xrightarrow{\sim} A_\infty$ be the pullback/tensor comparison in the first part of the completed construction, and let $t : D_\infty \xrightarrow{\sim} T$ be the inverse of direct-image/limit commutation. Define genuine completed Weil structures by

$$\Theta_S := s^{-1}\Theta_{A,\infty}\Phi_{Y'}(s) : \Phi_{Y'}S \xrightarrow{\sim} S,$$

$$\Theta_T := t\Theta_{D,\infty}\Phi_{Y'}(t^{-1}) : \Phi_{Y'}T \xrightarrow{\sim} T.$$

Under these identifications the completed base-change map is the ordered composite

$$\mathrm{Ex}^{\mathrm{comp}}_{\square,K} = tWs, \qquad W = V'H = H'V.$$

Lemma 4.11.18 (Well-definedness certificate for def:weil-pbc-completed-frobenius-structures)**.** *Lean:* `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-completed-frobenius-structures`

Uses: `def:weil-pbc-completed-frobenius-structures`

The pair (Θ_S, Θ_T) *belongs to*

$$\mathrm{Iso}(\Phi_{Y'}S, S) \times \mathrm{Iso}(\Phi_{Y'}T, T),$$

and tWs *belongs to* $\mathrm{Hom}(S, T)$*. This certificate records the completed types but not the equivariance equation for* tWs*.*

Proof. Uses: `lemma:weil-sf-well-defined-def-weil-pbc-completed-base-change-map,lemma:weil-pbc-frobenius-product-cone-limit`

The product-cone lemma supplies isomorphisms $\Theta_{A,\infty} : \Phi_{Y'}A_\infty \to A_\infty$ and $\Theta_{D,\infty} : \Phi_{Y'}D_\infty \to D_\infty$, together with $W : A_\infty \to D_\infty$. The preceding certificate gives the isomorphisms $s : S \to A_\infty$ and $t : D_\infty \to T$. Conjugating the two limit structures by s and t yields the displayed pair of isomorphism types, and the endpoints of s, W, t give $tWs : S \to T$. The next lemma proves the equivariance equation. □

Lemma 4.11.19 (Completed proper base change is Frobenius-equivariant)**.** *Lean:* `conditional-interface:lemma:weil-pbc-completed-frobenius-equivariance`

Uses: `def:weil-pbc-completed-frobenius-structures`

For every Frobenius-equivariant square and Weil tower datum,

$$\mathrm{Ex}^{\mathrm{comp}}_{\square,K}\,\Theta_S = \Theta_T\Phi_{Y'}(\mathrm{Ex}^{\mathrm{comp}}_{\square,K}) : \Phi_{Y'}S \longrightarrow T.$$

If all four Frobenius automorphisms and all pullback/exchange comparisons are identities, then ϑ_n *is an ordinary compatible endomorphism and this equation specializes exactly to the operator naturality of the finite rectangle and its product-cone limit. For nonidentity Frobenius, no identification of* $\Phi_X K_n \to K_n$ *with an endomorphism* $K_n \to K_n$ *is made.*

Proof. Uses: `lemma:weil-pbc-frobenius-product-cone-limit,lemma:weil-pbc-level-map-operator-naturality`

Using the displayed definitions and cancelling $s^{-1}s$ and $t^{-1}t$, the left side is

$$tW\Theta_{A,\infty}\Phi_{Y'}(s),$$

whereas the right side is

$$t\Theta_{D,\infty}\Phi_{Y'}(W)\Phi_{Y'}(s).$$

They are equal by the product-cone Frobenius equation. When every Φ_Z is the identity, the four formulas defining $\Theta_{Z,n}$ become the endomorphisms induced from ϑ_n and the equation just proved is the fibrewise limit of the equations in operator naturality. This proves the specialization statement without using it to type the genuine Frobenius case. □

Definition 4.11.20 (The four edges of the completed reduction square)**.** Lean: `conditional-interface:def:weil-pbc-completed-reduction-square`

Uses: `def:weil-pbc-completed-base-change-map`

Put

$$S = Lg^*_{\mathrm{comp}} Rf_* K, \qquad T = Rf'_* L(g')^*_{\mathrm{comp}} K.$$

Let

$$\sigma_n : S \otimes^L P_n \xrightarrow{\sim} A_n$$

be finite reduction of completed pullback. For $E = L(g')^*K$, let

$$\tau_n : T \otimes^L P_n \xrightarrow{\sim} C_n$$

be the inverse of $\widehat{Rf'_*E} \xrightarrow{\sim} Rf'_*\widehat{E} = T$, tensored with P_n, followed by finite reduction of $\widehat{Rf'_*E}$. Define

$$\alpha_n = h_n\sigma_n : S \otimes^L P_n \to B_n, \qquad \delta_n = h'_n\tau_n : T \otimes^L P_n \to D_n,$$

and let

$$\bar{\xi}_n = \mathrm{Ex}^{\mathrm{comp}}_{\square,K} \otimes^L 1_{P_n} : S \otimes^L P_n \longrightarrow T \otimes^L P_n.$$

Thus every arrow in the completed reduction square is named, typed, and expressed as an ordered composite of reduction, direct-image, projection- formula, or base-change maps.

Lemma 4.11.21 (Well-definedness certificate for def:weil-pbc-completed-reduction-square)**.** *Lean:* `conditional-interface:lemma:weil-sf-well-defined-def-weil-pbc-completed-reduction-square`

Uses: `def:weil-pbc-completed-reduction-square`

For $n \geq 1$, $\sigma_n : S \otimes^L P_n \to A_n$ *and* $\tau_n : T \otimes^L P_n \to C_n$ *are isomorphisms, while*

$$(\alpha_n, \delta_n, \bar{\xi}_n) \in \mathrm{Hom}(S \otimes^L P_n, B_n) \times \mathrm{Hom}(T \otimes^L P_n, D_n) \times \mathrm{Hom}(S \otimes^L P_n, T \otimes^L P_n).$$

The first two components of this triple are isomorphisms. No commutativity claim for the reduction square is part of this certificate.

Proof. Uses: `lemma:weil-sf-well-defined-def-weil-pbc-level-map-arrows,lemma:weil-sf-well-defined-def-weil-pbc-completed-base-change-map,lemma:weil-pbc-completed-pullback-reduction,lemma:weil-pbc-direct-image-derived-completion,lemma:weil-pbc-completed-tensor-reduction`

Finite reduction of completed pullback gives σ_n with the stated source and target. Direct-image completion gives $\widehat{Rf'_*E} \xrightarrow{\sim} Rf'_*(\widehat{E}) = T$; tensoring its inverse with P_n and applying finite reduction of completed tensor gives $\tau_n : T \otimes^L P_n \xrightarrow{\sim} C_n$.

The level-arrow certificate makes $h_n : A_n \to B_n$ and $h'_n : C_n \to D_n$ isomorphisms. Hence $\alpha_n = h_n\sigma_n$ and $\delta_n = h'_n\tau_n$ have the asserted types and are invertible. Finally the completed-map certificate gives $\mathrm{Ex}^{\mathrm{comp}}_{\square,K} : S \to T$, so derived tensor with 1_{P_n} gives the asserted type of $\overline{\xi}_n$. The next lemma contains the square's commutativity and its compatibility fields. □

Lemma 4.11.22 (Finite reduction of the completed map). *Lean:* `conditional-interface:lemma:weil-pbc-completed-map-reduction`

Uses: `def:weil-pbc-completed-reduction-square, def:weil-pbc-completed-frobenius-structures`

For every n, the diagram

$$\begin{array}{ccc} S \otimes^L P_n & \xrightarrow{\overline{\xi}_n} & T \otimes^L P_n \\ {\scriptstyle\alpha_n}\downarrow{\scriptstyle\sim} & & \downarrow{\scriptstyle\delta_n}\ {\scriptstyle\sim} \\ B_n & \xrightarrow{v'_n} & D_n \end{array}$$

commutes. Hence reduction of $\mathrm{Ex}^{\mathrm{comp}}_{\square,K}$ modulo I^n, conjugated by the displayed canonical reduction isomorphisms, is exactly the S8 transported finite-level exchange map $v'_n = \mathrm{Ex}_{\square,K_n}$. The squares for $n+1 \to n$ commute, and the assertion is natural for compatible tower endomorphisms. Under a Frobenius-equivariant datum, use $\Phi_{Y'}P_n = P_n$ to give the two reductions their induced Weil structures. Then

$$\alpha_n(\Theta_S \otimes 1) = \Theta_{B,n}\Phi_{Y'}(\alpha_n),$$
$$\delta_n(\Theta_T \otimes 1) = \Theta_{D,n}\Phi_{Y'}(\delta_n),$$

and the displayed reduction square is a square of Weil objects.

Proof. Uses: `lemma:weil-pbc-level-map-derived-limit,lemma:weil-pbc-frobenius-tower-compatibility,lemma:weil-pbc-completed-frobenius-equivariance`

Under $S = A_\infty$ and the direct-image completion identification $T \simeq C_\infty$, the completed exchange map is both limit composites

$$V'H = H'V : A_\infty \longrightarrow D_\infty,$$

followed by the inverse of direct-image/limit commutation. More explicitly, if $d : T \to D_\infty$ is direct-image/limit commutation and $c : C_\infty \to T$ is direct-image completion, then its construction gives $dc = H'$. Functorial finite reduction and the component equations of the limit rectangle give

$$\tau_n\overline{\xi}_n = v_n\sigma_n.$$

Consequently the two composites around the completed reduction square are

$$\delta_n\overline{\xi}_n = h'_n\tau_n\overline{\xi}_n = h'_nv_n\sigma_n$$

and

$$v'_n\alpha_n = v'_nh_n\sigma_n.$$

They are equal term by term because $h'_nv_n = v'_nh_n$ in the finite-level rectangle. Naturality of finite reduction makes σ_n and τ_n commute with $n+1 \to n$, and functoriality of tensor reduction does the same for $\overline{\xi}_n$. Together with the four transition equations, this shows that the completed squares commute with $K_{n+1} \to K_n$ on every corner. The limit construction and all four reduction edges are operator-equivariant, so the conjugated equality is operator-natural as well.

For Frobenius, the definition of Θ_S transports $\Theta_{A,\infty}$ across s, and the nth product-cone component followed by h_n is α_n. This gives the first displayed Weil equation. The definition of Θ_T transports $\Theta_{D,\infty}$ across t, and its nth component is δ_n, giving the second. The completed Frobenius equation for $\overline{\xi}_n$, together with finite-level equivariance of v'_n, now says that every arrow of the reduction square is a Weil morphism. □

Lemma 4.11.23 (Completed exchange coherence). *Lean:* `conditional-interface:lemma:weil-pbc-completed-exchange-coherence`

Uses: `def:weil-pbc-completed-base-change-map`

The completed exchange map is natural in K, is the identity for identity squares, and commutes with horizontal and vertical pasting.

Proof. Uses: `lemma:weil-pbc-completed-map-reduction,lemma:weil-pbc-r-lim-levelwise-isomorphism`

At every finite level these equations are the S0/S8 equations. They commute with the tower transitions by naturality of coefficient reduction. Apply $R\varprojlim$ to each equality and conjugate by the same reconstruction, projection-formula, and direct-image/limit isomorphisms used in the definition. The resulting equations are precisely the completed equations. □

Proposition 4.11.24 (Complete pro-etale proper base change). *Lean:* `conditional-interface:prop:weil-proet-complete-proper-base-change`

Uses: `def:weil-pbc-complete-coefficient-datum, def:weil-pbc-completed-base-change-map`

Let $f : X \to Y$ be proper and let $\square$ be any base change. Suppose $K \in D(X_{\mathrm{proet}}, \Lambda)$ is derived I-complete, each K_n is equipped with a displayed witness $\epsilon_X^{-1}L_n \simeq K_n$ for some $L_n \in D^+_c(X_{\mathrm{et}}, \Lambda/I^n)$, and every Λ/I^n is perfect. Then

$$Lg^*_{\mathrm{comp}}Rf_*K \xrightarrow{\sim} Rf'_*L(g')^*_{\mathrm{comp}}K.$$

The isomorphism is natural and satisfies identity and pasting coherence. If the square and the reductions carry a Frobenius-equivariant square and Weil tower datum, this is an isomorphism of the completed Weil objects (S, Θ_S) and (T, Θ_T).

Proof. Uses: def:weil-pbc-completed-base-change-map, prop:weil-pbc-proet-finite-level-transport, lemma:weil-pbc-r-lim-levelwise-isomorphism, lemma:weil-pbc-completed-exchange-coherence, lemma:weil-pbc-completed-frobenius-equivariance

Every finite-level middle arrow in Definition 4.11.15 is an isomorphism by S8. The derived limit preserves this compatible levelwise isomorphism. The outer arrows in the defining composite are reconstruction, perfect projection formula, and direct-image/limit isomorphisms. Hence the completed map is an isomorphism. Coherence is the preceding lemma. This proves the mathematical content located at Tag 09C9 without using that theorem as a leaf. The completed Frobenius-equivariance lemma gives the last assertion with the genuine pullback type $\Phi_{Y'} S \to T$. □

Definition 4.11.25 (A2 constructible rebinding certificate). Lean: conditional-interface:def:weil-pbc-a2-constructible-rebind

Uses: def:weil-a2-dbc-zl, def:weil-a2-dbc-torsion, def:weil-a2-proper-complete-base-change-datum, def:weil-a2-finite-level-etale-origin-realization, def:weil-a2-torsion-constructibility-witness, def:weil-a2-constructible-stratification-witness, def:weil-a1-torsion-derived-category, def:weil-a1-rfshriek-selected, def:weil-pbc-level-map-corners, def:weil-pbc-completed-reduction-square

Specialize A2 to $\Lambda = \mathbf{Z}_\ell$, $I = (\ell)$, and $P_n = \Lambda/\ell^n$. For a scheme Z write

$$\mathsf{C}(Z) := D_c^b(Z_{\mathrm{proet}}, \mathbf{Z}_\ell), \qquad \mathsf{C}_n(Z) := D_c^b(Z_{\mathrm{proet}}, \mathbf{Z}/\ell^n)$$

for the full subcategories defined, respectively, by def:weil-a2-dbc-zl and def:weil-a2-dbc-torsion.

Fix a proper Cartesian square □ as in S10. An A2 constructible rebinding certificate for $K \in \mathsf{C}(X)$ consists of the following displayed data.

1. An A2 proper complete-base-change datum, hence compatible etale-origin objects
$$L_n \in D_c^b(X_{\mathrm{et}}, \mathbf{Z}/\ell^n), \qquad \alpha_n : \epsilon_X^{-1} L_n \xrightarrow{\sim} K_n,$$
including their transition equations. The superscript b is part of the certificate: one interval $[d_0, d_1]$ works for every L_n and every K_n.
2. On each of X, X', Y, Y', one finite stratification, each stratum having finitely many connected components, and A2 finite-torsion constructibility witnesses, uniform in n, for every finite object occurring in S7–S10. On Y' these are, in particular, the four objects A_n, B_n, C_n, D_n of Definition 4.11.1; their witnesses use one common interval and one common refinement.
3. The finite exceptional-direct-image values for the identity compactifications of the proper maps,
$$Rf_!^{(1,f)} K_n \in \mathsf{C}_n(Y), \qquad Rf'^{(1,f')}_! E_n \in \mathsf{C}_n(Y'),$$
with the A2 torsion witnesses retained by the typed A1 input and with their transition squares. The certificate records their displayed identifications with $Rf_* K_n$ and $Rf'_* E_n$.
4. A2 constructible-stratification witnesses for
$$S(K) := Lg^*_{\mathrm{comp}} Rf_* K, \qquad T(K) := Rf'_* L(g')^*_{\mathrm{comp}} K$$
in $\mathsf{C}(Y')$. Their finite reductions are required to be the objects identified by the maps α_n and δ_n of Definition 4.11.20.

Let $\mathsf{C}_\square(X)$ be the full subcategory of $\mathsf{C}(X)$ on objects for which the type of such certificates is nonempty. Certificates are membership witnesses only; morphisms in $\mathsf{C}_\square(X)$ are the ambient A2 morphisms, so no selected witness enters their equality.

Lemma 4.11.26 (Reduction and completed-tensor transport to A2). Lean: conditional-interface:lemma:weil-pbc-a2-reduction-tensor-transport

Uses: def:weil-pbc-a2-constructible-rebind, def:weil-a2-finite-reduction, def:weil-a2-reduction-transition, def:weil-a2-completed-tensor, def:weil-pbc-completed-coefficient-datum, def:weil-pbc-completed-tensor-product

For every K and n, there is a canonical natural isomorphism

$$\chi_{K,n} : \mathrm{Red}_n^{\mathrm{A2}}(K) \xrightarrow{\sim} K \otimes^L_{\mathbf{Z}_\ell} \mathbf{Z}/\ell^n = K_n.$$

It intertwines the A2 transition with the quotient transition $q_{n+1,n}$ of S9. For complete A, B there is likewise a canonical natural isomorphism

$$\omega_{A,B} : A \widehat{\otimes}^{L,\mathrm{A2}} B \xrightarrow{\sim} A \widehat{\otimes}^{L,\mathrm{S9}} B,$$

and its n-th reduction is the identity of $A \otimes^L B \otimes^L \mathbf{Z}/\ell^n$ under χ. Both isomorphisms are independent of the K-flat, quotient-complex, and completion aliases.

Proof. Uses: lemma:weil-a2-finite-reduction-package-independent, prop:weil-a2-finite-reduction-functor, lemma:weil-pbc-zl-reduction-perfect, lemma:weil-pbc-completed-tensor-reduction

A2 represents $\mathbf{Z}/\ell^n$ by the two-term complex

$$L_n = [\mathbf{Z}_\ell \xrightarrow{\ell^n} \mathbf{Z}_\ell].$$

The augmentation $L_n \to \mathbf{Z}/\ell^n$ is a quasi-isomorphism and S9 uses the same complex as its strict perfect resolution. Tensor it with the A2 K-flat model of K and localize. This gives $\chi_{K,n}$. If the K-flat model is changed, the two maps are obtained from a common comparison roof; A2 package independence says that they are equal. Thus χ is canonical, and A2 functoriality makes it natural.

The A2 transition is represented by multiplication by ℓ in degree -1 and the identity in degree 0. After the augmentation it is the quotient map $\mathbf{Z}/\ell^{n+1} \to \mathbf{Z}/\ell^n$. Therefore

$$\chi_{K,n}\, r^{\mathrm{A2}}_{n+1,n} = q_{n+1,n}\, \chi_{K,n+1}.$$

Finally, A2 completes the localized K-flat tensor of A and B, whereas S9 takes the product-cone limit of its reductions. Apply χ levelwise to these two identical towers. The transition equation just proved gives a map of product-cone models and hence $\omega_{A,B}$. Reduction of completion and the S9 completed-tensor reduction identify its n-th component with the identity of the displayed triple tensor. Common K-flat roofs give identical level maps, and equality into the two products is componentwise; consequently the induced cone map is independent of every alias. □

Lemma 4.11.27 (Site-comparison and exceptional-image transport to A2). *Lean:* `conditional-interface:lemma:weil-pbc-a2-comparison-rfshriek-transport`

Uses: `def:weil-pbc-a2-constructible-rebind`, `def:weil-a2-etale-origin-datum`, `def:weil-a2-etale-proetale-pullback-comparison`, `def:weil-a2-etale-proetale-direct-image-comparison`, `def:weil-a2-six-operation-coefficient-target`, `def:weil-a1-torsion-derived-category`, `def:weil-a1-rfshriek-selected`, `def:weil-pbc-epsilon-comparison`, `def:weil-pbc-projection-compactification`

Under the identity-on-objects site functor of S8, the A2 notation ν_Z^* is the functor ϵ_Z^{-1}. The A2 pullback comparison c_g, direct-image comparison d_f, and etale-origin isomorphism therefore transport to the S8 maps with the same components. Moreover the A1 typed input category $\mathcal{D}_n(Z)$ transports by the identity on objects and morphisms to $\mathsf{C}_n(Z)$, and on a proper map

$$Rf_!^{(1,f)} = Rf_* : \mathsf{C}_n(X) \longrightarrow \mathsf{C}_n(Y).$$

Thus the $Rf_!$ field of the A2 six-operation target has exactly the displayed source and target on $\mathsf{C}_\square(X)$. These transports are independent of whether the A1, A2, S8, or S9 alias is used.

Proof. Uses: `lemma:weil-pbc-epsilon-pullback`, `lemma:weil-pbc-epsilon-direct-image`, `lemma:weil-pbc-projection-formula-identity-compactification`

Both ν_Z and ϵ_Z are the geometric morphism induced by the functor that sends an etale object to the same object of the pro-etale site. Their inverse images are the same sheafified left Kan extension, so the identity of the indexing categories identifies them. Expanding c_g gives the S8 pullback square, and expanding d_f gives the S8 unit followed by the same derived direct image. The component formulas and their naturality squares are therefore literally equal, not merely abstractly isomorphic.

A1 denoted by $\mathcal{D}_n(Z)$ the typed finite torsion constructible category whose construction was assigned to A2. A2 realizes that input as the full subcategory defined by its finite-torsion constructibility witness. Consequently the transport functor and its inverse are both the identity on underlying complexes and maps; their two composites are definitional identities. The selected A1 exceptional image is $Rp_*j_!$. This is the S9 definition of $Rf_!^{(j,p)}$. When f is proper, the selected compactification is $(1_X, f)$, so $j_! = 1$ and both expressions are Rf_*. The identity-compactification lemma shows that the projection-formula arrows also agree. Since all transports are identities on representatives, changing only the displayed alias cannot change an object, a morphism, or a coherence equation. □

Lemma 4.11.28 (A2 compatibility with every S7–S10 endpoint). *Lean:* `conditional-interface:lemma:weil-pbc-a2-s7-s10-endpoint-compatibility`

Uses: `def:weil-a2-finite-etale-proper-base-change-map`, `def:weil-a2-transported-finite-level-base-change-map`, `def:weil-a2-proetale-proper-base-change-map`

Let $K \in \mathsf{C}_\square(X)$. After the transports χ and ω:

1. *the A2 finite etale map b_n is the S7 unit–counit exchange map;*
2. *the A2 transported pro-etale map $\widetilde{\beta}_n$ is the S8 map v'_n;*
3. *the A2 product-cone map β_K^{proet} is the S10 completed map ξ_K;*
4. *the n-th reduction of either completed alias is exactly $\widetilde{\beta}_n$, and the equations are compatible with every tower transition, compatible tower endomorphism, identity square, and pasted square.*

All four equalities are independent of the rebinding certificate chosen for K.

Proof. Uses: `lemma:weil-pbc-a2-reduction-tensor-transport`, `lemma:weil-pbc-a2-comparison-rfshriek-transport`, `prop:weil-proet-finite-derived-proper-base-change`, `prop:weil-pbc-proet-finite-level-transport`, `lemma:weil-pbc-bounded-constructible-reconstruction`, `lemma:weil-pbc-completed-pullback-reduction`, `lemma:weil-pbc-completed-tensor-reduction`, `lemma:weil-pbc-projection-formula-identity-compactification`, `lemma:weil-pbc-completed-map-reduction`, `lemma:weil-pbc-completed-exchange-coherence`

The A2 definition of b_n and the S7 definition both use the adjunction unit, the chosen Cartesian equality $fg' = gf'$, and the adjunction counit, in that order. Hence their resolved representatives are the same chain map. By the preceding lemma, the four A2 etale/pro-etale comparison arrows are the four S8 arrows. Pasting the same comparison squares around the common middle map b_n therefore makes $\widetilde{\beta}_n = v'_n$.

The A2 completed map and the S10 completed map are both induced from the tower $(v'_n)_n$ on the product-cone model. The reduction and tensor transports commute with the transitions, so the induced maps on both products are componentwise equal and their cone maps are equal. The completed reduction square then gives

$$\delta_n \, \xi_K = v'_n \, \alpha_n,$$

which is precisely the fourth assertion under χ. S7 naturality, S8 comparison naturality, the S9 tower equations, and S10 completed coherence give the transition, operator, identity, and pasting equations. Finally, the certificate supplies only membership and bound witnesses: none occurs in the unit–counit formula or in the product-cone map. Changing it therefore leaves every displayed morphism unchanged. □

Lemma 4.11.29 (Constructible-subcategory restriction). *Lean:* `conditional-interface:lemma:weil-pbc-complete-constructible-restriction`

Uses: `def:weil-pbc-a2-constructible-rebind`

For every proper Cartesian square, the complete base-change isomorphism restricts to a natural isomorphism on the selected A2 full subcategory:

$$\xi_K : S(K) = Lg^*_{\mathrm{comp}} Rf_* K \xrightarrow{\sim} T(K) = Rf'_* L(g')^*_{\mathrm{comp}} K, \qquad K \in \mathsf{C}_\square(X),$$

where both endpoints lie in $\mathsf{C}(Y')$. Its finite reductions, $Rf_!$ proper aliases, etale/pro-etale transports, compatible tower endomorphisms, and identity and pasting equations are the A2 transports of the S7–S10 endpoints. The restricted isomorphism is independent of all certificate and notation choices.

Proof. Uses: `lemma:weil-pbc-a2-reduction-tensor-transport,lemma:weil-pbc-a2-comparison-rfshriek-transport,lemma:weil-pbc-a2-s7-s10-endpoint-compatibility,prop:weil-proet-complete-proper-base-change`

A rebinding certificate supplies the compatible bounded constructible etale witnesses required in the complete proposition. Its uniform intervals and stratifications are exactly the input of S9 bounded reconstruction, while the perfect coefficient is the two-term resolution used in the reduction transport lemma. Thus Proposition 4.11.24 gives an ambient isomorphism $\xi_K : S(K) \to T(K)$ and its explicit ambient inverse.

The last field of the certificate exhibits $S(K)$ and $T(K)$ as objects of the A2 full subcategory $\mathsf{C}(Y')$. Fullness puts ξ_K and its inverse in that subcategory, and faithfulness preserves their two inverse equations. The three transport lemmas identify all A2 and A3 aliases and prove every finite-reduction and coherence equation. Since the ambient map is constructed without consulting a membership witness, replacing the certificate changes neither the map nor its inverse. This proves the claimed restriction with no additional external hypothesis. □

Chapter 5

Compact-support continuation

Status. Ordinary proper base change is used here only as the proper part of the raw compact-support comparison. Densification, open exchange, refinement independence, and all four displayed coherence diagrams are additional inputs.

A43. Dense compactifications and composition

Let $f : X \to Y$ be separated and finitely presented. A dense compactification is data

$$C = (X \xrightarrow{j_C} \overline{X}_C \xrightarrow{p_C} Y, \epsilon_C : p_C \circ j_C = f), \tag{BA43-DENSE-DATA}$$

where j_C is a schematically dense open immersion and p_C is proper and finitely presented. The equality ϵ_C is stored structure data. A refinement $q : D \to C$ is a Y-map with $qj_D = j_C$ and $q^{-1}(X) = X$. Its properness is a separate geometric atom.

Before any constructible restriction, set

$$F_C^{\mathrm{amb}} = Rp_{C*}Rj_{C!} : D^+(X_{\acute{e}t}, \Lambda_n) \longrightarrow D^+(Y_{\acute{e}t}, \Lambda_n). \tag{BA43-AMBIENT-F}$$

All F_C and all selected $Rf_!$ in this fragment denote these ambient functors. A restriction to D_c^+ is a separate theorem. Consequently $Rg_!^{\mathrm{amb}}Rf_!^{\mathrm{amb}}$ is typed before any assertion that $Rf_!K$ is constructible.

The refinement middle owner constructs

$$\gamma_q : Rq_*Rj_{D!} \xrightarrow{\sim} Rj_{C!}, \qquad \rho_q : R(p_Cq)_*Rj_{D!} \xrightarrow{\sim} Rp_{C*}Rj_{C!}, \tag{BA43-RHO}$$

and proves identity, composition, and required naturality. It also constructs the same proper-support comparison ρ_q^{raw} for a proper map between typed raw models that is the identity over the source open; density is not used for that construction. Only dense models enter the following κ system. For a common dense refinement E, set

$$\kappa_{C,D} = \rho_{E\to D}\,\rho_{E\to C}^{-1}.$$

Schematic-density uniqueness, common refinement, and the equations for ρ must prove independence of E and the κ cocycle.

For $X \xrightarrow{f} Y \xrightarrow{g} Z$, a successive witness includes a Cartesian square

$$\begin{array}{ccc} \overline{X}_f & \xrightarrow{u \text{ schematically dense open}} & W \\ p_f \downarrow & & \downarrow \bar{p} \text{ proper} \\ Y & \xrightarrow{j_g} & \overline{Y}_g \end{array}$$

and a stored equality proving that $u \circ j_f : X \to W$ is schematically dense. The open–proper exchange

$$Rj_{g!}Rp_{f*} \xrightarrow{\sim} R\bar{p}_*Ru_! \tag{BA43-OPX}$$

must satisfy refinement naturality and typed two-square and three-square pasting. The separately constructed isomorphism

$$Ru_!Rj_{f!} \xrightarrow{\sim} R(u \circ j_f)_!$$

has unit, composition, and pentagon laws. Two-witness independence and triple pasting then reassemble the choice-independent comparison $Rg_!Rf_! \to R(gf)_!$. Thus `OBL-BA43-RELATIVE-COHERENT-CORE` remains `OPEN/INCOMPLETE`.

A43/33. Densification after base change

Let $g : Y' \to Y$ be an arbitrary morphism with both Y and Y' noetherian and finite-dimensional. Put $X' = X \times_Y Y'$ and

$$\overline{X}^{\mathrm{raw}}_{C,g} = \overline{X}_C \times_Y Y'.$$

The pulled-back open need not be dense. Define the dense model by schematic closure

$$\overline{X}^{\mathrm{den}}_{C,g} = \overline{X'}^{\mathrm{sch}} \subset \overline{X}^{\mathrm{raw}}_{C,g}, \qquad d_{C,g} : \overline{X}^{\mathrm{den}}_{C,g} \longrightarrow \overline{X}^{\mathrm{raw}}_{C,g}. \tag{BA43-DEN}$$

The schematic-closure leaf proves that $d_{C,g}$ is the resulting closed proper map and that it restricts to the identity on X'.

A raw compactification is data

$$R = (X' \xrightarrow{j_R} \overline{X}_R \xrightarrow{p_R} Y', \epsilon_R : p_R \circ j_R = f'), \tag{BA43-RAW-DATA}$$

with j_R open and p_R proper and finitely presented, but with no density condition. Its stored equation ϵ_R is part of the type, and $F^{\mathrm{amb}}_R = Rp_{R*}Rj_{R!}$ is a functor $D^+(X'_{\acute{e}t}, \Lambda_n) \to D^+(Y'_{\acute{e}t}, \Lambda_n)$. Its D^+_c restriction is asserted only after a constructibility theorem. Raw models are not objects of the dense refinement category and have no κ arrows.

The terminal leaf `CAT-GEO-BA43-DENSE-TO-RAW-SUPPORT` has exactly one output:

$$\gamma^{\mathrm{raw}}_{d_{C,g}} : Rd_{C,g*}Rj_{C^{\mathrm{den}}_g!} \xrightarrow{\sim} Rj_{C^{\mathrm{raw}}_g!}. \tag{BA43-GAMMA-RAW}$$

It outputs neither σ, nor refinement naturality, nor identity, iterated-base-change, or open-exchange coherence. Those are all conclusions of the middle reassembly `OBL-BA43-BASECHANGE-DENSIFICATION`. Using proper-direct-image composition, that middle owner first defines

$$\sigma_{C,g} : F_{C^{\mathrm{den}}_g} \xrightarrow{\sim} F_{C^{\mathrm{raw}}_g}. \tag{BA43-SIGMA}$$

It must then prove the following four separately owned diagrams. For a refinement $q : D \to C$, let $\rho^{\mathrm{den}}_{q,g}$ and $\rho^{\mathrm{raw}}_{q,g}$ be the proper-support comparisons already constructed by `OBL-BA43-REFINEMENT-EXCHANGE` on the densified and raw pullbacks. Refinement naturality is the commutative square

$$\begin{array}{ccc} F_{D^{\mathrm{den}}_g} & \xrightarrow{\sigma_{D,g}} & F_{D^{\mathrm{raw}}_g} \\ {\scriptstyle \rho^{\mathrm{den}}_{q,g}}\downarrow & & \downarrow{\scriptstyle \rho^{\mathrm{raw}}_{q,g}} \\ F_{C^{\mathrm{den}}_g} & \xrightarrow{\sigma_{C,g}} & F_{C^{\mathrm{raw}}_g}. \end{array} \tag{BA43-DEN-REF}$$

After the canonical identifications $C^{\mathrm{den}}_{\mathrm{id}} = C^{\mathrm{raw}}_{\mathrm{id}} = C$, identity coherence is the equation

$$\sigma_{C,\mathrm{id}_Y} = 1_{F_C}. \tag{BA43-DEN-ID}$$

For $Y'' \xrightarrow{h} Y' \xrightarrow{g} Y$, let $\delta_{C;g,h} : (C^{\mathrm{den}}_g)^{\mathrm{den}}_h \xrightarrow{\sim} C^{\mathrm{den}}_{gh}$ be the canonical comparison of the two schematic closures. Let $\tau_{C;g,h}$ be the proper-support comparison from $(C^{\mathrm{den}}_g)^{\mathrm{raw}}_h$ to C^{raw}_{gh} induced by the base change of $d_{C,g}$. Iterated coherence is

$$\begin{array}{ccc} F_{(C^{\mathrm{den}}_g)^{\mathrm{den}}_h} & \xrightarrow{F(\delta_{C;g,h})} & F_{C^{\mathrm{den}}_{gh}} \\ {\scriptstyle \sigma_{C^{\mathrm{den}}_g,h}}\downarrow & & \downarrow{\scriptstyle \sigma_{C,gh}} \\ F_{(C^{\mathrm{den}}_g)^{\mathrm{raw}}_h} & \xrightarrow{\tau_{C;g,h}} & F_{C^{\mathrm{raw}}_{gh}}. \end{array} \tag{BA43-DEN-ITER}$$

Finally let $A = L\bar{g}^*_{\mathrm{raw}}Rj_{C!}K$ and let $s_d : A \xrightarrow{\sim} Rd_{C,g*}Ld^*_{C,g}A$ be the closed-support identification. Construct the raw and dense open-exchange maps independently. Open-exchange coherence is precisely

$$\begin{array}{ccc} A & \xrightarrow{\beta^{\mathrm{raw}}_{jC,g,K}} & Rj_{C^{\mathrm{raw}}_g!}Lg'^*K \\ {\scriptstyle s_d}\downarrow & & \downarrow{\scriptstyle (\gamma^{\mathrm{raw}}_{d_{C,g}})^{-1}} \\ Rd_{C,g*}Ld^*_{C,g}A & \xrightarrow{Rd_{C,g*}(\beta^{\mathrm{den}}_{jC,g,K})} & Rd_{C,g*}Rj_{C^{\mathrm{den}}_g!}Lg'^*K. \end{array} \tag{BA43-DEN-OPX}$$

None of these equations is a field of (BA43-GAMMA-RAW).

A33. Finite compact-support base change

Continue with the same fixed Λ_n, ambient categories, and noetherian finite-dimensional Y, Y'. Let $f : X \to Y$ be separated and finitely presented, let the square in A31 be Cartesian, and let $K \in D_c^+(X, \Lambda_n)$. For the dense compactification C of (BA43-DENSE-DATA), construct the raw one-model map as the following two-arrow composite:

$$\begin{aligned} \mathrm{BC}^{C,\mathrm{raw}}_{f,g,K} : Lg^* F_C(K) &\xrightarrow{\mathrm{BC}_{p_C,g,Rj_{C!}K}} Rp^{\mathrm{raw}}_{C,g,*} L\bar{g}^* Rj_{C!}K \\ &\xrightarrow{Rp^{\mathrm{raw}}_{C,g,*}(\beta^{\mathrm{open}}_{j_C,\bar{g},K})} Rp^{\mathrm{raw}}_{C,g,*} Rj^{\mathrm{raw}}_{C,g,!} Lg'^* K. \end{aligned} \tag{BA33-RAW-BC}$$

Here the first arrow is the already constructed proper base-change mate and the second is the derived open-extension exchange. By definition the name $\mathrm{BC}^{C,\mathrm{raw}}$ denotes their composite, never the first arrow alone. Before finite PBC is applied, the separate leaf `CAT-GEO-BA33-OPEN-EXTENSION-CONSTRUCTIBILITY` must prove $Rj_{C!}K \in D_c^+(\overline{X}_C, \Lambda_n)$.

The dense representative is

$$\mathrm{BC}^{C,\mathrm{den}}_{f,g,K} = \sigma^{-1}_{C,g} \circ \mathrm{BC}^{C,\mathrm{raw}}_{f,g,K} : Lg^* F_C(K) \longrightarrow F_{C^{\mathrm{den}}_g}(Lg'^* K). \tag{BA33-DEN-BC}$$

For a refinement $q : D \to C$, `OBL-BA33-REFINEMENT-BASECHANGE-NATURALITY` proves equality of the two explicit composites built from $Lg^*(\rho_q)$, $\rho_{q^{\mathrm{den}}_g}$, the two maps (BA33-DEN-BC), and the densification comparison. The proof must expand proper-base-change naturality, open-exchange naturality, and all four densification diagrams; the equality is not supplied in a comparison record.

Only then may the κ cocycle descend the representatives to

$$Lg^* Rf_! K \longrightarrow Rf'_! Lg'^* K.$$

Identity and horizontal/vertical pasting consume the corresponding proper, open, densification, and κ equations. Therefore `OBL-BA33-FINITE-COMPACT-SUPPORT-BASE-CHANGE` remains `OPEN/INCOMPLETE`. Geometric-fibre, Frobenius-equivariant, tower, integral, and rational exports are outside this finite-level owner.

Appendix A

Exact unresolved interfaces

Status. Every entry in this appendix remains OPEN. The middle entries are reductions; the leaf entries are the exact terminal proof obligations. No entry is discharged by the extracted prose alone.

A.1 Middle interfaces

Open interface A.1.1 (Exact unresolved interface)**. Owner.** OBL-BA30-PROPER-FINITE-CONSTRUCTIBLE **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: OBL-BA30-A4-SEED-TO-PROPER-FINITE,EXT-BA30-RELATIVE-COHOMOLOGICAL-DIMENSION,FA-BA30-BOUNDED-POSTNIKOV-SERRE

Exact output required. For fixed Lambda_n, a proper finitely presented p:Z->Y of fibre dimension at most d sends D_c^[a,b](Z,Lambda_n) to D_c^[a,b+2d](Y,Lambda_n).

Open obligation: OBL-BA30-PROPER-FINITE-CONSTRUCTIBLE

Open interface A.1.2 (Exact unresolved interface)**. Owner.** OBL-BA30-A4-SEED-TO-PROPER-FINITE **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: EXT-BA30-A4-SEED-CONSTRUCTOR,CAT-BA30-RFSTAR-REBIND,EXT-BA30-RELATIVE-COHOMOLOGICAL-DIMENSION,FA-BA30-BOUNDED-POSTNIKOV-SERRE,OBL-A4-RC-FORMAL-SERRE,OBL-A4-RC-FORMAL-FINITE-DIRECT-IMAGE,OBL-A4-RC-FORMAL-SEED-LEAVES,OBL-A4-RC-FORMAL-AFFINE-LINE-DEVISSAGE,OBL-A4-RC-FORMAL-AFFINE-SPACE,OBL-A4-RC-FORMAL-CLOSED-IMMERSION,OBL-A4-RC-FORMAL-AFFINE-MORPHISM,OBL-A4-RC-FORMAL-PROPER-CECH

Exact output required. An actual literal A4 seed, the ambient Rf_* rebind, the formal A4 chain, the relative 2d bound, and finite Postnikov/Serre closure imply the finite proper theorem.

Open obligation: OBL-BA30-A4-SEED-TO-PROPER-FINITE

Open interface A.1.3 (Exact unresolved interface)**. Owner.** OBL-BA31-FINITE-PBC **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: OBL-BA31-S0-GLOBAL-MATE,OBL-BA31-S1-FINITE,OBL-BA31-S2-STABILITY,OBL-BA31-S3-PROJECTIVE-REDUCTION,OBL-BA31-S4-HENSELIAN-P1,OBL-BA31-S5-STALK-COMPARISON,OBL-BA31-S6-SHEAF-ASSEMBLY,OBL-BA31-S7-BOUNDED-BELOW

Exact output required. For fixed Lambda_n, the globally constructed unit-counit mate is invertible on D_c^+ for a proper finitely presented Cartesian square and obeys mate-calculus identity/horizontal/vertical coherence.

Open obligation: OBL-BA31-FINITE-PBC

Open interface A.1.4 (Exact unresolved interface)**. Owner.** OBL-BA31-S0-GLOBAL-MATE **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: CAT-BA31-SMALL-ETALE-MATE,CAT-BA31-MATE-REPLACEMENT-INDEPENDENCE,CAT-BA31-MATE-COHERENCE

Exact output required. Construct the single ambient mate and prove replacement independence, naturality, identity, and the two pasting equations.

Open obligation: OBL-BA31-S0-GLOBAL-MATE

Open interface A.1.5 (Exact unresolved interface)**. Owner.** OBL-BA31-S1-FINITE **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: OBL-BA31-S0-GLOBAL-MATE,CAT-GEO-BA31-FINITE-FIBRE-PRODUCT

Exact output required. Prove exact finite pushforward and the finite fibre product formula, and identify that map with the stalk of the fixed mate.

Open obligation: OBL-BA31-S1-FINITE

Open interface A.1.6 (Exact unresolved interface)**. Owner.** OBL-BA31-S2-STABILITY **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: OBL-BA31-S0-GLOBAL-MATE,CAT-BA30-RFSTAR-REBIND,FA-CAT-BA31-LERAY-CANCELLATION

Exact output required. Prove typed composition and proper-surjective cancellation for the fixed mate and actual `Rf_*`; no inverse is input data.

Open obligation: `OBL-BA31-S2-STABILITY`

Open interface A.1.7 (Exact unresolved interface)**. Owner.** `OBL-BA31-S3-PROJECTIVE-REDUCTION` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `OBL-BA31-S1-FINITE,OBL-BA31-S2-STABILITY,EXT-GEO-BA31-CHOW-PROPER-FP,GEO-BA31-SYMMETRIC-P1-FINITE-SURJECTIVE`

Exact output required. Give the proper-finitely-presented Chow datum and the finite-surjective retract/cancellation reduction through projective space to products of projective lines.

Open obligation: `OBL-BA31-S3-PROJECTIVE-REDUCTION`

Open interface A.1.8 (Exact unresolved interface)**. Owner.** `OBL-BA31-S4-HENSELIAN-P1` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `EXT-GEO-BA31-HENSELIAN-H0,EXT-GEO-BA31-GABBER-AFFINE,EXT-GEO-BA31-TORSOR-H1`

Exact output required. Assemble independently owned H0, affine, and torsor/H1 inputs into strict-henselian projective-line comparison.

Open obligation: `OBL-BA31-S4-HENSELIAN-P1`

Open interface A.1.9 (Exact unresolved interface)**. Owner.** `OBL-BA31-S5-STALK-COMPARISON` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `OBL-BA31-S0-GLOBAL-MATE,OBL-BA31-S4-HENSELIAN-P1,CAT-GEO-BA31-STRICT-HENSEL-STALK,EXT-GEO-BA31-CURVE-FIELD-INVARIANCE,CAT-BA31-SHEAF-STALK-FAITHFULNESS`

Exact output required. Identify the strict-henselian comparison with each `H^i(BC)`; use sheaf-stalk faithfulness only for equality of already constructed maps of cohomology sheaves.

Open obligation: `OBL-BA31-S5-STALK-COMPARISON`

Open interface A.1.10 (Exact unresolved interface)**. Owner.** `OBL-BA31-S6-SHEAF-ASSEMBLY` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `OBL-BA31-S0-GLOBAL-MATE,OBL-BA31-S1-FINITE,OBL-BA31-S2-STABILITY,OBL-BA31-S3-PROJECTIVE-REDUCTION,OBL-BA31-S5-STALK-COMPARISON,CAT-BA31-DERIVED-STALK-ISO-CONSERVATIVITY`

Exact output required. Apply derived-stalk isomorphism conservativity only to the fixed mate after all its cohomology-stalk maps are invertible.

Open obligation: `OBL-BA31-S6-SHEAF-ASSEMBLY`

Open interface A.1.11 (Exact unresolved interface)**. Owner.** `OBL-BA31-S7-BOUNDED-BELOW` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `OBL-BA31-S0-GLOBAL-MATE,OBL-BA31-S6-SHEAF-ASSEMBLY,FA-CAT-BA31-BOUNDED-BELOW-POSTNIKOV`

Exact output required. Use a selected bounded-below Postnikov/Cartan-Eilenberg construction, convergence, and comparison identification to upgrade the bounded result to `D_c^+`.

Open obligation: `OBL-BA31-S7-BOUNDED-BELOW`

Open interface A.1.12 (Exact unresolved interface)**. Owner.** `OBL-BA43-RELATIVE-COHERENT-CORE` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `GEO-BA43-REFINEMENT-PROPER,GEO-BA43-DENSE-REFINEMENT-UNIQUENESS,GEO-BA43-PAIRWISE-COMMON-REFINEMENT,OBL-BA43-REFINEMENT-EXCHANGE,CAT-BA30-RFSTAR-REBIND,EXT-GEO-BA43-DENSE-SUCCESSIVE-WITNESS,CAT-GEO-BA43-OPEN-PROPER-EXCHANGE,CAT-BA43-OPEN-EXTENSION-COMPOSITION,CAT-BA43-SUCCESSIVE-WITNESS-INDEPENDENCE,CAT-BA43-TRIPLE-PASTING`

Exact output required. For typed dense compactifications, build refinement comparisons, the choice-independent selected `Rf_!`, and unit/composition/associativity from successive witnesses.

Open obligation: `OBL-BA43-RELATIVE-COHERENT-CORE`

Open interface A.1.13 (Exact unresolved interface)**. Owner.** `OBL-BA43-REFINEMENT-EXCHANGE` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `OBL-BA31-FINITE-PBC,CAT-BA30-RFSTAR-REBIND,CAT-BA33-DERIVED-OPEN-EXCHANGE`

Exact output required. Construct `gamma_q` and `rho_q`, their inverses, identity/composition, and downstream naturality for dense refinements; also construct the proper-support `rho_raw` for a proper typed raw-model map that is the identity over the source open. This is a middle node, not a leaf.

Open obligation: `OBL-BA43-REFINEMENT-EXCHANGE`

Open interface A.1.14 (Exact unresolved interface)**. Owner.** `OBL-BA43-BASECHANGE-DENSIFICATION` **Status.** `OPEN`; **reduction status.** `INCOMPLETE`; **class.** `MIDDLE`.

Uses: `GEO-BA43-SCHEMATIC-CLOSURE-DENSIFICATION,CAT-GEO-BA43-DENSE-TO-RAW-SUPPORT,CAT-BA30-RFSTAR-REBIND,OBL-BA43-REFINEMENT-EXCHANGE,CAT-BA43-DENSIFICATION-REFINEMENT-NATURALITY,CAT-BA43-DENSIFICATION-IDENTITY,CAT-BA43-DENSIFICATION-ITERATED-COHERENCE,CAT-BA43-DENSIFICATION-OPEN-EXCHANGE`

Exact output required. For noetherian finite-dimensional `Y,Y'`, build `sigma` from schematic closure and the gamma-raw atom, then prove refinement, identity, iterated-base-change, and open-exchange coherence.

Open obligation: `OBL-BA43-BASECHANGE-DENSIFICATION`

Open interface A.1.15 (Exact unresolved interface)**. Owner.** OBL-BA33-REFINEMENT-BASECHANGE-NATURALITY **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: OBL-BA31-SO-GLOBAL-MATE, CAT-BA33-DERIVED-OPEN-EXCHANGE, CAT-GEO-BA33-OPEN-EXTENSION-CONSTRUCTIBILITY, OBL-BA43-REFINEMENT-EXCHANGE, OBL-BA43-BASECHANGE-DENSIFICATION

Exact output required. Prove equality of the two explicit densified one-model BC composites under refinement after expanding proper BC, open exchange, `rho`, and `sigma`.

Open obligation: OBL-BA33-REFINEMENT-BASECHANGE-NATURALITY

Open interface A.1.16 (Exact unresolved interface)**. Owner.** OBL-BA33-FINITE-COMPACT-SUPPORT-BASE-CHANGE **Status.** OPEN; **reduction status.** INCOMPLETE; **class.** MIDDLE.

Uses: OBL-BA31-FINITE-PBC, CAT-BA33-DERIVED-OPEN-EXCHANGE, CAT-GEO-BA33-OPEN-EXTENSION-CONSTRUCTIBILITY, OBL-BA43-RELATIVE-COHERENT-CORE, OBL-BA43-BASECHANGE-DENSIFICATION, OBL-BA33-REFINEMENT-BASECHANGE-NATURALITY, CAT-BA33-FINITE-BC-PASTING

Exact output required. The raw representative is proper BC followed by open exchange; densify, descend through `kappa`, and prove identity/horizontal/vertical pasting.

Open obligation: OBL-BA33-FINITE-COMPACT-SUPPORT-BASE-CHANGE

A.2 Terminal leaves

Open interface A.2.1 (Exact unresolved interface)**. Owner.** EXT-BA30-A4-SEED-CONSTRUCTOR **Status.** OPEN; **reduction status.** LEAF; **class.** MIXED.

Uses:

Exact output required. Construct the literal `RelativeCurveConstructibilitySeed(T,Lambda_n)` inhabitant: specialization, finite monodromy, compactification, descent/refinement independence, finite affine-line devissage tree, and all coherence fields.

Open obligation: EXT-BA30-A4-SEED-CONSTRUCTOR

Open interface A.2.2 (Exact unresolved interface)**. Owner.** CAT-BA30-RFSTAR-REBIND **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/FA.

Uses:

Exact output required. Identify every consumed historical direct image with the ambient K-injective `Rf_*`; prove naturality, unit, composition, and pentagon.

Open obligation: CAT-BA30-RFSTAR-REBIND

Open interface A.2.3 (Exact unresolved interface)**. Owner.** EXT-BA30-RELATIVE-COHOMOLOGICAL-DIMENSION **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Under the displayed noetherian finite-dimensional and relative-dimension hypotheses, prove the exact vanishing above degree `2d`.

Open obligation: EXT-BA30-RELATIVE-COHOMOLOGICAL-DIMENSION

Open interface A.2.4 (Exact unresolved interface)**. Owner.** FA-BA30-BOUNDED-POSTNIKOV-SERRE **Status.** OPEN; **reduction status.** LEAF; **class.** FA.

Uses:

Exact output required. From constructible cohomology sheaves and the independent degree bound, prove membership in `D_c^[a,b+2d]`.

Open obligation: FA-BA30-BOUNDED-POSTNIKOV-SERRE

Open interface A.2.5 (Exact unresolved interface)**. Owner.** CAT-BA31-SMALL-ETALE-MATE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Construct the unit-cartesian-identification-counit mate in the fixed ambient etale derived categories and prove naturality in `K`.

Open obligation: CAT-BA31-SMALL-ETALE-MATE

Open interface A.2.6 (Exact unresolved interface)**. Owner.** CAT-BA31-MATE-REPLACEMENT-INDEPENDENCE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/FA.

Uses:

Exact output required. Prove the mate is independent of the selected K-injective/K-flat representatives.

Open obligation: CAT-BA31-MATE-REPLACEMENT-INDEPENDENCE

Open interface A.2.7 (Exact unresolved interface)**. Owner.** CAT-BA31-MATE-COHERENCE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Prove identity and horizontal/vertical pasting for the fixed global mate by adjunction calculus.

Open obligation: CAT-BA31-MATE-COHERENCE

Open interface A.2.8 (Exact unresolved interface)**. Owner.** CAT-GEO-BA31-FINITE-FIBRE-PRODUCT **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/GEO.

Uses:

Exact output required. For finite `f`, prove exact pushforward, the finite geometric-fibre product formula, and equality with the stalk map of the global mate.

Open obligation: CAT-GEO-BA31-FINITE-FIBRE-PRODUCT

Open interface A.2.9 (Exact unresolved interface)**. Owner.** FA-CAT-BA31-LERAY-CANCELLATION **Status.** OPEN; **reduction status.** LEAF; **class.** FA/CAT.

Uses:

Exact output required. State and prove the exact Leray/composition cancellation diagram used in proper-surjective descent.

Open obligation: FA-CAT-BA31-LERAY-CANCELLATION

Open interface A.2.10 (Exact unresolved interface)**. Owner.** EXT-GEO-BA31-CHOW-PROPER-FP **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Supply the exact Chow datum for proper finitely presented maps in the stated noetherian finite-dimensional class.

Open obligation: EXT-GEO-BA31-CHOW-PROPER-FP

Open interface A.2.11 (Exact unresolved interface)**. Owner.** GEO-BA31-SYMMETRIC-P1-FINITE-SURJECTIVE **Status.** OPEN; **reduction status.** LEAF; **class.** GEO.

Uses:

Exact output required. Construct the finite-surjective symmetric cover reducing the projective-space case to a product of projective lines, with the required retraction datum.

Open obligation: GEO-BA31-SYMMETRIC-P1-FINITE-SURJECTIVE

Open interface A.2.12 (Exact unresolved interface)**. Owner.** EXT-GEO-BA31-HENSELIAN-H0 **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Prove the degree-zero strict-henselian projective-line comparison.

Open obligation: EXT-GEO-BA31-HENSELIAN-H0

Open interface A.2.13 (Exact unresolved interface)**. Owner.** EXT-GEO-BA31-GABBER-AFFINE **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Prove the exact affine effacement/comparison statement consumed by S4.

Open obligation: EXT-GEO-BA31-GABBER-AFFINE

Open interface A.2.14 (Exact unresolved interface)**. Owner.** EXT-GEO-BA31-TORSOR-H1 **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Prove the finite-etale torsor lifting and degree-one comparison consumed by S4.

Open obligation: EXT-GEO-BA31-TORSOR-H1

Open interface A.2.15 (Exact unresolved interface)**. Owner.** CAT-GEO-BA31-STRICT-HENSEL-STALK **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/GEO.

Uses:

Exact output required. Identify the geometric stalk of `Rf_*K` with cohomology over the strict-henselization, naturally in the fixed mate.

Open obligation: CAT-GEO-BA31-STRICT-HENSEL-STALK

Open interface A.2.16 (Exact unresolved interface)**. Owner.** EXT-GEO-BA31-CURVE-FIELD-INVARIANCE **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Prove the proper-curve separably-closed field-extension invariance needed by S5.

Open obligation: EXT-GEO-BA31-CURVE-FIELD-INVARIANCE

Open interface A.2.17 (Exact unresolved interface)**. Owner.** CAT-BA31-SHEAF-STALK-FAITHFULNESS **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Equality of morphisms in the abelian category of `Lambda_n`-sheaves is detected on all geometric stalks; no derived equality claim.

Open obligation: CAT-BA31-SHEAF-STALK-FAITHFULNESS

Open interface A.2.18 (Exact unresolved interface)**. Owner.** CAT-BA31-DERIVED-STALK-ISO-CONSERVATIVITY **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. An already constructed map in D^+ is invertible if every cohomology-sheaf stalk map is invertible; no fullness or inverse construction.

Open obligation: CAT-BA31-DERIVED-STALK-ISO-CONSERVATIVITY

Open interface A.2.19 (Exact unresolved interface)**. Owner.** FA-CAT-BA31-BOUNDED-BELOW-POSTNIKOV **Status.** OPEN; **reduction status.** LEAF; **class.** FA/CAT.

Uses:

Exact output required. Select bounded truncation/Postnikov or Cartan-Eilenberg models, prove convergence, independence, and identify the induced comparison with the fixed global mate.

Open obligation: FA-CAT-BA31-BOUNDED-BELOW-POSTNIKOV

Open interface A.2.20 (Exact unresolved interface)**. Owner.** GEO-BA43-REFINEMENT-PROPER **Status.** OPEN; **reduction status.** LEAF; **class.** GEO.

Uses:

Exact output required. A refinement between the typed models is proper because its source is proper and its target separated over the base.

Open obligation: GEO-BA43-REFINEMENT-PROPER

Open interface A.2.21 (Exact unresolved interface)**. Owner.** GEO-BA43-DENSE-REFINEMENT-UNIQUENESS **Status.** OPEN; **reduction status.** LEAF; **class.** GEO.

Uses:

Exact output required. Two refinement maps agreeing on the schematically dense source open agree globally when the target is separated.

Open obligation: GEO-BA43-DENSE-REFINEMENT-UNIQUENESS

Open interface A.2.22 (Exact unresolved interface)**. Owner.** GEO-BA43-PAIRWISE-COMMON-REFINEMENT **Status.** OPEN; **reduction status.** LEAF; **class.** GEO.

Uses:

Exact output required. Construct a typed common dense refinement with finitely presented proper structure map.

Open obligation: GEO-BA43-PAIRWISE-COMMON-REFINEMENT

Open interface A.2.23 (Exact unresolved interface)**. Owner.** CAT-GEO-BA43-OPEN-PROPER-EXCHANGE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/GEO.

Uses:

Exact output required. Construct Rj_!Rp_*~=Rbarp_*Ru_! for the stated Cartesian open-proper square and prove refinement naturality and two-/three-square pasting.

Open obligation: CAT-GEO-BA43-OPEN-PROPER-EXCHANGE

Open interface A.2.24 (Exact unresolved interface)**. Owner.** EXT-GEO-BA43-DENSE-SUCCESSIVE-WITNESS **Status.** OPEN; **reduction status.** LEAF; **class.** EXT-GEO.

Uses:

Exact output required. Produce the typed Cartesian successive witness with schematically dense u and schematically dense composite uj_f.

Open obligation: EXT-GEO-BA43-DENSE-SUCCESSIVE-WITNESS

Open interface A.2.25 (Exact unresolved interface)**. Owner.** CAT-BA43-OPEN-EXTENSION-COMPOSITION **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Construct Ru_!Rj_!~=R(uj)_! and prove unit, composition, and pentagon.

Open obligation: CAT-BA43-OPEN-EXTENSION-COMPOSITION

Open interface A.2.26 (Exact unresolved interface)**. Owner.** CAT-BA43-SUCCESSIVE-WITNESS-INDEPENDENCE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Compare two typed successive witnesses through the refinement system and prove the explicit comparison square.

Open obligation: CAT-BA43-SUCCESSIVE-WITNESS-INDEPENDENCE

Open interface A.2.27 (Exact unresolved interface)**. Owner.** CAT-BA43-TRIPLE-PASTING **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Prove the explicit triple-composition pasting equation yielding associativity.

Open obligation: CAT-BA43-TRIPLE-PASTING

Open interface A.2.28 (Exact unresolved interface)**. Owner.** CAT-BA33-DERIVED-OPEN-EXCHANGE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Construct the derived open-extension base-change map Lbarg^*Rj_!->Rj'_!Lg'^* and prove naturality and its identity/horizontal/vertical pasting equations.

Open obligation: CAT-BA33-DERIVED-OPEN-EXCHANGE

Open interface A.2.29 (Exact unresolved interface)**. Owner.** GEO-BA43-SCHEMATIC-CLOSURE-DENSIFICATION **Status.** OPEN; **reduction status.** LEAF; **class.** GEO.

Uses:

Exact output required. For noetherian finite-dimensional Y,Y', the schematic closure of X' in the raw pullback is proper finitely presented over Y', has dense source open, and maps closed-properly to the raw model.

Open obligation: GEO-BA43-SCHEMATIC-CLOSURE-DENSIFICATION

Open interface A.2.30 (Exact unresolved interface)**. Owner.** CAT-GEO-BA43-DENSE-TO-RAW-SUPPORT **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/GEO.

Uses:

Exact output required. Construct only gamma_raw:Rd_*Rj_den!~=Rj_raw! for the schematic-closure closed immersion d_(C,g), which is the identity over the source open. It supplies no sigma and no naturality/coherence fields.

Open obligation: CAT-GEO-BA43-DENSE-TO-RAW-SUPPORT

Open interface A.2.31 (Exact unresolved interface)**. Owner.** CAT-BA43-DENSIFICATION-REFINEMENT-NATURALITY **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Prove square (BA43-DEN-REF): rho_rawsigma_D=sigma_Crho_den for every typed refinement q:D->C.

Open obligation: CAT-BA43-DENSIFICATION-REFINEMENT-NATURALITY

Open interface A.2.32 (Exact unresolved interface)**. Owner.** CAT-BA43-DENSIFICATION-IDENTITY **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Prove equation (BA43-DEN-ID): sigma_(C,id_Y)=1_(F_C) under the canonical raw/dense identity identifications.

Open obligation: CAT-BA43-DENSIFICATION-IDENTITY

Open interface A.2.33 (Exact unresolved interface)**. Owner.** CAT-BA43-DENSIFICATION-ITERATED-COHERENCE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/GEO.

Uses:

Exact output required. Construct delta_(C;g,h) and tau_(C;g,h) with the displayed endpoints and prove square (BA43-DEN-ITER).

Open obligation: CAT-BA43-DENSIFICATION-ITERATED-COHERENCE

Open interface A.2.34 (Exact unresolved interface)**. Owner.** CAT-BA43-DENSIFICATION-OPEN-EXCHANGE **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. For A=Lbar(g)_raw^*Rj_C!K, construct the closed-support map s_d and independently constructed raw/dense beta maps, then prove square (BA43-DEN-OPX).

Open obligation: CAT-BA43-DENSIFICATION-OPEN-EXCHANGE

Open interface A.2.35 (Exact unresolved interface)**. Owner.** CAT-GEO-BA33-OPEN-EXTENSION-CONSTRUCTIBILITY **Status.** OPEN; **reduction status.** LEAF; **class.** CAT/GEO.

Uses:

Exact output required. For the typed open immersion and KinD_c^+, prove Rj_!KinD_c^+ in the exact Lambda_n site consumed by finite PBC.

Open obligation: CAT-GEO-BA33-OPEN-EXTENSION-CONSTRUCTIBILITY

Open interface A.2.36 (Exact unresolved interface)**. Owner.** CAT-BA33-FINITE-BC-PASTING **Status.** OPEN; **reduction status.** LEAF; **class.** CAT.

Uses:

Exact output required. Prove identity and horizontal/vertical pasting for the final compact-support map by combining the already proved proper, open, densification, and kappa diagrams.

Open obligation: CAT-BA33-FINITE-BC-PASTING

Appendix B

Imported dependency registry

Status. These labels occur in an original \usesor cross-reference but are defined outside the extracted S0–S10 block. A red status is an additional explicit proof gap; a green status means only that the frozen manifest records the imported declaration as defined or proved.

Imported interface I.1. `def:weil-a1-rfshriek-selected` **Kind.** `definition`; **manifest status.** extttCONDITIONAL-ON-OPEN-ANCESTOR.

Imported interface I.2. `def:weil-a1-torsion-derived-category` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.3. `def:weil-a2-completed-tensor` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.4. `def:weil-a2-constructible-stratification-witness` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.5. `def:weil-a2-dbc-torsion` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.6. `def:weil-a2-dbc-zl` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.7. `def:weil-a2-derived-category` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.8. `def:weil-a2-etale-origin-datum` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.9. `def:weil-a2-etale-proetale-direct-image-comparison` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.10. `def:weil-a2-etale-proetale-pullback-comparison` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.11. `def:weil-a2-finite-etale-proper-base-change-map` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.12. `def:weil-a2-finite-level-etale-origin-realization` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.13. `def:weil-a2-finite-reduction` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.14. `def:weil-a2-proetale-proper-base-change-map` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.15. `def:weil-a2-proper-complete-base-change-datum` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.16. `def:weil-a2-reduction-transition` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.17. `def:weil-a2-six-operation-coefficient-target` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.18. `def:weil-a2-torsion-constructibility-witness` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.19. `def:weil-a2-transported-finite-level-base-change-map` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.20. `def:weil-small-etale-pretopology` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.21. `def:weil-small-etale-topology` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.22. `def:weil-small-proetale-category` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.23. `def:weil-small-proetale-precoverage` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.24. `def:weil-small-proetale-topology` **Kind.** `definition`; **manifest status.** extttDEFINED-IN-PREFIX.

Imported interface I.25. `lemma:weil-a2-finite-reduction-package-independent` **Kind.** `lemma`; **manifest status.** extttPROVED-IN-PREFIX.

Imported interface I.26. `prop:weil-a2-finite-reduction-functor` **Kind.** `proposition`; **manifest status.** extttPROVED-IN-PREFIX.

Appendix C

Retired or invalid proof routes

Warning C.0.1 (Stalk-family descent is not a proof of PBC). The historical certificate that starts with arbitrary specialization-compatible stalk maps and asks them to descend to a derived morphism is not valid. Geometric stalks detect whether an already constructed morphism is an isomorphism; they are not a fullness theorem for the derived category.

Retired invalid route: `OBL-PBC-FINITE-STALK-DESCENT`

Warning C.0.2 (Certificate projection is conditional only). An input record containing an inverse to the Beck–Chevalley map, both inverse equations, and all coherence fields merely repackages the theorem. Projection from such a record proves only a conditional implication and cannot discharge `OBL-A1-PROPER-BASE-CHANGE-FINITE-TORSION`.

Warning C.0.3 (Completed coefficient shortcut). Coefficient reduction does not commute with proper direct image merely by functoriality. The completed proof must explicitly use a perfect quotient, the projection-formula comparison, compatible finite-level maps, the product-cone model for $R\varprojlim$, and reconstruction. The older A5 shortcut is not an admissible proof source.

> **Status.** The detailed body was retained because it exposes the intended argument. The warnings above and the gap ledger govern its logical status.